In [1]:
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import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import random
import math

# Load dataset
df = pd.read_csv("Clinical_Trial.csv")

# Define reward function
def get_reward(row):
    return 1 if (row['label'] == 0 and row['cd420'] > row['cd40']) else 0

df['reward'] = df.apply(get_reward, axis=1)

arms = df['trt'].unique()
num_arms = len(arms)
iterations = 1000

# Store results
policies_rewards = {}
arm_counts = {}

# --- RANDOM POLICY ---
print("\n--- Random Policy ---")
random_rewards = []
random_counts = np.zeros(num_arms)
for i in range(iterations):
    arm = random.choice(arms)
    reward = df[df['trt'] == arm].sample(1).iloc[0]['reward']
    random_rewards.append(reward)
    random_counts[arm] += 1
    print(f"Iteration {i+1}: Arm = {arm}, Reward = {reward}")
policies_rewards["Random"] = random_rewards
arm_counts["Random"] = random_counts
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--- Random Policy ---
Iteration 1: Arm = 1, Reward = 1.0
Iteration 2: Arm = 3, Reward = 1.0
Iteration 3: Arm = 2, Reward = 0.0
Iteration 4: Arm = 3, Reward = 1.0
Iteration 5: Arm = 0, Reward = 0.0
Iteration 6: Arm = 1, Reward = 1.0
Iteration 7: Arm = 2, Reward = 1.0
Iteration 8: Arm = 2, Reward = 1.0
Iteration 9: Arm = 3, Reward = 1.0
Iteration 10: Arm = 2, Reward = 0.0
Iteration 11: Arm = 0, Reward = 0.0
Iteration 12: Arm = 3, Reward = 1.0
Iteration 13: Arm = 2, Reward = 0.0
Iteration 14: Arm = 1, Reward = 1.0
Iteration 15: Arm = 1, Reward = 0.0
Iteration 16: Arm = 1, Reward = 0.0
Iteration 17: Arm = 3, Reward = 1.0
Iteration 18: Arm = 2, Reward = 0.0
Iteration 19: Arm = 0, Reward = 0.0
Iteration 20: Arm = 2, Reward = 1.0
Iteration 21: Arm = 3, Reward = 0.0
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Iteration 37: Arm = 1, Reward = 1.0
Iteration 38: Arm = 3, Reward = 1.0
Iteration 39: Arm = 2, Reward = 0.0
Iteration 40: Arm = 3, Reward = 0.0
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Iteration 41: Arm = 3, Reward = 0.0
Iteration 42: Arm = 3, Reward = 0.0
Iteration 43: Arm = 2, Reward = 0.0
Iteration 44: Arm = 1, Reward = 1.0
Iteration 45: Arm = 2, Reward = 0.0
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Iteration 290: Arm = 2, Reward = 1.0
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Iteration 478: Arm = 3, Reward = 0.0
Iteration 479: Arm = 3, Reward = 1.0
Iteration 480: Arm = 0, Reward = 0.0
Iteration 481: Arm = 2, Reward = 0.0
Iteration 482: Arm = 1, Reward = 1.0
Iteration 483: Arm = 3, Reward = 0.0
Iteration 484: Arm = 0, Reward = 1.0
Iteration 485: Arm = 1, Reward = 0.0
Iteration 486: Arm = 1, Reward = 0.0
Iteration 487: Arm = 2, Reward = 0.0
Iteration 488: Arm = 0, Reward = 0.0
Iteration 489: Arm = 3, Reward = 1.0
Iteration 490: Arm = 0, Reward = 0.0
Iteration 491: Arm = 0, Reward = 0.0
Iteration 492: Arm = 2, Reward = 0.0
Iteration 493: Arm = 3, Reward = 1.0
Iteration 494: Arm = 2, Reward = 1.0
Iteration 495: Arm = 3, Reward = 1.0
Iteration 496: Arm = 0, Reward = 1.0
Iteration 497: Arm = 2, Reward = 1.0
Iteration 498: Arm = 1, Reward = 1.0
Iteration 499: Arm = 2, Reward = 1.0
Iteration 500: Arm = 3, Reward = 1.0
Iteration 501: Arm = 2, Reward = 1.0
Iteration 502: Arm = 0, Reward = 0.0
Iteration 503: Arm = 0, Reward = 1.0
Iteration 504: Arm = 3, Reward = 1.0
Iteration 505: Arm = 3, Reward = 1.0
Iteration 506: Arm = 1, Reward = 0.0
Iteration 507: Arm = 3, Reward = 0.0
Iteration 508: Arm = 3, Reward = 0.0
Iteration 509: Arm = 3, Reward = 0.0
Iteration 510: Arm = 1, Reward = 1.0
Iteration 511: Arm = 0, Reward = 0.0
Iteration 512: Arm = 3, Reward = 0.0
Iteration 513: Arm = 0, Reward = 0.0
Iteration 514: Arm = 0, Reward = 0.0
Iteration 515: Arm = 2, Reward = 0.0
Iteration 516: Arm = 3, Reward = 0.0
Iteration 517: Arm = 3, Reward = 0.0
Iteration 518: Arm = 1, Reward = 1.0
Iteration 519: Arm = 3, Reward = 1.0
Iteration 520: Arm = 2, Reward = 0.0
Iteration 521: Arm = 0, Reward = 0.0
Iteration 522: Arm = 1, Reward = 1.0
Iteration 523: Arm = 0, Reward = 1.0
Iteration 524: Arm = 3, Reward = 1.0
Iteration 525: Arm = 2, Reward = 1.0
Iteration 526: Arm = 0, Reward = 1.0
Iteration 527: Arm = 0, Reward = 0.0
Iteration 528: Arm = 0, Reward = 0.0
Iteration 529: Arm = 1, Reward = 0.0
Iteration 530: Arm = 2, Reward = 1.0
Iteration 531: Arm = 1, Reward = 0.0
Iteration 532: Arm = 2, Reward = 0.0
Iteration 533: Arm = 3, Reward = 0.0
Iteration 534: Arm = 1, Reward = 0.0
Iteration 535: Arm = 2, Reward = 1.0
Iteration 536: Arm = 3, Reward = 1.0
Iteration 537: Arm = 3, Reward = 0.0
Iteration 538: Arm = 1, Reward = 0.0
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Iteration 539: Arm = 3, Reward = 0.0
Iteration 540: Arm = 3, Reward = 0.0
Iteration 541: Arm = 1, Reward = 0.0
Iteration 542: Arm = 1, Reward = 0.0
Iteration 543: Arm = 0, Reward = 1.0
Iteration 544: Arm = 1, Reward = 0.0
Iteration 545: Arm = 0, Reward = 0.0
Iteration 546: Arm = 1, Reward = 1.0
Iteration 547: Arm = 3, Reward = 0.0
Iteration 548: Arm = 2, Reward = 1.0
Iteration 549: Arm = 1, Reward = 1.0
Iteration 550: Arm = 0, Reward = 1.0
Iteration 551: Arm = 0, Reward = 1.0
Iteration 552: Arm = 3, Reward = 1.0
Iteration 553: Arm = 0, Reward = 0.0
Iteration 554: Arm = 3, Reward = 1.0
Iteration 555: Arm = 0, Reward = 0.0
Iteration 556: Arm = 3, Reward = 1.0
Iteration 557: Arm = 0, Reward = 0.0
Iteration 558: Arm = 1, Reward = 0.0
Iteration 559: Arm = 3, Reward = 1.0
Iteration 560: Arm = 1, Reward = 1.0
Iteration 561: Arm = 3, Reward = 1.0
Iteration 562: Arm = 1, Reward = 0.0
Iteration 563: Arm = 3, Reward = 0.0
Iteration 564: Arm = 2, Reward = 1.0
Iteration 565: Arm = 3, Reward = 0.0
Iteration 566: Arm = 0, Reward = 0.0
Iteration 567: Arm = 2, Reward = 1.0
Iteration 568: Arm = 3, Reward = 0.0
Iteration 569: Arm = 0, Reward = 0.0
Iteration 570: Arm = 0, Reward = 1.0
Iteration 571: Arm = 0, Reward = 0.0
Iteration 572: Arm = 3, Reward = 0.0
Iteration 573: Arm = 1, Reward = 0.0
Iteration 574: Arm = 1, Reward = 0.0
Iteration 575: Arm = 2, Reward = 0.0
Iteration 576: Arm = 1, Reward = 1.0
Iteration 577: Arm = 2, Reward = 0.0
Iteration 578: Arm = 1, Reward = 1.0
Iteration 579: Arm = 3, Reward = 1.0
Iteration 580: Arm = 2, Reward = 0.0
Iteration 581: Arm = 2, Reward = 1.0
Iteration 582: Arm = 0, Reward = 0.0
Iteration 583: Arm = 1, Reward = 0.0
Iteration 584: Arm = 3, Reward = 0.0
Iteration 585: Arm = 0, Reward = 1.0
Iteration 586: Arm = 2, Reward = 0.0
Iteration 587: Arm = 3, Reward = 0.0
Iteration 588: Arm = 1, Reward = 1.0
Iteration 589: Arm = 1, Reward = 1.0
Iteration 590: Arm = 1, Reward = 0.0
Iteration 591: Arm = 1, Reward = 0.0
Iteration 592: Arm = 2, Reward = 0.0
Iteration 593: Arm = 2, Reward = 0.0
Iteration 594: Arm = 3, Reward = 1.0
Iteration 595: Arm = 0, Reward = 0.0
Iteration 596: Arm = 0, Reward = 0.0
Iteration 597: Arm = 0, Reward = 0.0
Iteration 598: Arm = 2, Reward = 0.0
Iteration 599: Arm = 0, Reward = 0.0
Iteration 600: Arm = 1, Reward = 0.0
Iteration 601: Arm = 2, Reward = 0.0
Iteration 602: Arm = 1, Reward = 0.0
Iteration 603: Arm = 2, Reward = 1.0
Iteration 604: Arm = 0, Reward = 0.0
Iteration 605: Arm = 2, Reward = 0.0
Iteration 606: Arm = 0, Reward = 0.0
Iteration 607: Arm = 0, Reward = 1.0
Iteration 608: Arm = 1, Reward = 1.0
Iteration 609: Arm = 3, Reward = 0.0
Iteration 610: Arm = 0, Reward = 0.0
Iteration 611: Arm = 2, Reward = 0.0
Iteration 612: Arm = 0, Reward = 0.0
Iteration 613: Arm = 0, Reward = 0.0
Iteration 614: Arm = 0, Reward = 0.0
Iteration 615: Arm = 1, Reward = 1.0
Iteration 616: Arm = 1, Reward = 0.0
Iteration 617: Arm = 1, Reward = 0.0
Iteration 618: Arm = 3, Reward = 1.0
Iteration 619: Arm = 1, Reward = 0.0
Iteration 620: Arm = 3, Reward = 1.0
Iteration 621: Arm = 0, Reward = 0.0
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Iteration 622: Arm = 0, Reward = 1.0
Iteration 623: Arm = 0, Reward = 1.0
Iteration 624: Arm = 3, Reward = 0.0
Iteration 625: Arm = 3, Reward = 0.0
Iteration 626: Arm = 2, Reward = 0.0
Iteration 627: Arm = 3, Reward = 1.0
Iteration 628: Arm = 2, Reward = 1.0
Iteration 629: Arm = 0, Reward = 0.0
Iteration 630: Arm = 2, Reward = 1.0
Iteration 631: Arm = 1, Reward = 1.0
Iteration 632: Arm = 2, Reward = 0.0
Iteration 633: Arm = 0, Reward = 1.0
Iteration 634: Arm = 3, Reward = 0.0
Iteration 635: Arm = 1, Reward = 0.0
Iteration 636: Arm = 3, Reward = 1.0
Iteration 637: Arm = 3, Reward = 1.0
Iteration 638: Arm = 3, Reward = 0.0
Iteration 639: Arm = 3, Reward = 0.0
Iteration 640: Arm = 2, Reward = 1.0
Iteration 641: Arm = 2, Reward = 0.0
Iteration 642: Arm = 2, Reward = 0.0
Iteration 643: Arm = 3, Reward = 1.0
Iteration 644: Arm = 0, Reward = 0.0
Iteration 645: Arm = 3, Reward = 1.0
Iteration 646: Arm = 1, Reward = 1.0
Iteration 647: Arm = 2, Reward = 0.0
Iteration 648: Arm = 3, Reward = 0.0
Iteration 649: Arm = 3, Reward = 1.0
Iteration 650: Arm = 1, Reward = 0.0
Iteration 651: Arm = 2, Reward = 1.0
Iteration 652: Arm = 2, Reward = 1.0
Iteration 653: Arm = 2, Reward = 0.0
Iteration 654: Arm = 3, Reward = 1.0
Iteration 655: Arm = 3, Reward = 1.0
Iteration 656: Arm = 3, Reward = 1.0
Iteration 657: Arm = 1, Reward = 1.0
Iteration 658: Arm = 1, Reward = 0.0
Iteration 659: Arm = 3, Reward = 0.0
Iteration 660: Arm = 1, Reward = 1.0
Iteration 661: Arm = 0, Reward = 0.0
Iteration 662: Arm = 2, Reward = 1.0
Iteration 663: Arm = 0, Reward = 1.0
Iteration 664: Arm = 1, Reward = 1.0
Iteration 665: Arm = 0, Reward = 0.0
Iteration 666: Arm = 1, Reward = 1.0
Iteration 667: Arm = 2, Reward = 1.0
Iteration 668: Arm = 0, Reward = 0.0
Iteration 669: Arm = 3, Reward = 0.0
Iteration 670: Arm = 0, Reward = 1.0
Iteration 671: Arm = 2, Reward = 0.0
Iteration 672: Arm = 0, Reward = 1.0
Iteration 673: Arm = 3, Reward = 1.0
Iteration 674: Arm = 1, Reward = 0.0
Iteration 675: Arm = 2, Reward = 1.0
Iteration 676: Arm = 1, Reward = 1.0
Iteration 677: Arm = 0, Reward = 1.0
Iteration 678: Arm = 3, Reward = 0.0
Iteration 679: Arm = 1, Reward = 1.0
Iteration 680: Arm = 2, Reward = 0.0
Iteration 681: Arm = 1, Reward = 1.0
Iteration 682: Arm = 0, Reward = 0.0
Iteration 683: Arm = 2, Reward = 0.0
Iteration 684: Arm = 3, Reward = 1.0
Iteration 685: Arm = 1, Reward = 1.0
Iteration 686: Arm = 2, Reward = 1.0
Iteration 687: Arm = 3, Reward = 0.0
Iteration 688: Arm = 3, Reward = 1.0
Iteration 689: Arm = 2, Reward = 1.0
Iteration 690: Arm = 2, Reward = 1.0
Iteration 691: Arm = 0, Reward = 0.0
Iteration 692: Arm = 3, Reward = 1.0
Iteration 693: Arm = 0, Reward = 0.0
Iteration 694: Arm = 0, Reward = 1.0
Iteration 695: Arm = 2, Reward = 0.0
Iteration 696: Arm = 0, Reward = 1.0
Iteration 697: Arm = 2, Reward = 1.0
Iteration 698: Arm = 3, Reward = 1.0
Iteration 699: Arm = 3, Reward = 0.0
Iteration 700: Arm = 1, Reward = 0.0
Iteration 701: Arm = 0, Reward = 0.0
Iteration 702: Arm = 0, Reward = 0.0
Iteration 703: Arm = 0, Reward = 0.0
Iteration 704: Arm = 3, Reward = 0.0
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Iteration 705: Arm = 3, Reward = 0.0
Iteration 706: Arm = 0, Reward = 1.0
Iteration 707: Arm = 3, Reward = 1.0
Iteration 708: Arm = 1, Reward = 1.0
Iteration 709: Arm = 3, Reward = 0.0
Iteration 710: Arm = 1, Reward = 0.0
Iteration 711: Arm = 1, Reward = 1.0
Iteration 712: Arm = 1, Reward = 1.0
Iteration 713: Arm = 3, Reward = 0.0
Iteration 714: Arm = 2, Reward = 1.0
Iteration 715: Arm = 2, Reward = 0.0
Iteration 716: Arm = 1, Reward = 1.0
Iteration 717: Arm = 0, Reward = 0.0
Iteration 718: Arm = 1, Reward = 0.0
Iteration 719: Arm = 3, Reward = 0.0
Iteration 720: Arm = 2, Reward = 1.0
Iteration 721: Arm = 2, Reward = 1.0
Iteration 722: Arm = 0, Reward = 1.0
Iteration 723: Arm = 3, Reward = 1.0
Iteration 724: Arm = 0, Reward = 0.0
Iteration 725: Arm = 2, Reward = 0.0
Iteration 726: Arm = 3, Reward = 1.0
Iteration 727: Arm = 1, Reward = 0.0
Iteration 728: Arm = 3, Reward = 1.0
Iteration 729: Arm = 0, Reward = 1.0
Iteration 730: Arm = 2, Reward = 1.0
Iteration 731: Arm = 0, Reward = 0.0
Iteration 732: Arm = 3, Reward = 0.0
Iteration 733: Arm = 2, Reward = 1.0
Iteration 734: Arm = 1, Reward = 0.0
Iteration 735: Arm = 3, Reward = 1.0
Iteration 736: Arm = 1, Reward = 1.0
Iteration 737: Arm = 1, Reward = 0.0
Iteration 738: Arm = 0, Reward = 0.0
Iteration 739: Arm = 0, Reward = 1.0
Iteration 740: Arm = 3, Reward = 0.0
Iteration 741: Arm = 1, Reward = 0.0
Iteration 742: Arm = 0, Reward = 1.0
Iteration 743: Arm = 0, Reward = 0.0
Iteration 744: Arm = 0, Reward = 0.0
Iteration 745: Arm = 1, Reward = 1.0
Iteration 746: Arm = 0, Reward = 0.0
Iteration 747: Arm = 2, Reward = 1.0
Iteration 748: Arm = 3, Reward = 0.0
Iteration 749: Arm = 3, Reward = 1.0
Iteration 750: Arm = 1, Reward = 0.0
Iteration 751: Arm = 3, Reward = 0.0
Iteration 752: Arm = 0, Reward = 0.0
Iteration 753: Arm = 1, Reward = 1.0
Iteration 754: Arm = 0, Reward = 0.0
Iteration 755: Arm = 3, Reward = 1.0
Iteration 756: Arm = 1, Reward = 1.0
Iteration 757: Arm = 0, Reward = 1.0
Iteration 758: Arm = 1, Reward = 0.0
Iteration 759: Arm = 3, Reward = 1.0
Iteration 760: Arm = 3, Reward = 1.0
Iteration 761: Arm = 0, Reward = 0.0
Iteration 762: Arm = 1, Reward = 0.0
Iteration 763: Arm = 1, Reward = 0.0
Iteration 764: Arm = 3, Reward = 1.0
Iteration 765: Arm = 0, Reward = 0.0
Iteration 766: Arm = 2, Reward = 0.0
Iteration 767: Arm = 2, Reward = 1.0
Iteration 768: Arm = 0, Reward = 0.0
Iteration 769: Arm = 3, Reward = 0.0
Iteration 770: Arm = 2, Reward = 1.0
Iteration 771: Arm = 2, Reward = 1.0
Iteration 772: Arm = 1, Reward = 0.0
Iteration 773: Arm = 0, Reward = 0.0
Iteration 774: Arm = 1, Reward = 1.0
Iteration 775: Arm = 0, Reward = 1.0
Iteration 776: Arm = 2, Reward = 1.0
Iteration 777: Arm = 1, Reward = 1.0
Iteration 778: Arm = 0, Reward = 1.0
Iteration 779: Arm = 3, Reward = 1.0
Iteration 780: Arm = 2, Reward = 0.0
Iteration 781: Arm = 1, Reward = 0.0
Iteration 782: Arm = 1, Reward = 1.0
Iteration 783: Arm = 2, Reward = 1.0
Iteration 784: Arm = 0, Reward = 1.0
Iteration 785: Arm = 3, Reward = 0.0
Iteration 786: Arm = 1, Reward = 1.0
Iteration 787: Arm = 3, Reward = 0.0
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Iteration 788: Arm = 3, Reward = 1.0
Iteration 789: Arm = 1, Reward = 1.0
Iteration 790: Arm = 3, Reward = 0.0
Iteration 791: Arm = 3, Reward = 1.0
Iteration 792: Arm = 0, Reward = 0.0
Iteration 793: Arm = 2, Reward = 1.0
Iteration 794: Arm = 3, Reward = 1.0
Iteration 795: Arm = 0, Reward = 0.0
Iteration 796: Arm = 3, Reward = 1.0
Iteration 797: Arm = 3, Reward = 1.0
Iteration 798: Arm = 2, Reward = 1.0
Iteration 799: Arm = 3, Reward = 1.0
Iteration 800: Arm = 2, Reward = 0.0
Iteration 801: Arm = 1, Reward = 1.0
Iteration 802: Arm = 0, Reward = 0.0
Iteration 803: Arm = 2, Reward = 0.0
Iteration 804: Arm = 2, Reward = 1.0
Iteration 805: Arm = 2, Reward = 1.0
Iteration 806: Arm = 3, Reward = 1.0
Iteration 807: Arm = 2, Reward = 0.0
Iteration 808: Arm = 3, Reward = 1.0
Iteration 809: Arm = 1, Reward = 1.0
Iteration 810: Arm = 2, Reward = 0.0
Iteration 811: Arm = 2, Reward = 0.0
Iteration 812: Arm = 2, Reward = 1.0
Iteration 813: Arm = 1, Reward = 0.0
Iteration 814: Arm = 2, Reward = 0.0
Iteration 815: Arm = 0, Reward = 0.0
Iteration 816: Arm = 2, Reward = 0.0
Iteration 817: Arm = 2, Reward = 1.0
Iteration 818: Arm = 1, Reward = 0.0
Iteration 819: Arm = 0, Reward = 1.0
Iteration 820: Arm = 2, Reward = 1.0
Iteration 821: Arm = 3, Reward = 0.0
Iteration 822: Arm = 2, Reward = 0.0
Iteration 823: Arm = 3, Reward = 0.0
Iteration 824: Arm = 1, Reward = 0.0
Iteration 825: Arm = 1, Reward = 0.0
Iteration 826: Arm = 2, Reward = 0.0
Iteration 827: Arm = 3, Reward = 0.0
Iteration 828: Arm = 1, Reward = 1.0
Iteration 829: Arm = 0, Reward = 1.0
Iteration 830: Arm = 1, Reward = 1.0
Iteration 831: Arm = 0, Reward = 1.0
Iteration 832: Arm = 2, Reward = 1.0
Iteration 833: Arm = 1, Reward = 0.0
Iteration 834: Arm = 2, Reward = 1.0
Iteration 835: Arm = 1, Reward = 0.0
Iteration 836: Arm = 3, Reward = 1.0
Iteration 837: Arm = 2, Reward = 0.0
Iteration 838: Arm = 3, Reward = 1.0
Iteration 839: Arm = 1, Reward = 0.0
Iteration 840: Arm = 3, Reward = 1.0
Iteration 841: Arm = 0, Reward = 1.0
Iteration 842: Arm = 0, Reward = 0.0
Iteration 843: Arm = 0, Reward = 0.0
Iteration 844: Arm = 3, Reward = 1.0
Iteration 845: Arm = 2, Reward = 1.0
Iteration 846: Arm = 1, Reward = 1.0
Iteration 847: Arm = 3, Reward = 0.0
Iteration 848: Arm = 3, Reward = 0.0
Iteration 849: Arm = 2, Reward = 1.0
Iteration 850: Arm = 3, Reward = 1.0
Iteration 851: Arm = 3, Reward = 1.0
Iteration 852: Arm = 2, Reward = 0.0
Iteration 853: Arm = 1, Reward = 1.0
Iteration 854: Arm = 1, Reward = 0.0
Iteration 855: Arm = 0, Reward = 1.0
Iteration 856: Arm = 0, Reward = 0.0
Iteration 857: Arm = 2, Reward = 1.0
Iteration 858: Arm = 1, Reward = 1.0
Iteration 859: Arm = 1, Reward = 1.0
Iteration 860: Arm = 1, Reward = 1.0
Iteration 861: Arm = 2, Reward = 0.0
Iteration 862: Arm = 0, Reward = 0.0
Iteration 863: Arm = 3, Reward = 0.0
Iteration 864: Arm = 1, Reward = 0.0
Iteration 865: Arm = 2, Reward = 1.0
Iteration 866: Arm = 0, Reward = 0.0
Iteration 867: Arm = 2, Reward = 0.0
Iteration 868: Arm = 0, Reward = 1.0
Iteration 869: Arm = 1, Reward = 0.0
Iteration 870: Arm = 3, Reward = 0.0
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Iteration 871: Arm = 3, Reward = 0.0
Iteration 872: Arm = 0, Reward = 0.0
Iteration 873: Arm = 3, Reward = 1.0
Iteration 874: Arm = 3, Reward = 1.0
Iteration 875: Arm = 2, Reward = 0.0
Iteration 876: Arm = 1, Reward = 1.0
Iteration 877: Arm = 1, Reward = 0.0
Iteration 878: Arm = 2, Reward = 0.0
Iteration 879: Arm = 2, Reward = 1.0
Iteration 880: Arm = 1, Reward = 0.0
Iteration 881: Arm = 0, Reward = 0.0
Iteration 882: Arm = 2, Reward = 1.0
Iteration 883: Arm = 0, Reward = 0.0
Iteration 884: Arm = 0, Reward = 1.0
Iteration 885: Arm = 2, Reward = 1.0
Iteration 886: Arm = 1, Reward = 0.0
Iteration 887: Arm = 0, Reward = 0.0
Iteration 888: Arm = 0, Reward = 0.0
Iteration 889: Arm = 0, Reward = 0.0
Iteration 890: Arm = 0, Reward = 0.0
Iteration 891: Arm = 0, Reward = 1.0
Iteration 892: Arm = 3, Reward = 0.0
Iteration 893: Arm = 0, Reward = 1.0
Iteration 894: Arm = 2, Reward = 0.0
Iteration 895: Arm = 0, Reward = 1.0
Iteration 896: Arm = 3, Reward = 0.0
Iteration 897: Arm = 3, Reward = 0.0
Iteration 898: Arm = 1, Reward = 1.0
Iteration 899: Arm = 2, Reward = 0.0
Iteration 900: Arm = 2, Reward = 0.0
Iteration 901: Arm = 2, Reward = 1.0
Iteration 902: Arm = 0, Reward = 1.0
Iteration 903: Arm = 1, Reward = 0.0
Iteration 904: Arm = 3, Reward = 1.0
Iteration 905: Arm = 2, Reward = 1.0
Iteration 906: Arm = 1, Reward = 0.0
Iteration 907: Arm = 2, Reward = 1.0
Iteration 908: Arm = 0, Reward = 0.0
Iteration 909: Arm = 2, Reward = 1.0
Iteration 910: Arm = 3, Reward = 1.0
Iteration 911: Arm = 0, Reward = 1.0
Iteration 912: Arm = 0, Reward = 1.0
Iteration 913: Arm = 0, Reward = 0.0
Iteration 914: Arm = 1, Reward = 0.0
Iteration 915: Arm = 2, Reward = 0.0
Iteration 916: Arm = 1, Reward = 1.0
Iteration 917: Arm = 3, Reward = 1.0
Iteration 918: Arm = 0, Reward = 0.0
Iteration 919: Arm = 0, Reward = 0.0
Iteration 920: Arm = 1, Reward = 1.0
Iteration 921: Arm = 2, Reward = 1.0
Iteration 922: Arm = 2, Reward = 0.0
Iteration 923: Arm = 2, Reward = 1.0
Iteration 924: Arm = 3, Reward = 0.0
Iteration 925: Arm = 3, Reward = 1.0
Iteration 926: Arm = 3, Reward = 0.0
Iteration 927: Arm = 0, Reward = 0.0
Iteration 928: Arm = 3, Reward = 0.0
Iteration 929: Arm = 0, Reward = 0.0
Iteration 930: Arm = 2, Reward = 0.0
Iteration 931: Arm = 1, Reward = 1.0
Iteration 932: Arm = 2, Reward = 1.0
Iteration 933: Arm = 3, Reward = 1.0
Iteration 934: Arm = 3, Reward = 1.0
Iteration 935: Arm = 3, Reward = 1.0
Iteration 936: Arm = 1, Reward = 0.0
Iteration 937: Arm = 0, Reward = 1.0
Iteration 938: Arm = 0, Reward = 0.0
Iteration 939: Arm = 0, Reward = 0.0
Iteration 940: Arm = 1, Reward = 0.0
Iteration 941: Arm = 2, Reward = 1.0
Iteration 942: Arm = 3, Reward = 1.0
Iteration 943: Arm = 3, Reward = 1.0
Iteration 944: Arm = 0, Reward = 0.0
Iteration 945: Arm = 3, Reward = 1.0
Iteration 946: Arm = 0, Reward = 0.0
Iteration 947: Arm = 0, Reward = 1.0
Iteration 948: Arm = 0, Reward = 0.0
Iteration 949: Arm = 2, Reward = 0.0
Iteration 950: Arm = 2, Reward = 1.0
Iteration 951: Arm = 0, Reward = 1.0
Iteration 952: Arm = 3, Reward = 1.0
Iteration 953: Arm = 1, Reward = 1.0
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Iteration 954: Arm = 2, Reward = 1.0
Iteration 955: Arm = 0, Reward = 0.0
Iteration 956: Arm = 3, Reward = 1.0
Iteration 957: Arm = 3, Reward = 1.0
Iteration 958: Arm = 3, Reward = 0.0
Iteration 959: Arm = 0, Reward = 0.0
Iteration 960: Arm = 1, Reward = 1.0
Iteration 961: Arm = 0, Reward = 1.0
Iteration 962: Arm = 0, Reward = 1.0
Iteration 963: Arm = 1, Reward = 1.0
Iteration 964: Arm = 3, Reward = 0.0
Iteration 965: Arm = 1, Reward = 1.0
Iteration 966: Arm = 1, Reward = 0.0
Iteration 967: Arm = 1, Reward = 1.0
Iteration 968: Arm = 1, Reward = 1.0
Iteration 969: Arm = 1, Reward = 1.0
Iteration 970: Arm = 0, Reward = 0.0
Iteration 971: Arm = 0, Reward = 0.0
Iteration 972: Arm = 2, Reward = 1.0
Iteration 973: Arm = 0, Reward = 0.0
Iteration 974: Arm = 0, Reward = 0.0
Iteration 975: Arm = 0, Reward = 0.0
Iteration 976: Arm = 2, Reward = 1.0
Iteration 977: Arm = 1, Reward = 0.0
Iteration 978: Arm = 1, Reward = 1.0
Iteration 979: Arm = 1, Reward = 1.0
Iteration 980: Arm = 3, Reward = 0.0
Iteration 981: Arm = 0, Reward = 0.0
Iteration 982: Arm = 2, Reward = 0.0
Iteration 983: Arm = 3, Reward = 0.0
Iteration 984: Arm = 1, Reward = 1.0
Iteration 985: Arm = 0, Reward = 1.0
Iteration 986: Arm = 0, Reward = 0.0
Iteration 987: Arm = 0, Reward = 0.0
Iteration 988: Arm = 0, Reward = 0.0
Iteration 989: Arm = 3, Reward = 0.0
Iteration 990: Arm = 2, Reward = 1.0
Iteration 991: Arm = 2, Reward = 0.0
Iteration 992: Arm = 1, Reward = 1.0
Iteration 993: Arm = 0, Reward = 0.0
Iteration 994: Arm = 0, Reward = 0.0
Iteration 995: Arm = 0, Reward = 0.0
Iteration 996: Arm = 2, Reward = 1.0
Iteration 997: Arm = 2, Reward = 0.0
Iteration 998: Arm = 0, Reward = 0.0
Iteration 999: Arm = 1, Reward = 0.0
Iteration 1000: Arm = 3, Reward = 1.0
```

In [2]:

```python
# --- GREEDY POLICY ---
print("\n--- Greedy Policy ---")
greedy_rewards = []
greedy_estimates = np.zeros(num_arms)
greedy_counts = np.zeros(num_arms)
for i in range(iterations):
    if i < num_arms:
        arm = i
    else:
        arm = np.argmax(greedy_estimates)
    reward = df[df['trt'] == arm].sample(1).iloc[0]['reward']
    greedy_counts[arm] += 1
    greedy_estimates[arm] += (reward - greedy_estimates[arm]) / greedy_counts[arm]
    greedy_rewards.append(reward)
    print(f"Iteration {i+1}: Arm = {arm}, Reward = {reward}")
policies_rewards["Greedy"] = greedy_rewards
arm_counts["Greedy"] = greedy_counts
```

```
--- Greedy Policy ---
Iteration 1: Arm = 0, Reward = 1.0
Iteration 2: Arm = 1, Reward = 0.0
Iteration 3: Arm = 2, Reward = 0.0
Iteration 4: Arm = 3, Reward = 1.0
Iteration 5: Arm = 0, Reward = 1.0
Iteration 6: Arm = 0, Reward = 0.0
Iteration 7: Arm = 3, Reward = 1.0
Iteration 8: Arm = 3, Reward = 0.0
Iteration 9: Arm = 0, Reward = 1.0
Iteration 10: Arm = 0, Reward = 1.0
Iteration 11: Arm = 0, Reward = 1.0
Iteration 12: Arm = 0, Reward = 1.0
Iteration 13: Arm = 0, Reward = 1.0
Iteration 14: Arm = 0, Reward = 0.0
Iteration 15: Arm = 0, Reward = 0.0
Iteration 16: Arm = 0, Reward = 0.0
Iteration 17: Arm = 3, Reward = 0.0
```

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Iteration 18: Arm = 0, Reward = 0.0
Iteration 19: Arm = 0, Reward = 0.0
Iteration 20: Arm = 0, Reward = 1.0
Iteration 21: Arm = 0, Reward = 0.0
Iteration 22: Arm = 0, Reward = 1.0
Iteration 23: Arm = 0, Reward = 0.0
Iteration 24: Arm = 0, Reward = 0.0
Iteration 25: Arm = 0, Reward = 1.0
Iteration 26: Arm = 0, Reward = 1.0
Iteration 27: Arm = 0, Reward = 0.0
Iteration 28: Arm = 0, Reward = 0.0
Iteration 29: Arm = 0, Reward = 0.0
Iteration 30: Arm = 3, Reward = 1.0
Iteration 31: Arm = 3, Reward = 1.0
Iteration 32: Arm = 3, Reward = 1.0
Iteration 33: Arm = 3, Reward = 1.0
Iteration 34: Arm = 3, Reward = 1.0
Iteration 35: Arm = 3, Reward = 0.0
Iteration 36: Arm = 3, Reward = 0.0
Iteration 37: Arm = 3, Reward = 0.0
Iteration 38: Arm = 3, Reward = 0.0
Iteration 39: Arm = 3, Reward = 0.0
Iteration 40: Arm = 3, Reward = 1.0
Iteration 41: Arm = 3, Reward = 1.0
Iteration 42: Arm = 3, Reward = 1.0
Iteration 43: Arm = 3, Reward = 1.0
Iteration 44: Arm = 3, Reward = 0.0
Iteration 45: Arm = 3, Reward = 0.0
Iteration 46: Arm = 3, Reward = 0.0
Iteration 47: Arm = 3, Reward = 1.0
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Iteration 50: Arm = 3, Reward = 0.0
Iteration 51: Arm = 3, Reward = 1.0
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Iteration 55: Arm = 3, Reward = 0.0
Iteration 56: Arm = 3, Reward = 1.0
Iteration 57: Arm = 3, Reward = 0.0
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Iteration 60: Arm = 0, Reward = 0.0
Iteration 61: Arm = 3, Reward = 1.0
Iteration 62: Arm = 3, Reward = 0.0
Iteration 63: Arm = 3, Reward = 1.0
Iteration 64: Arm = 3, Reward = 0.0
Iteration 65: Arm = 3, Reward = 0.0
Iteration 66: Arm = 3, Reward = 1.0
Iteration 67: Arm = 3, Reward = 0.0
Iteration 68: Arm = 3, Reward = 1.0
Iteration 69: Arm = 3, Reward = 0.0
Iteration 70: Arm = 3, Reward = 0.0
Iteration 71: Arm = 0, Reward = 0.0
Iteration 72: Arm = 3, Reward = 1.0
Iteration 73: Arm = 3, Reward = 0.0
Iteration 74: Arm = 3, Reward = 0.0
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Iteration 76: Arm = 0, Reward = 0.0
Iteration 77: Arm = 3, Reward = 1.0
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Iteration 79: Arm = 3, Reward = 0.0
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Iteration 94: Arm = 3, Reward = 1.0
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Iteration 96: Arm = 3, Reward = 1.0
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Iteration 98: Arm = 3, Reward = 1.0
Iteration 99: Arm = 3, Reward = 0.0
Iteration 100: Arm = 3, Reward = 0.0
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Iteration 101: Arm = 3, Reward = 1.0
Iteration 102: Arm = 3, Reward = 0.0
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Iteration 106: Arm = 3, Reward = 1.0
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Iteration 350: Arm = 3, Reward = 1.0
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Iteration 513: Arm = 3, Reward = 1.0
Iteration 514: Arm = 3, Reward = 0.0
Iteration 515: Arm = 3, Reward = 1.0
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Iteration 516: Arm = 3, Reward = 1.0
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Iteration 580: Arm = 3, Reward = 0.0
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Iteration 584: Arm = 3, Reward = 0.0
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Iteration 597: Arm = 3, Reward = 1.0
Iteration 598: Arm = 3, Reward = 0.0
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Iteration 599: Arm = 3, Reward = 0.0
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Iteration 603: Arm = 3, Reward = 0.0
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Iteration 617: Arm = 3, Reward = 0.0
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Iteration 633: Arm = 3, Reward = 1.0
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Iteration 640: Arm = 3, Reward = 0.0
Iteration 641: Arm = 3, Reward = 1.0
Iteration 642: Arm = 3, Reward = 0.0
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Iteration 645: Arm = 3, Reward = 1.0
Iteration 646: Arm = 3, Reward = 1.0
Iteration 647: Arm = 3, Reward = 0.0
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Iteration 668: Arm = 3, Reward = 1.0
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Iteration 671: Arm = 3, Reward = 1.0
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Iteration 674: Arm = 3, Reward = 0.0
Iteration 675: Arm = 3, Reward = 0.0
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Iteration 677: Arm = 3, Reward = 0.0
Iteration 678: Arm = 3, Reward = 0.0
Iteration 679: Arm = 3, Reward = 1.0
Iteration 680: Arm = 3, Reward = 0.0
Iteration 681: Arm = 3, Reward = 1.0
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Iteration 682: Arm = 3, Reward = 0.0
Iteration 683: Arm = 3, Reward = 0.0
Iteration 684: Arm = 3, Reward = 1.0
Iteration 685: Arm = 3, Reward = 1.0
Iteration 686: Arm = 3, Reward = 0.0
Iteration 687: Arm = 3, Reward = 1.0
Iteration 688: Arm = 3, Reward = 0.0
Iteration 689: Arm = 3, Reward = 0.0
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Iteration 691: Arm = 3, Reward = 0.0
Iteration 692: Arm = 3, Reward = 1.0
Iteration 693: Arm = 3, Reward = 0.0
Iteration 694: Arm = 3, Reward = 1.0
Iteration 695: Arm = 3, Reward = 1.0
Iteration 696: Arm = 3, Reward = 0.0
Iteration 697: Arm = 3, Reward = 0.0
Iteration 698: Arm = 3, Reward = 0.0
Iteration 699: Arm = 3, Reward = 0.0
Iteration 700: Arm = 3, Reward = 1.0
Iteration 701: Arm = 3, Reward = 1.0
Iteration 702: Arm = 3, Reward = 0.0
Iteration 703: Arm = 3, Reward = 0.0
Iteration 704: Arm = 3, Reward = 1.0
Iteration 705: Arm = 3, Reward = 0.0
Iteration 706: Arm = 3, Reward = 1.0
Iteration 707: Arm = 3, Reward = 0.0
Iteration 708: Arm = 3, Reward = 1.0
Iteration 709: Arm = 3, Reward = 0.0
Iteration 710: Arm = 3, Reward = 1.0
Iteration 711: Arm = 3, Reward = 0.0
Iteration 712: Arm = 3, Reward = 0.0
Iteration 713: Arm = 3, Reward = 1.0
Iteration 714: Arm = 3, Reward = 0.0
Iteration 715: Arm = 3, Reward = 0.0
Iteration 716: Arm = 3, Reward = 1.0
Iteration 717: Arm = 3, Reward = 0.0
Iteration 718: Arm = 3, Reward = 1.0
Iteration 719: Arm = 3, Reward = 0.0
Iteration 720: Arm = 3, Reward = 1.0
Iteration 721: Arm = 3, Reward = 0.0
Iteration 722: Arm = 3, Reward = 0.0
Iteration 723: Arm = 3, Reward = 0.0
Iteration 724: Arm = 3, Reward = 0.0
Iteration 725: Arm = 3, Reward = 1.0
Iteration 726: Arm = 3, Reward = 0.0
Iteration 727: Arm = 3, Reward = 1.0
Iteration 728: Arm = 3, Reward = 0.0
Iteration 729: Arm = 3, Reward = 0.0
Iteration 730: Arm = 3, Reward = 0.0
Iteration 731: Arm = 3, Reward = 0.0
Iteration 732: Arm = 3, Reward = 1.0
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Iteration 734: Arm = 3, Reward = 1.0
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Iteration 737: Arm = 3, Reward = 1.0
Iteration 738: Arm = 3, Reward = 0.0
Iteration 739: Arm = 3, Reward = 0.0
Iteration 740: Arm = 3, Reward = 1.0
Iteration 741: Arm = 3, Reward = 1.0
Iteration 742: Arm = 3, Reward = 1.0
Iteration 743: Arm = 3, Reward = 0.0
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Iteration 745: Arm = 3, Reward = 1.0
Iteration 746: Arm = 3, Reward = 0.0
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Iteration 748: Arm = 3, Reward = 1.0
Iteration 749: Arm = 3, Reward = 1.0
Iteration 750: Arm = 3, Reward = 1.0
Iteration 751: Arm = 3, Reward = 0.0
Iteration 752: Arm = 3, Reward = 1.0
Iteration 753: Arm = 3, Reward = 0.0
Iteration 754: Arm = 3, Reward = 0.0
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Iteration 757: Arm = 3, Reward = 1.0
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Iteration 759: Arm = 3, Reward = 1.0
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Iteration 762: Arm = 3, Reward = 1.0
Iteration 763: Arm = 3, Reward = 0.0
Iteration 764: Arm = 3, Reward = 1.0
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Iteration 765: Arm = 3, Reward = 0.0
Iteration 766: Arm = 3, Reward = 0.0
Iteration 767: Arm = 3, Reward = 1.0
Iteration 768: Arm = 3, Reward = 0.0
Iteration 769: Arm = 3, Reward = 0.0
Iteration 770: Arm = 3, Reward = 0.0
Iteration 771: Arm = 3, Reward = 1.0
Iteration 772: Arm = 3, Reward = 0.0
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Iteration 777: Arm = 3, Reward = 1.0
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Iteration 780: Arm = 3, Reward = 1.0
Iteration 781: Arm = 3, Reward = 0.0
Iteration 782: Arm = 3, Reward = 1.0
Iteration 783: Arm = 3, Reward = 0.0
Iteration 784: Arm = 3, Reward = 0.0
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Iteration 791: Arm = 3, Reward = 1.0
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Iteration 797: Arm = 3, Reward = 1.0
Iteration 798: Arm = 3, Reward = 0.0
Iteration 799: Arm = 3, Reward = 1.0
Iteration 800: Arm = 3, Reward = 0.0
Iteration 801: Arm = 3, Reward = 1.0
Iteration 802: Arm = 3, Reward = 1.0
Iteration 803: Arm = 3, Reward = 1.0
Iteration 804: Arm = 3, Reward = 0.0
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Iteration 806: Arm = 3, Reward = 1.0
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Iteration 819: Arm = 3, Reward = 0.0
Iteration 820: Arm = 3, Reward = 1.0
Iteration 821: Arm = 3, Reward = 1.0
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Iteration 834: Arm = 3, Reward = 1.0
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Iteration 836: Arm = 3, Reward = 1.0
Iteration 837: Arm = 3, Reward = 0.0
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Iteration 839: Arm = 3, Reward = 0.0
Iteration 840: Arm = 3, Reward = 0.0
Iteration 841: Arm = 3, Reward = 0.0
Iteration 842: Arm = 3, Reward = 0.0
Iteration 843: Arm = 3, Reward = 0.0
Iteration 844: Arm = 3, Reward = 1.0
Iteration 845: Arm = 3, Reward = 0.0
Iteration 846: Arm = 3, Reward = 1.0
Iteration 847: Arm = 3, Reward = 1.0
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Iteration 848: Arm = 3, Reward = 0.0
Iteration 849: Arm = 3, Reward = 1.0
Iteration 850: Arm = 3, Reward = 1.0
Iteration 851: Arm = 3, Reward = 0.0
Iteration 852: Arm = 3, Reward = 0.0
Iteration 853: Arm = 3, Reward = 1.0
Iteration 854: Arm = 3, Reward = 1.0
Iteration 855: Arm = 3, Reward = 0.0
Iteration 856: Arm = 3, Reward = 0.0
Iteration 857: Arm = 3, Reward = 1.0
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Iteration 860: Arm = 3, Reward = 1.0
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Iteration 863: Arm = 3, Reward = 1.0
Iteration 864: Arm = 3, Reward = 1.0
Iteration 865: Arm = 3, Reward = 1.0
Iteration 866: Arm = 3, Reward = 0.0
Iteration 867: Arm = 3, Reward = 0.0
Iteration 868: Arm = 3, Reward = 1.0
Iteration 869: Arm = 3, Reward = 0.0
Iteration 870: Arm = 3, Reward = 1.0
Iteration 871: Arm = 3, Reward = 1.0
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Iteration 880: Arm = 3, Reward = 0.0
Iteration 881: Arm = 3, Reward = 1.0
Iteration 882: Arm = 3, Reward = 0.0
Iteration 883: Arm = 3, Reward = 1.0
Iteration 884: Arm = 3, Reward = 0.0
Iteration 885: Arm = 3, Reward = 1.0
Iteration 886: Arm = 3, Reward = 1.0
Iteration 887: Arm = 3, Reward = 0.0
Iteration 888: Arm = 3, Reward = 0.0
Iteration 889: Arm = 3, Reward = 0.0
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Iteration 891: Arm = 3, Reward = 1.0
Iteration 892: Arm = 3, Reward = 0.0
Iteration 893: Arm = 3, Reward = 1.0
Iteration 894: Arm = 3, Reward = 0.0
Iteration 895: Arm = 3, Reward = 0.0
Iteration 896: Arm = 3, Reward = 0.0
Iteration 897: Arm = 3, Reward = 1.0
Iteration 898: Arm = 3, Reward = 0.0
Iteration 899: Arm = 3, Reward = 1.0
Iteration 900: Arm = 3, Reward = 0.0
Iteration 901: Arm = 3, Reward = 1.0
Iteration 902: Arm = 3, Reward = 0.0
Iteration 903: Arm = 3, Reward = 0.0
Iteration 904: Arm = 3, Reward = 1.0
Iteration 905: Arm = 3, Reward = 0.0
Iteration 906: Arm = 3, Reward = 0.0
Iteration 907: Arm = 3, Reward = 1.0
Iteration 908: Arm = 3, Reward = 1.0
Iteration 909: Arm = 3, Reward = 0.0
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Iteration 911: Arm = 3, Reward = 1.0
Iteration 912: Arm = 3, Reward = 0.0
Iteration 913: Arm = 3, Reward = 1.0
Iteration 914: Arm = 3, Reward = 0.0
Iteration 915: Arm = 3, Reward = 0.0
Iteration 916: Arm = 3, Reward = 1.0
Iteration 917: Arm = 3, Reward = 0.0
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Iteration 919: Arm = 3, Reward = 1.0
Iteration 920: Arm = 3, Reward = 1.0
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Iteration 922: Arm = 3, Reward = 1.0
Iteration 923: Arm = 3, Reward = 0.0
Iteration 924: Arm = 3, Reward = 1.0
Iteration 925: Arm = 3, Reward = 0.0
Iteration 926: Arm = 3, Reward = 1.0
Iteration 927: Arm = 3, Reward = 1.0
Iteration 928: Arm = 3, Reward = 1.0
Iteration 929: Arm = 3, Reward = 1.0
Iteration 930: Arm = 3, Reward = 0.0
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Iteration 931: Arm = 3, Reward = 1.0
Iteration 932: Arm = 3, Reward = 0.0
Iteration 933: Arm = 3, Reward = 0.0
Iteration 934: Arm = 3, Reward = 1.0
Iteration 935: Arm = 3, Reward = 1.0
Iteration 936: Arm = 3, Reward = 0.0
Iteration 937: Arm = 3, Reward = 1.0
Iteration 938: Arm = 3, Reward = 0.0
Iteration 939: Arm = 3, Reward = 1.0
Iteration 940: Arm = 3, Reward = 1.0
Iteration 941: Arm = 3, Reward = 1.0
Iteration 942: Arm = 3, Reward = 0.0
Iteration 943: Arm = 3, Reward = 0.0
Iteration 944: Arm = 3, Reward = 1.0
Iteration 945: Arm = 3, Reward = 0.0
Iteration 946: Arm = 3, Reward = 1.0
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Iteration 948: Arm = 3, Reward = 0.0
Iteration 949: Arm = 3, Reward = 0.0
Iteration 950: Arm = 3, Reward = 1.0
Iteration 951: Arm = 3, Reward = 0.0
Iteration 952: Arm = 3, Reward = 0.0
Iteration 953: Arm = 3, Reward = 1.0
Iteration 954: Arm = 3, Reward = 0.0
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Iteration 976: Arm = 3, Reward = 1.0
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Iteration 993: Arm = 3, Reward = 1.0
Iteration 994: Arm = 3, Reward = 0.0
Iteration 995: Arm = 3, Reward = 1.0
Iteration 996: Arm = 3, Reward = 0.0
Iteration 997: Arm = 3, Reward = 1.0
Iteration 998: Arm = 3, Reward = 1.0
Iteration 999: Arm = 3, Reward = 1.0
Iteration 1000: Arm = 3, Reward = 1.0
```

In [3]:

```python
# --- EPSILON-GREEDY POLICIES ---
def run_epsilon_greedy(epsilon):
    print(f"\n--- Epsilon-Greedy Policy (ε = {epsilon}) ---")
    estimates = np.zeros(num_arms)
    counts = np.zeros(num_arms)
    rewards = []
    for i in range(iterations):
        if random.random() < epsilon:
            arm = random.choice(arms)
        else:
            arm = np.argmax(estimates)
        reward = df[df['trt'] == arm].sample(1).iloc[0]['reward']
```

```
        counts[arm] += 1
        estimates[arm] += (reward - estimates[arm]) / counts[arm]
        rewards.append(reward)
        print(f"Iteration {i+1}: Arm = {arm}, Reward = {reward}")
    return rewards, counts

for eps in [0.1, 0.2, 0.5]:
    r, c = run_epsilon_greedy(eps)
    policies_rewards[f"Epsilon-Greedy-{eps}"] = r
    arm_counts[f"Epsilon-Greedy-{eps}"] = c
```

```
--- Epsilon-Greedy Policy (ε = 0.1) ---
Iteration 1: Arm = 0, Reward = 1.0
Iteration 2: Arm = 0, Reward = 0.0
Iteration 3: Arm = 0, Reward = 0.0
Iteration 4: Arm = 0, Reward = 1.0
Iteration 5: Arm = 0, Reward = 0.0
Iteration 6: Arm = 0, Reward = 0.0
Iteration 7: Arm = 0, Reward = 0.0
Iteration 8: Arm = 0, Reward = 0.0
Iteration 9: Arm = 0, Reward = 0.0
Iteration 10: Arm = 0, Reward = 1.0
Iteration 11: Arm = 0, Reward = 0.0
Iteration 12: Arm = 0, Reward = 1.0
Iteration 13: Arm = 0, Reward = 0.0
Iteration 14: Arm = 3, Reward = 1.0
Iteration 15: Arm = 3, Reward = 0.0
Iteration 16: Arm = 3, Reward = 1.0
Iteration 17: Arm = 3, Reward = 1.0
Iteration 18: Arm = 3, Reward = 0.0
Iteration 19: Arm = 3, Reward = 1.0
Iteration 20: Arm = 3, Reward = 1.0
Iteration 21: Arm = 3, Reward = 0.0
Iteration 22: Arm = 3, Reward = 0.0
Iteration 23: Arm = 3, Reward = 0.0
Iteration 24: Arm = 3, Reward = 1.0
Iteration 25: Arm = 1, Reward = 1.0
Iteration 26: Arm = 1, Reward = 1.0
Iteration 27: Arm = 1, Reward = 0.0
Iteration 28: Arm = 2, Reward = 1.0
Iteration 29: Arm = 0, Reward = 0.0
Iteration 30: Arm = 2, Reward = 1.0
Iteration 31: Arm = 2, Reward = 0.0
Iteration 32: Arm = 1, Reward = 1.0
Iteration 33: Arm = 1, Reward = 1.0
Iteration 34: Arm = 1, Reward = 0.0
Iteration 35: Arm = 1, Reward = 0.0
Iteration 36: Arm = 0, Reward = 1.0
Iteration 37: Arm = 2, Reward = 0.0
Iteration 38: Arm = 1, Reward = 0.0
Iteration 39: Arm = 3, Reward = 1.0
Iteration 40: Arm = 3, Reward = 0.0
Iteration 41: Arm = 3, Reward = 1.0
Iteration 42: Arm = 3, Reward = 0.0
Iteration 43: Arm = 3, Reward = 1.0
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Iteration 45: Arm = 3, Reward = 1.0
Iteration 46: Arm = 3, Reward = 0.0
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Iteration 52: Arm = 1, Reward = 1.0
Iteration 53: Arm = 3, Reward = 1.0
Iteration 54: Arm = 3, Reward = 0.0
Iteration 55: Arm = 1, Reward = 1.0
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Iteration 57: Arm = 1, Reward = 1.0
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Iteration 69: Arm = 1, Reward = 1.0
Iteration 70: Arm = 1, Reward = 0.0
Iteration 71: Arm = 1, Reward = 1.0
Iteration 72: Arm = 1, Reward = 0.0
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Iteration 73: Arm = 1, Reward = 0.0
Iteration 74: Arm = 1, Reward = 0.0
Iteration 75: Arm = 0, Reward = 1.0
Iteration 76: Arm = 1, Reward = 0.0
Iteration 77: Arm = 3, Reward = 0.0
Iteration 78: Arm = 1, Reward = 0.0
Iteration 79: Arm = 3, Reward = 0.0
Iteration 80: Arm = 1, Reward = 0.0
Iteration 81: Arm = 3, Reward = 1.0
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Iteration 119: Arm = 3, Reward = 1.0
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Iteration 154: Arm = 3, Reward = 1.0
Iteration 155: Arm = 3, Reward = 0.0
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Iteration 156: Arm = 3, Reward = 0.0
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Iteration 239: Arm = 3, Reward = 1.0
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Iteration 322: Arm = 3, Reward = 1.0
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Iteration 349: Arm = 3, Reward = 1.0
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Iteration 421: Arm = 3, Reward = 0.0
Iteration 422: Arm = 3, Reward = 1.0
Iteration 423: Arm = 3, Reward = 1.0
Iteration 424: Arm = 3, Reward = 1.0
Iteration 425: Arm = 3, Reward = 0.0
Iteration 426: Arm = 3, Reward = 1.0
Iteration 427: Arm = 3, Reward = 0.0
Iteration 428: Arm = 3, Reward = 1.0
Iteration 429: Arm = 3, Reward = 0.0
Iteration 430: Arm = 3, Reward = 0.0
Iteration 431: Arm = 3, Reward = 1.0
Iteration 432: Arm = 3, Reward = 0.0
Iteration 433: Arm = 3, Reward = 1.0
Iteration 434: Arm = 3, Reward = 1.0
Iteration 435: Arm = 3, Reward = 1.0
Iteration 436: Arm = 3, Reward = 0.0
Iteration 437: Arm = 3, Reward = 0.0
Iteration 438: Arm = 3, Reward = 0.0
Iteration 439: Arm = 3, Reward = 1.0
Iteration 440: Arm = 3, Reward = 1.0
Iteration 441: Arm = 3, Reward = 1.0
Iteration 442: Arm = 3, Reward = 1.0
Iteration 443: Arm = 3, Reward = 0.0
Iteration 444: Arm = 3, Reward = 0.0
Iteration 445: Arm = 3, Reward = 0.0
Iteration 446: Arm = 3, Reward = 0.0
Iteration 447: Arm = 3, Reward = 1.0
Iteration 448: Arm = 3, Reward = 0.0
Iteration 449: Arm = 3, Reward = 1.0
Iteration 450: Arm = 3, Reward = 1.0
Iteration 451: Arm = 3, Reward = 1.0
Iteration 452: Arm = 3, Reward = 1.0
Iteration 453: Arm = 3, Reward = 1.0
Iteration 454: Arm = 3, Reward = 0.0
Iteration 455: Arm = 0, Reward = 0.0
Iteration 456: Arm = 3, Reward = 1.0
Iteration 457: Arm = 3, Reward = 0.0
Iteration 458: Arm = 3, Reward = 0.0
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Iteration 460: Arm = 3, Reward = 0.0
Iteration 461: Arm = 3, Reward = 0.0
Iteration 462: Arm = 3, Reward = 0.0
Iteration 463: Arm = 3, Reward = 1.0
Iteration 464: Arm = 3, Reward = 0.0
Iteration 465: Arm = 3, Reward = 0.0
Iteration 466: Arm = 3, Reward = 0.0
Iteration 467: Arm = 3, Reward = 0.0
Iteration 468: Arm = 3, Reward = 1.0
Iteration 469: Arm = 3, Reward = 1.0
Iteration 470: Arm = 3, Reward = 0.0
Iteration 471: Arm = 3, Reward = 0.0
Iteration 472: Arm = 3, Reward = 0.0
Iteration 473: Arm = 3, Reward = 0.0
Iteration 474: Arm = 3, Reward = 0.0
Iteration 475: Arm = 3, Reward = 0.0
Iteration 476: Arm = 2, Reward = 0.0
Iteration 477: Arm = 3, Reward = 1.0
Iteration 478: Arm = 3, Reward = 1.0
Iteration 479: Arm = 3, Reward = 0.0
Iteration 480: Arm = 3, Reward = 0.0
Iteration 481: Arm = 3, Reward = 0.0
Iteration 482: Arm = 3, Reward = 0.0
Iteration 483: Arm = 3, Reward = 0.0
Iteration 484: Arm = 3, Reward = 1.0
Iteration 485: Arm = 3, Reward = 1.0
Iteration 486: Arm = 3, Reward = 1.0
Iteration 487: Arm = 3, Reward = 1.0
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Iteration 488: Arm = 3, Reward = 1.0
Iteration 489: Arm = 3, Reward = 1.0
Iteration 490: Arm = 3, Reward = 0.0
Iteration 491: Arm = 3, Reward = 1.0
Iteration 492: Arm = 2, Reward = 1.0
Iteration 493: Arm = 3, Reward = 1.0
Iteration 494: Arm = 3, Reward = 0.0
Iteration 495: Arm = 3, Reward = 0.0
Iteration 496: Arm = 3, Reward = 0.0
Iteration 497: Arm = 3, Reward = 1.0
Iteration 498: Arm = 3, Reward = 1.0
Iteration 499: Arm = 3, Reward = 0.0
Iteration 500: Arm = 3, Reward = 1.0
Iteration 501: Arm = 3, Reward = 1.0
Iteration 502: Arm = 3, Reward = 1.0
Iteration 503: Arm = 3, Reward = 0.0
Iteration 504: Arm = 3, Reward = 0.0
Iteration 505: Arm = 3, Reward = 1.0
Iteration 506: Arm = 3, Reward = 1.0
Iteration 507: Arm = 3, Reward = 0.0
Iteration 508: Arm = 3, Reward = 0.0
Iteration 509: Arm = 3, Reward = 1.0
Iteration 510: Arm = 3, Reward = 1.0
Iteration 511: Arm = 3, Reward = 1.0
Iteration 512: Arm = 3, Reward = 0.0
Iteration 513: Arm = 3, Reward = 1.0
Iteration 514: Arm = 3, Reward = 1.0
Iteration 515: Arm = 3, Reward = 1.0
Iteration 516: Arm = 3, Reward = 0.0
Iteration 517: Arm = 3, Reward = 0.0
Iteration 518: Arm = 3, Reward = 1.0
Iteration 519: Arm = 3, Reward = 1.0
Iteration 520: Arm = 3, Reward = 0.0
Iteration 521: Arm = 3, Reward = 0.0
Iteration 522: Arm = 3, Reward = 1.0
Iteration 523: Arm = 3, Reward = 0.0
Iteration 524: Arm = 3, Reward = 0.0
Iteration 525: Arm = 3, Reward = 1.0
Iteration 526: Arm = 3, Reward = 0.0
Iteration 527: Arm = 3, Reward = 1.0
Iteration 528: Arm = 3, Reward = 1.0
Iteration 529: Arm = 3, Reward = 0.0
Iteration 530: Arm = 3, Reward = 0.0
Iteration 531: Arm = 3, Reward = 0.0
Iteration 532: Arm = 3, Reward = 1.0
Iteration 533: Arm = 3, Reward = 1.0
Iteration 534: Arm = 3, Reward = 0.0
Iteration 535: Arm = 0, Reward = 0.0
Iteration 536: Arm = 3, Reward = 0.0
Iteration 537: Arm = 3, Reward = 0.0
Iteration 538: Arm = 3, Reward = 0.0
Iteration 539: Arm = 3, Reward = 0.0
Iteration 540: Arm = 3, Reward = 0.0
Iteration 541: Arm = 3, Reward = 0.0
Iteration 542: Arm = 3, Reward = 0.0
Iteration 543: Arm = 3, Reward = 0.0
Iteration 544: Arm = 3, Reward = 0.0
Iteration 545: Arm = 3, Reward = 1.0
Iteration 546: Arm = 3, Reward = 0.0
Iteration 547: Arm = 3, Reward = 0.0
Iteration 548: Arm = 3, Reward = 0.0
Iteration 549: Arm = 1, Reward = 0.0
Iteration 550: Arm = 3, Reward = 0.0
Iteration 551: Arm = 3, Reward = 0.0
Iteration 552: Arm = 3, Reward = 0.0
Iteration 553: Arm = 3, Reward = 0.0
Iteration 554: Arm = 1, Reward = 0.0
Iteration 555: Arm = 3, Reward = 0.0
Iteration 556: Arm = 3, Reward = 0.0
Iteration 557: Arm = 3, Reward = 0.0
Iteration 558: Arm = 3, Reward = 0.0
Iteration 559: Arm = 3, Reward = 0.0
Iteration 560: Arm = 3, Reward = 1.0
Iteration 561: Arm = 3, Reward = 0.0
Iteration 562: Arm = 3, Reward = 0.0
Iteration 563: Arm = 3, Reward = 1.0
Iteration 564: Arm = 3, Reward = 0.0
Iteration 565: Arm = 3, Reward = 0.0
Iteration 566: Arm = 3, Reward = 1.0
Iteration 567: Arm = 3, Reward = 0.0
Iteration 568: Arm = 3, Reward = 1.0
Iteration 569: Arm = 3, Reward = 1.0
Iteration 570: Arm = 2, Reward = 1.0
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Iteration 571: Arm = 2, Reward = 1.0
Iteration 572: Arm = 1, Reward = 1.0
Iteration 573: Arm = 2, Reward = 1.0
Iteration 574: Arm = 2, Reward = 0.0
Iteration 575: Arm = 2, Reward = 0.0
Iteration 576: Arm = 2, Reward = 0.0
Iteration 577: Arm = 3, Reward = 0.0
Iteration 578: Arm = 3, Reward = 0.0
Iteration 579: Arm = 3, Reward = 1.0
Iteration 580: Arm = 3, Reward = 0.0
Iteration 581: Arm = 3, Reward = 1.0
Iteration 582: Arm = 3, Reward = 0.0
Iteration 583: Arm = 3, Reward = 0.0
Iteration 584: Arm = 3, Reward = 1.0
Iteration 585: Arm = 3, Reward = 0.0
Iteration 586: Arm = 3, Reward = 0.0
Iteration 587: Arm = 3, Reward = 0.0
Iteration 588: Arm = 1, Reward = 0.0
Iteration 589: Arm = 3, Reward = 1.0
Iteration 590: Arm = 3, Reward = 0.0
Iteration 591: Arm = 3, Reward = 1.0
Iteration 592: Arm = 3, Reward = 1.0
Iteration 593: Arm = 3, Reward = 0.0
Iteration 594: Arm = 3, Reward = 0.0
Iteration 595: Arm = 3, Reward = 1.0
Iteration 596: Arm = 3, Reward = 1.0
Iteration 597: Arm = 3, Reward = 0.0
Iteration 598: Arm = 3, Reward = 1.0
Iteration 599: Arm = 3, Reward = 1.0
Iteration 600: Arm = 3, Reward = 0.0
Iteration 601: Arm = 3, Reward = 0.0
Iteration 602: Arm = 2, Reward = 0.0
Iteration 603: Arm = 3, Reward = 0.0
Iteration 604: Arm = 3, Reward = 1.0
Iteration 605: Arm = 3, Reward = 1.0
Iteration 606: Arm = 3, Reward = 1.0
Iteration 607: Arm = 3, Reward = 0.0
Iteration 608: Arm = 3, Reward = 0.0
Iteration 609: Arm = 3, Reward = 1.0
Iteration 610: Arm = 3, Reward = 0.0
Iteration 611: Arm = 3, Reward = 1.0
Iteration 612: Arm = 3, Reward = 1.0
Iteration 613: Arm = 3, Reward = 0.0
Iteration 614: Arm = 3, Reward = 0.0
Iteration 615: Arm = 3, Reward = 1.0
Iteration 616: Arm = 3, Reward = 0.0
Iteration 617: Arm = 3, Reward = 0.0
Iteration 618: Arm = 3, Reward = 0.0
Iteration 619: Arm = 3, Reward = 1.0
Iteration 620: Arm = 3, Reward = 0.0
Iteration 621: Arm = 3, Reward = 1.0
Iteration 622: Arm = 3, Reward = 1.0
Iteration 623: Arm = 2, Reward = 0.0
Iteration 624: Arm = 3, Reward = 1.0
Iteration 625: Arm = 3, Reward = 1.0
Iteration 626: Arm = 3, Reward = 0.0
Iteration 627: Arm = 3, Reward = 0.0
Iteration 628: Arm = 3, Reward = 0.0
Iteration 629: Arm = 3, Reward = 0.0
Iteration 630: Arm = 3, Reward = 0.0
Iteration 631: Arm = 3, Reward = 1.0
Iteration 632: Arm = 3, Reward = 0.0
Iteration 633: Arm = 3, Reward = 1.0
Iteration 634: Arm = 3, Reward = 1.0
Iteration 635: Arm = 3, Reward = 1.0
Iteration 636: Arm = 3, Reward = 0.0
Iteration 637: Arm = 3, Reward = 1.0
Iteration 638: Arm = 3, Reward = 0.0
Iteration 639: Arm = 3, Reward = 1.0
Iteration 640: Arm = 1, Reward = 0.0
Iteration 641: Arm = 3, Reward = 0.0
Iteration 642: Arm = 3, Reward = 1.0
Iteration 643: Arm = 3, Reward = 0.0
Iteration 644: Arm = 3, Reward = 0.0
Iteration 645: Arm = 3, Reward = 0.0
Iteration 646: Arm = 3, Reward = 1.0
Iteration 647: Arm = 3, Reward = 0.0
Iteration 648: Arm = 3, Reward = 0.0
Iteration 649: Arm = 2, Reward = 1.0
Iteration 650: Arm = 3, Reward = 1.0
Iteration 651: Arm = 3, Reward = 0.0
Iteration 652: Arm = 3, Reward = 1.0
Iteration 653: Arm = 3, Reward = 1.0
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Iteration 654: Arm = 3, Reward = 0.0
Iteration 655: Arm = 3, Reward = 0.0
Iteration 656: Arm = 3, Reward = 1.0
Iteration 657: Arm = 3, Reward = 0.0
Iteration 658: Arm = 3, Reward = 1.0
Iteration 659: Arm = 3, Reward = 0.0
Iteration 660: Arm = 3, Reward = 1.0
Iteration 661: Arm = 3, Reward = 0.0
Iteration 662: Arm = 3, Reward = 1.0
Iteration 663: Arm = 2, Reward = 0.0
Iteration 664: Arm = 3, Reward = 1.0
Iteration 665: Arm = 3, Reward = 1.0
Iteration 666: Arm = 3, Reward = 0.0
Iteration 667: Arm = 3, Reward = 1.0
Iteration 668: Arm = 3, Reward = 1.0
Iteration 669: Arm = 0, Reward = 0.0
Iteration 670: Arm = 3, Reward = 1.0
Iteration 671: Arm = 3, Reward = 0.0
Iteration 672: Arm = 3, Reward = 0.0
Iteration 673: Arm = 3, Reward = 0.0
Iteration 674: Arm = 3, Reward = 1.0
Iteration 675: Arm = 1, Reward = 0.0
Iteration 676: Arm = 3, Reward = 1.0
Iteration 677: Arm = 3, Reward = 1.0
Iteration 678: Arm = 3, Reward = 0.0
Iteration 679: Arm = 3, Reward = 1.0
Iteration 680: Arm = 3, Reward = 0.0
Iteration 681: Arm = 1, Reward = 1.0
Iteration 682: Arm = 3, Reward = 0.0
Iteration 683: Arm = 3, Reward = 0.0
Iteration 684: Arm = 3, Reward = 0.0
Iteration 685: Arm = 3, Reward = 0.0
Iteration 686: Arm = 3, Reward = 1.0
Iteration 687: Arm = 3, Reward = 1.0
Iteration 688: Arm = 3, Reward = 1.0
Iteration 689: Arm = 3, Reward = 0.0
Iteration 690: Arm = 3, Reward = 0.0
Iteration 691: Arm = 3, Reward = 0.0
Iteration 692: Arm = 3, Reward = 1.0
Iteration 693: Arm = 3, Reward = 1.0
Iteration 694: Arm = 3, Reward = 0.0
Iteration 695: Arm = 2, Reward = 0.0
Iteration 696: Arm = 1, Reward = 1.0
Iteration 697: Arm = 3, Reward = 1.0
Iteration 698: Arm = 3, Reward = 0.0
Iteration 699: Arm = 3, Reward = 1.0
Iteration 700: Arm = 3, Reward = 1.0
Iteration 701: Arm = 3, Reward = 0.0
Iteration 702: Arm = 3, Reward = 0.0
Iteration 703: Arm = 3, Reward = 0.0
Iteration 704: Arm = 3, Reward = 0.0
Iteration 705: Arm = 3, Reward = 1.0
Iteration 706: Arm = 3, Reward = 1.0
Iteration 707: Arm = 3, Reward = 0.0
Iteration 708: Arm = 3, Reward = 0.0
Iteration 709: Arm = 3, Reward = 0.0
Iteration 710: Arm = 3, Reward = 1.0
Iteration 711: Arm = 3, Reward = 0.0
Iteration 712: Arm = 3, Reward = 0.0
Iteration 713: Arm = 3, Reward = 1.0
Iteration 714: Arm = 3, Reward = 0.0
Iteration 715: Arm = 3, Reward = 1.0
Iteration 716: Arm = 3, Reward = 0.0
Iteration 717: Arm = 0, Reward = 0.0
Iteration 718: Arm = 3, Reward = 1.0
Iteration 719: Arm = 3, Reward = 0.0
Iteration 720: Arm = 3, Reward = 1.0
Iteration 721: Arm = 3, Reward = 1.0
Iteration 722: Arm = 3, Reward = 0.0
Iteration 723: Arm = 3, Reward = 1.0
Iteration 724: Arm = 3, Reward = 1.0
Iteration 725: Arm = 3, Reward = 0.0
Iteration 726: Arm = 3, Reward = 1.0
Iteration 727: Arm = 3, Reward = 0.0
Iteration 728: Arm = 3, Reward = 0.0
Iteration 729: Arm = 3, Reward = 1.0
Iteration 730: Arm = 3, Reward = 0.0
Iteration 731: Arm = 3, Reward = 1.0
Iteration 732: Arm = 3, Reward = 0.0
Iteration 733: Arm = 3, Reward = 0.0
Iteration 734: Arm = 3, Reward = 0.0
Iteration 735: Arm = 3, Reward = 0.0
Iteration 736: Arm = 3, Reward = 1.0
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Iteration 737: Arm = 3, Reward = 0.0
Iteration 738: Arm = 3, Reward = 1.0
Iteration 739: Arm = 2, Reward = 0.0
Iteration 740: Arm = 3, Reward = 0.0
Iteration 741: Arm = 3, Reward = 0.0
Iteration 742: Arm = 3, Reward = 1.0
Iteration 743: Arm = 3, Reward = 1.0
Iteration 744: Arm = 3, Reward = 1.0
Iteration 745: Arm = 3, Reward = 0.0
Iteration 746: Arm = 3, Reward = 0.0
Iteration 747: Arm = 3, Reward = 1.0
Iteration 748: Arm = 3, Reward = 1.0
Iteration 749: Arm = 3, Reward = 1.0
Iteration 750: Arm = 3, Reward = 0.0
Iteration 751: Arm = 3, Reward = 0.0
Iteration 752: Arm = 3, Reward = 0.0
Iteration 753: Arm = 3, Reward = 0.0
Iteration 754: Arm = 3, Reward = 0.0
Iteration 755: Arm = 3, Reward = 0.0
Iteration 756: Arm = 3, Reward = 0.0
Iteration 757: Arm = 3, Reward = 1.0
Iteration 758: Arm = 3, Reward = 1.0
Iteration 759: Arm = 3, Reward = 0.0
Iteration 760: Arm = 2, Reward = 0.0
Iteration 761: Arm = 3, Reward = 0.0
Iteration 762: Arm = 3, Reward = 0.0
Iteration 763: Arm = 3, Reward = 0.0
Iteration 764: Arm = 3, Reward = 0.0
Iteration 765: Arm = 1, Reward = 0.0
Iteration 766: Arm = 3, Reward = 0.0
Iteration 767: Arm = 3, Reward = 1.0
Iteration 768: Arm = 3, Reward = 0.0
Iteration 769: Arm = 3, Reward = 0.0
Iteration 770: Arm = 3, Reward = 1.0
Iteration 771: Arm = 3, Reward = 0.0
Iteration 772: Arm = 3, Reward = 1.0
Iteration 773: Arm = 3, Reward = 0.0
Iteration 774: Arm = 3, Reward = 1.0
Iteration 775: Arm = 3, Reward = 1.0
Iteration 776: Arm = 3, Reward = 0.0
Iteration 777: Arm = 3, Reward = 1.0
Iteration 778: Arm = 3, Reward = 0.0
Iteration 779: Arm = 3, Reward = 1.0
Iteration 780: Arm = 3, Reward = 1.0
Iteration 781: Arm = 3, Reward = 1.0
Iteration 782: Arm = 3, Reward = 1.0
Iteration 783: Arm = 3, Reward = 1.0
Iteration 784: Arm = 3, Reward = 0.0
Iteration 785: Arm = 3, Reward = 0.0
Iteration 786: Arm = 3, Reward = 1.0
Iteration 787: Arm = 3, Reward = 1.0
Iteration 788: Arm = 3, Reward = 1.0
Iteration 789: Arm = 3, Reward = 0.0
Iteration 790: Arm = 3, Reward = 1.0
Iteration 791: Arm = 3, Reward = 1.0
Iteration 792: Arm = 3, Reward = 0.0
Iteration 793: Arm = 3, Reward = 1.0
Iteration 794: Arm = 3, Reward = 1.0
Iteration 795: Arm = 3, Reward = 1.0
Iteration 796: Arm = 3, Reward = 1.0
Iteration 797: Arm = 3, Reward = 0.0
Iteration 798: Arm = 3, Reward = 1.0
Iteration 799: Arm = 2, Reward = 1.0
Iteration 800: Arm = 3, Reward = 1.0
Iteration 801: Arm = 1, Reward = 0.0
Iteration 802: Arm = 3, Reward = 0.0
Iteration 803: Arm = 1, Reward = 1.0
Iteration 804: Arm = 3, Reward = 1.0
Iteration 805: Arm = 2, Reward = 0.0
Iteration 806: Arm = 3, Reward = 1.0
Iteration 807: Arm = 3, Reward = 0.0
Iteration 808: Arm = 3, Reward = 0.0
Iteration 809: Arm = 3, Reward = 1.0
Iteration 810: Arm = 3, Reward = 1.0
Iteration 811: Arm = 3, Reward = 1.0
Iteration 812: Arm = 3, Reward = 1.0
Iteration 813: Arm = 3, Reward = 0.0
Iteration 814: Arm = 3, Reward = 0.0
Iteration 815: Arm = 3, Reward = 1.0
Iteration 816: Arm = 3, Reward = 0.0
Iteration 817: Arm = 3, Reward = 1.0
Iteration 818: Arm = 3, Reward = 0.0
Iteration 819: Arm = 2, Reward = 0.0
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Iteration 820: Arm = 3, Reward = 1.0
Iteration 821: Arm = 3, Reward = 0.0
Iteration 822: Arm = 1, Reward = 0.0
Iteration 823: Arm = 3, Reward = 1.0
Iteration 824: Arm = 3, Reward = 1.0
Iteration 825: Arm = 3, Reward = 0.0
Iteration 826: Arm = 3, Reward = 0.0
Iteration 827: Arm = 3, Reward = 0.0
Iteration 828: Arm = 3, Reward = 1.0
Iteration 829: Arm = 3, Reward = 1.0
Iteration 830: Arm = 0, Reward = 0.0
Iteration 831: Arm = 3, Reward = 1.0
Iteration 832: Arm = 3, Reward = 0.0
Iteration 833: Arm = 3, Reward = 1.0
Iteration 834: Arm = 3, Reward = 1.0
Iteration 835: Arm = 3, Reward = 0.0
Iteration 836: Arm = 3, Reward = 1.0
Iteration 837: Arm = 3, Reward = 0.0
Iteration 838: Arm = 3, Reward = 0.0
Iteration 839: Arm = 3, Reward = 1.0
Iteration 840: Arm = 1, Reward = 1.0
Iteration 841: Arm = 3, Reward = 0.0
Iteration 842: Arm = 3, Reward = 0.0
Iteration 843: Arm = 3, Reward = 1.0
Iteration 844: Arm = 3, Reward = 0.0
Iteration 845: Arm = 3, Reward = 1.0
Iteration 846: Arm = 3, Reward = 1.0
Iteration 847: Arm = 3, Reward = 1.0
Iteration 848: Arm = 3, Reward = 1.0
Iteration 849: Arm = 3, Reward = 0.0
Iteration 850: Arm = 3, Reward = 1.0
Iteration 851: Arm = 3, Reward = 0.0
Iteration 852: Arm = 3, Reward = 0.0
Iteration 853: Arm = 3, Reward = 1.0
Iteration 854: Arm = 3, Reward = 1.0
Iteration 855: Arm = 3, Reward = 0.0
Iteration 856: Arm = 3, Reward = 0.0
Iteration 857: Arm = 3, Reward = 1.0
Iteration 858: Arm = 3, Reward = 0.0
Iteration 859: Arm = 3, Reward = 1.0
Iteration 860: Arm = 3, Reward = 1.0
Iteration 861: Arm = 3, Reward = 1.0
Iteration 862: Arm = 3, Reward = 0.0
Iteration 863: Arm = 3, Reward = 0.0
Iteration 864: Arm = 3, Reward = 1.0
Iteration 865: Arm = 3, Reward = 0.0
Iteration 866: Arm = 3, Reward = 0.0
Iteration 867: Arm = 3, Reward = 1.0
Iteration 868: Arm = 3, Reward = 1.0
Iteration 869: Arm = 3, Reward = 1.0
Iteration 870: Arm = 2, Reward = 1.0
Iteration 871: Arm = 3, Reward = 0.0
Iteration 872: Arm = 3, Reward = 1.0
Iteration 873: Arm = 3, Reward = 0.0
Iteration 874: Arm = 3, Reward = 1.0
Iteration 875: Arm = 3, Reward = 1.0
Iteration 876: Arm = 2, Reward = 0.0
Iteration 877: Arm = 3, Reward = 1.0
Iteration 878: Arm = 3, Reward = 1.0
Iteration 879: Arm = 3, Reward = 1.0
Iteration 880: Arm = 3, Reward = 1.0
Iteration 881: Arm = 3, Reward = 1.0
Iteration 882: Arm = 3, Reward = 1.0
Iteration 883: Arm = 3, Reward = 0.0
Iteration 884: Arm = 3, Reward = 1.0
Iteration 885: Arm = 3, Reward = 0.0
Iteration 886: Arm = 3, Reward = 1.0
Iteration 887: Arm = 0, Reward = 0.0
Iteration 888: Arm = 3, Reward = 0.0
Iteration 889: Arm = 3, Reward = 1.0
Iteration 890: Arm = 3, Reward = 1.0
Iteration 891: Arm = 3, Reward = 0.0
Iteration 892: Arm = 3, Reward = 1.0
Iteration 893: Arm = 3, Reward = 1.0
Iteration 894: Arm = 0, Reward = 1.0
Iteration 895: Arm = 1, Reward = 1.0
Iteration 896: Arm = 3, Reward = 0.0
Iteration 897: Arm = 3, Reward = 1.0
Iteration 898: Arm = 3, Reward = 1.0
Iteration 899: Arm = 3, Reward = 0.0
Iteration 900: Arm = 3, Reward = 0.0
Iteration 901: Arm = 3, Reward = 1.0
Iteration 902: Arm = 3, Reward = 0.0
```

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Iteration 903: Arm = 3, Reward = 0.0
Iteration 904: Arm = 1, Reward = 0.0
Iteration 905: Arm = 3, Reward = 0.0
Iteration 906: Arm = 3, Reward = 0.0
Iteration 907: Arm = 3, Reward = 1.0
Iteration 908: Arm = 3, Reward = 0.0
Iteration 909: Arm = 3, Reward = 0.0
Iteration 910: Arm = 3, Reward = 1.0
Iteration 911: Arm = 3, Reward = 0.0
Iteration 912: Arm = 3, Reward = 1.0
Iteration 913: Arm = 3, Reward = 0.0
Iteration 914: Arm = 0, Reward = 1.0
Iteration 915: Arm = 3, Reward = 0.0
Iteration 916: Arm = 3, Reward = 0.0
Iteration 917: Arm = 3, Reward = 1.0
Iteration 918: Arm = 3, Reward = 0.0
Iteration 919: Arm = 3, Reward = 0.0
Iteration 920: Arm = 3, Reward = 1.0
Iteration 921: Arm = 3, Reward = 0.0
Iteration 922: Arm = 3, Reward = 1.0
Iteration 923: Arm = 3, Reward = 1.0
Iteration 924: Arm = 3, Reward = 1.0
Iteration 925: Arm = 3, Reward = 1.0
Iteration 926: Arm = 3, Reward = 0.0
Iteration 927: Arm = 3, Reward = 1.0
Iteration 928: Arm = 3, Reward = 0.0
Iteration 929: Arm = 3, Reward = 1.0
Iteration 930: Arm = 3, Reward = 0.0
Iteration 931: Arm = 3, Reward = 1.0
Iteration 932: Arm = 3, Reward = 1.0
Iteration 933: Arm = 3, Reward = 1.0
Iteration 934: Arm = 3, Reward = 0.0
Iteration 935: Arm = 3, Reward = 0.0
Iteration 936: Arm = 3, Reward = 0.0
Iteration 937: Arm = 1, Reward = 0.0
Iteration 938: Arm = 3, Reward = 1.0
Iteration 939: Arm = 3, Reward = 1.0
Iteration 940: Arm = 3, Reward = 1.0
Iteration 941: Arm = 3, Reward = 0.0
Iteration 942: Arm = 3, Reward = 1.0
Iteration 943: Arm = 3, Reward = 1.0
Iteration 944: Arm = 3, Reward = 1.0
Iteration 945: Arm = 3, Reward = 1.0
Iteration 946: Arm = 3, Reward = 0.0
Iteration 947: Arm = 3, Reward = 1.0
Iteration 948: Arm = 3, Reward = 1.0
Iteration 949: Arm = 3, Reward = 1.0
Iteration 950: Arm = 3, Reward = 1.0
Iteration 951: Arm = 3, Reward = 1.0
Iteration 952: Arm = 3, Reward = 0.0
Iteration 953: Arm = 3, Reward = 1.0
Iteration 954: Arm = 3, Reward = 0.0
Iteration 955: Arm = 3, Reward = 1.0
Iteration 956: Arm = 3, Reward = 0.0
Iteration 957: Arm = 3, Reward = 0.0
Iteration 958: Arm = 3, Reward = 0.0
Iteration 959: Arm = 3, Reward = 0.0
Iteration 960: Arm = 3, Reward = 0.0
Iteration 961: Arm = 3, Reward = 1.0
Iteration 962: Arm = 3, Reward = 1.0
Iteration 963: Arm = 3, Reward = 1.0
Iteration 964: Arm = 3, Reward = 0.0
Iteration 965: Arm = 3, Reward = 0.0
Iteration 966: Arm = 3, Reward = 1.0
Iteration 967: Arm = 3, Reward = 1.0
Iteration 968: Arm = 3, Reward = 1.0
Iteration 969: Arm = 3, Reward = 0.0
Iteration 970: Arm = 3, Reward = 1.0
Iteration 971: Arm = 3, Reward = 1.0
Iteration 972: Arm = 3, Reward = 0.0
Iteration 973: Arm = 3, Reward = 0.0
Iteration 974: Arm = 3, Reward = 0.0
Iteration 975: Arm = 3, Reward = 0.0
Iteration 976: Arm = 3, Reward = 0.0
Iteration 977: Arm = 3, Reward = 1.0
Iteration 978: Arm = 3, Reward = 0.0
Iteration 979: Arm = 3, Reward = 0.0
Iteration 980: Arm = 3, Reward = 0.0
Iteration 981: Arm = 3, Reward = 1.0
Iteration 982: Arm = 3, Reward = 0.0
Iteration 983: Arm = 0, Reward = 0.0
Iteration 984: Arm = 3, Reward = 0.0
Iteration 985: Arm = 3, Reward = 0.0
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Iteration 986: Arm = 3, Reward = 0.0
Iteration 987: Arm = 3, Reward = 0.0
Iteration 988: Arm = 2, Reward = 0.0
Iteration 989: Arm = 3, Reward = 0.0
Iteration 990: Arm = 3, Reward = 0.0
Iteration 991: Arm = 3, Reward = 0.0
Iteration 992: Arm = 0, Reward = 0.0
Iteration 993: Arm = 3, Reward = 1.0
Iteration 994: Arm = 3, Reward = 0.0
Iteration 995: Arm = 3, Reward = 1.0
Iteration 996: Arm = 3, Reward = 1.0
Iteration 997: Arm = 3, Reward = 0.0
Iteration 998: Arm = 3, Reward = 1.0
Iteration 999: Arm = 3, Reward = 0.0
Iteration 1000: Arm = 3, Reward = 0.0

--- Epsilon-Greedy Policy (ε = 0.2) ---
Iteration 1: Arm = 0, Reward = 0.0
Iteration 2: Arm = 0, Reward = 1.0
Iteration 3: Arm = 0, Reward = 1.0
Iteration 4: Arm = 0, Reward = 1.0
Iteration 5: Arm = 0, Reward = 0.0
Iteration 6: Arm = 0, Reward = 1.0
Iteration 7: Arm = 0, Reward = 1.0
Iteration 8: Arm = 0, Reward = 0.0
Iteration 9: Arm = 0, Reward = 0.0
Iteration 10: Arm = 0, Reward = 0.0
Iteration 11: Arm = 0, Reward = 0.0
Iteration 12: Arm = 0, Reward = 1.0
Iteration 13: Arm = 0, Reward = 0.0
Iteration 14: Arm = 0, Reward = 0.0
Iteration 15: Arm = 1, Reward = 0.0
Iteration 16: Arm = 0, Reward = 1.0
Iteration 17: Arm = 0, Reward = 0.0
Iteration 18: Arm = 0, Reward = 0.0
Iteration 19: Arm = 0, Reward = 1.0
Iteration 20: Arm = 0, Reward = 1.0
Iteration 21: Arm = 0, Reward = 0.0
Iteration 22: Arm = 0, Reward = 1.0
Iteration 23: Arm = 3, Reward = 0.0
Iteration 24: Arm = 0, Reward = 0.0
Iteration 25: Arm = 0, Reward = 0.0
Iteration 26: Arm = 2, Reward = 1.0
Iteration 27: Arm = 2, Reward = 1.0
Iteration 28: Arm = 2, Reward = 1.0
Iteration 29: Arm = 2, Reward = 0.0
Iteration 30: Arm = 2, Reward = 0.0
Iteration 31: Arm = 2, Reward = 0.0
Iteration 32: Arm = 2, Reward = 1.0
Iteration 33: Arm = 2, Reward = 1.0
Iteration 34: Arm = 3, Reward = 1.0
Iteration 35: Arm = 2, Reward = 1.0
Iteration 36: Arm = 2, Reward = 1.0
Iteration 37: Arm = 0, Reward = 1.0
Iteration 38: Arm = 2, Reward = 0.0
Iteration 39: Arm = 2, Reward = 0.0
Iteration 40: Arm = 2, Reward = 0.0
Iteration 41: Arm = 2, Reward = 1.0
Iteration 42: Arm = 2, Reward = 0.0
Iteration 43: Arm = 2, Reward = 1.0
Iteration 44: Arm = 2, Reward = 1.0
Iteration 45: Arm = 2, Reward = 0.0
Iteration 46: Arm = 2, Reward = 0.0
Iteration 47: Arm = 2, Reward = 1.0
Iteration 48: Arm = 2, Reward = 1.0
Iteration 49: Arm = 2, Reward = 0.0
Iteration 50: Arm = 2, Reward = 0.0
Iteration 51: Arm = 2, Reward = 1.0
Iteration 52: Arm = 1, Reward = 1.0
Iteration 53: Arm = 2, Reward = 1.0
Iteration 54: Arm = 2, Reward = 0.0
Iteration 55: Arm = 2, Reward = 1.0
Iteration 56: Arm = 3, Reward = 0.0
Iteration 57: Arm = 2, Reward = 0.0
Iteration 58: Arm = 2, Reward = 1.0
Iteration 59: Arm = 2, Reward = 1.0
Iteration 60: Arm = 2, Reward = 0.0
Iteration 61: Arm = 2, Reward = 0.0
Iteration 62: Arm = 2, Reward = 0.0
Iteration 63: Arm = 2, Reward = 1.0
Iteration 64: Arm = 2, Reward = 0.0
Iteration 65: Arm = 2, Reward = 1.0
Iteration 66: Arm = 2, Reward = 0.0
```

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Iteration 67: Arm = 2, Reward = 0.0
Iteration 68: Arm = 1, Reward = 1.0
Iteration 69: Arm = 1, Reward = 1.0
Iteration 70: Arm = 1, Reward = 0.0
Iteration 71: Arm = 3, Reward = 0.0
Iteration 72: Arm = 1, Reward = 1.0
Iteration 73: Arm = 1, Reward = 0.0
Iteration 74: Arm = 3, Reward = 1.0
Iteration 75: Arm = 1, Reward = 0.0
Iteration 76: Arm = 1, Reward = 1.0
Iteration 77: Arm = 1, Reward = 0.0
Iteration 78: Arm = 1, Reward = 1.0
Iteration 79: Arm = 2, Reward = 1.0
Iteration 80: Arm = 1, Reward = 0.0
Iteration 81: Arm = 2, Reward = 0.0
Iteration 82: Arm = 1, Reward = 0.0
Iteration 83: Arm = 0, Reward = 0.0
Iteration 84: Arm = 2, Reward = 1.0
Iteration 85: Arm = 2, Reward = 1.0
Iteration 86: Arm = 2, Reward = 0.0
Iteration 87: Arm = 3, Reward = 0.0
Iteration 88: Arm = 2, Reward = 0.0
Iteration 89: Arm = 2, Reward = 0.0
Iteration 90: Arm = 2, Reward = 0.0
Iteration 91: Arm = 2, Reward = 0.0
Iteration 92: Arm = 2, Reward = 0.0
Iteration 93: Arm = 1, Reward = 1.0
Iteration 94: Arm = 2, Reward = 0.0
Iteration 95: Arm = 1, Reward = 0.0
Iteration 96: Arm = 1, Reward = 0.0
Iteration 97: Arm = 3, Reward = 0.0
Iteration 98: Arm = 2, Reward = 1.0
Iteration 99: Arm = 2, Reward = 0.0
Iteration 100: Arm = 2, Reward = 0.0
Iteration 101: Arm = 2, Reward = 0.0
Iteration 102: Arm = 0, Reward = 1.0
Iteration 103: Arm = 0, Reward = 0.0
Iteration 104: Arm = 0, Reward = 1.0
Iteration 105: Arm = 0, Reward = 0.0
Iteration 106: Arm = 0, Reward = 0.0
Iteration 107: Arm = 1, Reward = 0.0
Iteration 108: Arm = 2, Reward = 0.0
Iteration 109: Arm = 0, Reward = 0.0
Iteration 110: Arm = 2, Reward = 0.0
Iteration 111: Arm = 2, Reward = 1.0
Iteration 112: Arm = 2, Reward = 0.0
Iteration 113: Arm = 3, Reward = 0.0
Iteration 114: Arm = 2, Reward = 1.0
Iteration 115: Arm = 2, Reward = 0.0
Iteration 116: Arm = 2, Reward = 0.0
Iteration 117: Arm = 0, Reward = 0.0
Iteration 118: Arm = 2, Reward = 0.0
Iteration 119: Arm = 1, Reward = 0.0
Iteration 120: Arm = 0, Reward = 1.0
Iteration 121: Arm = 0, Reward = 0.0
Iteration 122: Arm = 0, Reward = 0.0
Iteration 123: Arm = 2, Reward = 0.0
Iteration 124: Arm = 3, Reward = 1.0
Iteration 125: Arm = 2, Reward = 0.0
Iteration 126: Arm = 0, Reward = 1.0
Iteration 127: Arm = 0, Reward = 0.0
Iteration 128: Arm = 0, Reward = 0.0
Iteration 129: Arm = 2, Reward = 1.0
Iteration 130: Arm = 2, Reward = 0.0
Iteration 131: Arm = 1, Reward = 0.0
Iteration 132: Arm = 2, Reward = 1.0
Iteration 133: Arm = 0, Reward = 0.0
Iteration 134: Arm = 2, Reward = 1.0
Iteration 135: Arm = 2, Reward = 1.0
Iteration 136: Arm = 2, Reward = 0.0
Iteration 137: Arm = 3, Reward = 0.0
Iteration 138: Arm = 2, Reward = 1.0
Iteration 139: Arm = 2, Reward = 1.0
Iteration 140: Arm = 2, Reward = 0.0
Iteration 141: Arm = 2, Reward = 1.0
Iteration 142: Arm = 2, Reward = 0.0
Iteration 143: Arm = 2, Reward = 0.0
Iteration 144: Arm = 2, Reward = 0.0
Iteration 145: Arm = 3, Reward = 1.0
Iteration 146: Arm = 2, Reward = 0.0
Iteration 147: Arm = 3, Reward = 0.0
Iteration 148: Arm = 2, Reward = 0.0
Iteration 149: Arm = 2, Reward = 1.0
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Iteration 150: Arm = 2, Reward = 0.0
Iteration 151: Arm = 2, Reward = 0.0
Iteration 152: Arm = 2, Reward = 0.0
Iteration 153: Arm = 2, Reward = 1.0
Iteration 154: Arm = 2, Reward = 0.0
Iteration 155: Arm = 2, Reward = 1.0
Iteration 156: Arm = 2, Reward = 0.0
Iteration 157: Arm = 2, Reward = 0.0
Iteration 158: Arm = 2, Reward = 0.0
Iteration 159: Arm = 2, Reward = 1.0
Iteration 160: Arm = 1, Reward = 0.0
Iteration 161: Arm = 2, Reward = 1.0
Iteration 162: Arm = 2, Reward = 1.0
Iteration 163: Arm = 2, Reward = 0.0
Iteration 164: Arm = 2, Reward = 0.0
Iteration 165: Arm = 2, Reward = 0.0
Iteration 166: Arm = 2, Reward = 1.0
Iteration 167: Arm = 0, Reward = 0.0
Iteration 168: Arm = 2, Reward = 0.0
Iteration 169: Arm = 2, Reward = 0.0
Iteration 170: Arm = 2, Reward = 0.0
Iteration 171: Arm = 2, Reward = 0.0
Iteration 172: Arm = 2, Reward = 1.0
Iteration 173: Arm = 2, Reward = 1.0
Iteration 174: Arm = 2, Reward = 0.0
Iteration 175: Arm = 2, Reward = 0.0
Iteration 176: Arm = 2, Reward = 0.0
Iteration 177: Arm = 2, Reward = 1.0
Iteration 178: Arm = 2, Reward = 1.0
Iteration 179: Arm = 2, Reward = 0.0
Iteration 180: Arm = 1, Reward = 1.0
Iteration 181: Arm = 2, Reward = 1.0
Iteration 182: Arm = 2, Reward = 1.0
Iteration 183: Arm = 2, Reward = 0.0
Iteration 184: Arm = 2, Reward = 1.0
Iteration 185: Arm = 2, Reward = 0.0
Iteration 186: Arm = 2, Reward = 0.0
Iteration 187: Arm = 0, Reward = 0.0
Iteration 188: Arm = 2, Reward = 1.0
Iteration 189: Arm = 2, Reward = 0.0
Iteration 190: Arm = 2, Reward = 1.0
Iteration 191: Arm = 2, Reward = 1.0
Iteration 192: Arm = 0, Reward = 0.0
Iteration 193: Arm = 2, Reward = 0.0
Iteration 194: Arm = 2, Reward = 0.0
Iteration 195: Arm = 2, Reward = 1.0
Iteration 196: Arm = 2, Reward = 0.0
Iteration 197: Arm = 2, Reward = 1.0
Iteration 198: Arm = 2, Reward = 1.0
Iteration 199: Arm = 2, Reward = 1.0
Iteration 200: Arm = 0, Reward = 1.0
Iteration 201: Arm = 2, Reward = 1.0
Iteration 202: Arm = 2, Reward = 1.0
Iteration 203: Arm = 2, Reward = 0.0
Iteration 204: Arm = 2, Reward = 0.0
Iteration 205: Arm = 3, Reward = 0.0
Iteration 206: Arm = 2, Reward = 0.0
Iteration 207: Arm = 3, Reward = 0.0
Iteration 208: Arm = 1, Reward = 0.0
Iteration 209: Arm = 2, Reward = 0.0
Iteration 210: Arm = 2, Reward = 0.0
Iteration 211: Arm = 2, Reward = 0.0
Iteration 212: Arm = 0, Reward = 0.0
Iteration 213: Arm = 2, Reward = 0.0
Iteration 214: Arm = 2, Reward = 1.0
Iteration 215: Arm = 2, Reward = 0.0
Iteration 216: Arm = 2, Reward = 1.0
Iteration 217: Arm = 2, Reward = 1.0
Iteration 218: Arm = 2, Reward = 1.0
Iteration 219: Arm = 0, Reward = 0.0
Iteration 220: Arm = 2, Reward = 0.0
Iteration 221: Arm = 2, Reward = 1.0
Iteration 222: Arm = 2, Reward = 1.0
Iteration 223: Arm = 0, Reward = 1.0
Iteration 224: Arm = 2, Reward = 1.0
Iteration 225: Arm = 2, Reward = 0.0
Iteration 226: Arm = 2, Reward = 0.0
Iteration 227: Arm = 2, Reward = 1.0
Iteration 228: Arm = 2, Reward = 1.0
Iteration 229: Arm = 3, Reward = 0.0
Iteration 230: Arm = 2, Reward = 0.0
Iteration 231: Arm = 2, Reward = 0.0
Iteration 232: Arm = 2, Reward = 0.0
```

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Iteration 233: Arm = 2, Reward = 1.0
Iteration 234: Arm = 2, Reward = 0.0
Iteration 235: Arm = 2, Reward = 0.0
Iteration 236: Arm = 2, Reward = 1.0
Iteration 237: Arm = 2, Reward = 1.0
Iteration 238: Arm = 2, Reward = 0.0
Iteration 239: Arm = 2, Reward = 0.0
Iteration 240: Arm = 2, Reward = 1.0
Iteration 241: Arm = 2, Reward = 1.0
Iteration 242: Arm = 2, Reward = 0.0
Iteration 243: Arm = 2, Reward = 0.0
Iteration 244: Arm = 2, Reward = 0.0
Iteration 245: Arm = 2, Reward = 1.0
Iteration 246: Arm = 2, Reward = 0.0
Iteration 247: Arm = 2, Reward = 0.0
Iteration 248: Arm = 2, Reward = 0.0
Iteration 249: Arm = 2, Reward = 0.0
Iteration 250: Arm = 3, Reward = 1.0
Iteration 251: Arm = 2, Reward = 1.0
Iteration 252: Arm = 2, Reward = 0.0
Iteration 253: Arm = 2, Reward = 1.0
Iteration 254: Arm = 2, Reward = 1.0
Iteration 255: Arm = 0, Reward = 0.0
Iteration 256: Arm = 2, Reward = 1.0
Iteration 257: Arm = 1, Reward = 0.0
Iteration 258: Arm = 2, Reward = 1.0
Iteration 259: Arm = 0, Reward = 1.0
Iteration 260: Arm = 2, Reward = 0.0
Iteration 261: Arm = 2, Reward = 0.0
Iteration 262: Arm = 2, Reward = 1.0
Iteration 263: Arm = 2, Reward = 1.0
Iteration 264: Arm = 2, Reward = 0.0
Iteration 265: Arm = 2, Reward = 1.0
Iteration 266: Arm = 2, Reward = 0.0
Iteration 267: Arm = 2, Reward = 0.0
Iteration 268: Arm = 2, Reward = 0.0
Iteration 269: Arm = 2, Reward = 0.0
Iteration 270: Arm = 2, Reward = 0.0
Iteration 271: Arm = 2, Reward = 0.0
Iteration 272: Arm = 2, Reward = 0.0
Iteration 273: Arm = 2, Reward = 1.0
Iteration 274: Arm = 2, Reward = 0.0
Iteration 275: Arm = 1, Reward = 0.0
Iteration 276: Arm = 2, Reward = 1.0
Iteration 277: Arm = 2, Reward = 1.0
Iteration 278: Arm = 2, Reward = 0.0
Iteration 279: Arm = 2, Reward = 1.0
Iteration 280: Arm = 2, Reward = 1.0
Iteration 281: Arm = 2, Reward = 0.0
Iteration 282: Arm = 2, Reward = 0.0
Iteration 283: Arm = 0, Reward = 0.0
Iteration 284: Arm = 2, Reward = 1.0
Iteration 285: Arm = 2, Reward = 0.0
Iteration 286: Arm = 2, Reward = 0.0
Iteration 287: Arm = 2, Reward = 0.0
Iteration 288: Arm = 2, Reward = 1.0
Iteration 289: Arm = 2, Reward = 1.0
Iteration 290: Arm = 2, Reward = 1.0
Iteration 291: Arm = 2, Reward = 0.0
Iteration 292: Arm = 0, Reward = 1.0
Iteration 293: Arm = 2, Reward = 1.0
Iteration 294: Arm = 2, Reward = 0.0
Iteration 295: Arm = 0, Reward = 0.0
Iteration 296: Arm = 2, Reward = 1.0
Iteration 297: Arm = 2, Reward = 1.0
Iteration 298: Arm = 2, Reward = 1.0
Iteration 299: Arm = 2, Reward = 1.0
Iteration 300: Arm = 2, Reward = 1.0
Iteration 301: Arm = 0, Reward = 0.0
Iteration 302: Arm = 2, Reward = 1.0
Iteration 303: Arm = 3, Reward = 0.0
Iteration 304: Arm = 2, Reward = 0.0
Iteration 305: Arm = 2, Reward = 1.0
Iteration 306: Arm = 2, Reward = 1.0
Iteration 307: Arm = 2, Reward = 1.0
Iteration 308: Arm = 2, Reward = 1.0
Iteration 309: Arm = 2, Reward = 0.0
Iteration 310: Arm = 2, Reward = 1.0
Iteration 311: Arm = 2, Reward = 1.0
Iteration 312: Arm = 2, Reward = 0.0
Iteration 313: Arm = 2, Reward = 0.0
Iteration 314: Arm = 2, Reward = 0.0
Iteration 315: Arm = 2, Reward = 0.0
```

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Iteration 316: Arm = 2, Reward = 1.0
Iteration 317: Arm = 2, Reward = 1.0
Iteration 318: Arm = 2, Reward = 1.0
Iteration 319: Arm = 2, Reward = 1.0
Iteration 320: Arm = 2, Reward = 0.0
Iteration 321: Arm = 2, Reward = 1.0
Iteration 322: Arm = 2, Reward = 0.0
Iteration 323: Arm = 2, Reward = 1.0
Iteration 324: Arm = 2, Reward = 1.0
Iteration 325: Arm = 2, Reward = 1.0
Iteration 326: Arm = 2, Reward = 0.0
Iteration 327: Arm = 1, Reward = 1.0
Iteration 328: Arm = 2, Reward = 0.0
Iteration 329: Arm = 2, Reward = 1.0
Iteration 330: Arm = 2, Reward = 1.0
Iteration 331: Arm = 2, Reward = 0.0
Iteration 332: Arm = 2, Reward = 1.0
Iteration 333: Arm = 2, Reward = 1.0
Iteration 334: Arm = 3, Reward = 1.0
Iteration 335: Arm = 2, Reward = 0.0
Iteration 336: Arm = 2, Reward = 0.0
Iteration 337: Arm = 2, Reward = 0.0
Iteration 338: Arm = 2, Reward = 0.0
Iteration 339: Arm = 2, Reward = 0.0
Iteration 340: Arm = 2, Reward = 1.0
Iteration 341: Arm = 2, Reward = 1.0
Iteration 342: Arm = 2, Reward = 0.0
Iteration 343: Arm = 2, Reward = 1.0
Iteration 344: Arm = 2, Reward = 0.0
Iteration 345: Arm = 2, Reward = 1.0
Iteration 346: Arm = 2, Reward = 0.0
Iteration 347: Arm = 2, Reward = 1.0
Iteration 348: Arm = 2, Reward = 1.0
Iteration 349: Arm = 2, Reward = 0.0
Iteration 350: Arm = 2, Reward = 0.0
Iteration 351: Arm = 2, Reward = 1.0
Iteration 352: Arm = 0, Reward = 1.0
Iteration 353: Arm = 2, Reward = 0.0
Iteration 354: Arm = 2, Reward = 1.0
Iteration 355: Arm = 2, Reward = 1.0
Iteration 356: Arm = 0, Reward = 1.0
Iteration 357: Arm = 2, Reward = 0.0
Iteration 358: Arm = 2, Reward = 0.0
Iteration 359: Arm = 2, Reward = 0.0
Iteration 360: Arm = 2, Reward = 0.0
Iteration 361: Arm = 1, Reward = 0.0
Iteration 362: Arm = 0, Reward = 0.0
Iteration 363: Arm = 2, Reward = 1.0
Iteration 364: Arm = 2, Reward = 0.0
Iteration 365: Arm = 2, Reward = 1.0
Iteration 366: Arm = 2, Reward = 0.0
Iteration 367: Arm = 2, Reward = 1.0
Iteration 368: Arm = 2, Reward = 0.0
Iteration 369: Arm = 2, Reward = 1.0
Iteration 370: Arm = 2, Reward = 1.0
Iteration 371: Arm = 3, Reward = 0.0
Iteration 372: Arm = 2, Reward = 0.0
Iteration 373: Arm = 2, Reward = 0.0
Iteration 374: Arm = 2, Reward = 1.0
Iteration 375: Arm = 2, Reward = 0.0
Iteration 376: Arm = 2, Reward = 0.0
Iteration 377: Arm = 2, Reward = 0.0
Iteration 378: Arm = 2, Reward = 1.0
Iteration 379: Arm = 2, Reward = 0.0
Iteration 380: Arm = 3, Reward = 0.0
Iteration 381: Arm = 2, Reward = 0.0
Iteration 382: Arm = 2, Reward = 1.0
Iteration 383: Arm = 2, Reward = 1.0
Iteration 384: Arm = 2, Reward = 1.0
Iteration 385: Arm = 2, Reward = 0.0
Iteration 386: Arm = 2, Reward = 0.0
Iteration 387: Arm = 2, Reward = 1.0
Iteration 388: Arm = 2, Reward = 0.0
Iteration 389: Arm = 2, Reward = 0.0
Iteration 390: Arm = 2, Reward = 1.0
Iteration 391: Arm = 2, Reward = 0.0
Iteration 392: Arm = 2, Reward = 0.0
Iteration 393: Arm = 2, Reward = 1.0
Iteration 394: Arm = 2, Reward = 1.0
Iteration 395: Arm = 2, Reward = 1.0
Iteration 396: Arm = 2, Reward = 0.0
Iteration 397: Arm = 2, Reward = 1.0
Iteration 398: Arm = 1, Reward = 0.0
```

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Iteration 399: Arm = 2, Reward = 0.0
Iteration 400: Arm = 3, Reward = 1.0
Iteration 401: Arm = 2, Reward = 1.0
Iteration 402: Arm = 2, Reward = 0.0
Iteration 403: Arm = 2, Reward = 0.0
Iteration 404: Arm = 2, Reward = 0.0
Iteration 405: Arm = 3, Reward = 1.0
Iteration 406: Arm = 2, Reward = 1.0
Iteration 407: Arm = 2, Reward = 1.0
Iteration 408: Arm = 2, Reward = 0.0
Iteration 409: Arm = 2, Reward = 1.0
Iteration 410: Arm = 2, Reward = 0.0
Iteration 411: Arm = 2, Reward = 1.0
Iteration 412: Arm = 0, Reward = 1.0
Iteration 413: Arm = 2, Reward = 0.0
Iteration 414: Arm = 2, Reward = 1.0
Iteration 415: Arm = 2, Reward = 1.0
Iteration 416: Arm = 2, Reward = 1.0
Iteration 417: Arm = 2, Reward = 1.0
Iteration 418: Arm = 2, Reward = 1.0
Iteration 419: Arm = 2, Reward = 1.0
Iteration 420: Arm = 2, Reward = 1.0
Iteration 421: Arm = 2, Reward = 1.0
Iteration 422: Arm = 2, Reward = 0.0
Iteration 423: Arm = 2, Reward = 1.0
Iteration 424: Arm = 3, Reward = 1.0
Iteration 425: Arm = 2, Reward = 0.0
Iteration 426: Arm = 2, Reward = 1.0
Iteration 427: Arm = 2, Reward = 1.0
Iteration 428: Arm = 2, Reward = 0.0
Iteration 429: Arm = 2, Reward = 1.0
Iteration 430: Arm = 2, Reward = 1.0
Iteration 431: Arm = 2, Reward = 0.0
Iteration 432: Arm = 2, Reward = 1.0
Iteration 433: Arm = 2, Reward = 1.0
Iteration 434: Arm = 2, Reward = 0.0
Iteration 435: Arm = 2, Reward = 0.0
Iteration 436: Arm = 2, Reward = 0.0
Iteration 437: Arm = 2, Reward = 0.0
Iteration 438: Arm = 2, Reward = 0.0
Iteration 439: Arm = 0, Reward = 0.0
Iteration 440: Arm = 2, Reward = 0.0
Iteration 441: Arm = 2, Reward = 0.0
Iteration 442: Arm = 2, Reward = 0.0
Iteration 443: Arm = 3, Reward = 1.0
Iteration 444: Arm = 2, Reward = 0.0
Iteration 445: Arm = 2, Reward = 0.0
Iteration 446: Arm = 2, Reward = 1.0
Iteration 447: Arm = 2, Reward = 1.0
Iteration 448: Arm = 2, Reward = 1.0
Iteration 449: Arm = 1, Reward = 0.0
Iteration 450: Arm = 2, Reward = 0.0
Iteration 451: Arm = 2, Reward = 1.0
Iteration 452: Arm = 2, Reward = 1.0
Iteration 453: Arm = 2, Reward = 0.0
Iteration 454: Arm = 2, Reward = 0.0
Iteration 455: Arm = 2, Reward = 0.0
Iteration 456: Arm = 2, Reward = 0.0
Iteration 457: Arm = 2, Reward = 0.0
Iteration 458: Arm = 2, Reward = 0.0
Iteration 459: Arm = 2, Reward = 1.0
Iteration 460: Arm = 2, Reward = 0.0
Iteration 461: Arm = 2, Reward = 0.0
Iteration 462: Arm = 3, Reward = 0.0
Iteration 463: Arm = 3, Reward = 1.0
Iteration 464: Arm = 2, Reward = 1.0
Iteration 465: Arm = 2, Reward = 0.0
Iteration 466: Arm = 2, Reward = 1.0
Iteration 467: Arm = 2, Reward = 0.0
Iteration 468: Arm = 2, Reward = 1.0
Iteration 469: Arm = 2, Reward = 1.0
Iteration 470: Arm = 2, Reward = 0.0
Iteration 471: Arm = 2, Reward = 0.0
Iteration 472: Arm = 2, Reward = 0.0
Iteration 473: Arm = 2, Reward = 0.0
Iteration 474: Arm = 2, Reward = 1.0
Iteration 475: Arm = 2, Reward = 0.0
Iteration 476: Arm = 1, Reward = 1.0
Iteration 477: Arm = 2, Reward = 0.0
Iteration 478: Arm = 2, Reward = 1.0
Iteration 479: Arm = 2, Reward = 0.0
Iteration 480: Arm = 2, Reward = 1.0
Iteration 481: Arm = 2, Reward = 0.0
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Iteration 482: Arm = 3, Reward = 1.0
Iteration 483: Arm = 0, Reward = 0.0
Iteration 484: Arm = 0, Reward = 0.0
Iteration 485: Arm = 2, Reward = 0.0
Iteration 486: Arm = 2, Reward = 0.0
Iteration 487: Arm = 3, Reward = 0.0
Iteration 488: Arm = 2, Reward = 0.0
Iteration 489: Arm = 2, Reward = 0.0
Iteration 490: Arm = 1, Reward = 0.0
Iteration 491: Arm = 2, Reward = 1.0
Iteration 492: Arm = 2, Reward = 0.0
Iteration 493: Arm = 2, Reward = 1.0
Iteration 494: Arm = 1, Reward = 0.0
Iteration 495: Arm = 2, Reward = 0.0
Iteration 496: Arm = 2, Reward = 0.0
Iteration 497: Arm = 1, Reward = 0.0
Iteration 498: Arm = 2, Reward = 0.0
Iteration 499: Arm = 2, Reward = 0.0
Iteration 500: Arm = 1, Reward = 1.0
Iteration 501: Arm = 2, Reward = 0.0
Iteration 502: Arm = 2, Reward = 1.0
Iteration 503: Arm = 2, Reward = 1.0
Iteration 504: Arm = 1, Reward = 1.0
Iteration 505: Arm = 2, Reward = 0.0
Iteration 506: Arm = 2, Reward = 0.0
Iteration 507: Arm = 2, Reward = 1.0
Iteration 508: Arm = 2, Reward = 0.0
Iteration 509: Arm = 2, Reward = 1.0
Iteration 510: Arm = 2, Reward = 0.0
Iteration 511: Arm = 2, Reward = 0.0
Iteration 512: Arm = 2, Reward = 1.0
Iteration 513: Arm = 2, Reward = 0.0
Iteration 514: Arm = 2, Reward = 1.0
Iteration 515: Arm = 2, Reward = 0.0
Iteration 516: Arm = 3, Reward = 0.0
Iteration 517: Arm = 2, Reward = 0.0
Iteration 518: Arm = 0, Reward = 0.0
Iteration 519: Arm = 2, Reward = 1.0
Iteration 520: Arm = 2, Reward = 1.0
Iteration 521: Arm = 2, Reward = 1.0
Iteration 522: Arm = 2, Reward = 0.0
Iteration 523: Arm = 2, Reward = 0.0
Iteration 524: Arm = 2, Reward = 0.0
Iteration 525: Arm = 1, Reward = 1.0
Iteration 526: Arm = 2, Reward = 0.0
Iteration 527: Arm = 1, Reward = 1.0
Iteration 528: Arm = 2, Reward = 1.0
Iteration 529: Arm = 2, Reward = 0.0
Iteration 530: Arm = 3, Reward = 1.0
Iteration 531: Arm = 0, Reward = 0.0
Iteration 532: Arm = 2, Reward = 1.0
Iteration 533: Arm = 1, Reward = 1.0
Iteration 534: Arm = 2, Reward = 1.0
Iteration 535: Arm = 2, Reward = 1.0
Iteration 536: Arm = 2, Reward = 1.0
Iteration 537: Arm = 2, Reward = 0.0
Iteration 538: Arm = 2, Reward = 1.0
Iteration 539: Arm = 2, Reward = 1.0
Iteration 540: Arm = 2, Reward = 1.0
Iteration 541: Arm = 2, Reward = 0.0
Iteration 542: Arm = 0, Reward = 0.0
Iteration 543: Arm = 2, Reward = 1.0
Iteration 544: Arm = 2, Reward = 1.0
Iteration 545: Arm = 3, Reward = 1.0
Iteration 546: Arm = 0, Reward = 1.0
Iteration 547: Arm = 2, Reward = 1.0
Iteration 548: Arm = 2, Reward = 1.0
Iteration 549: Arm = 2, Reward = 1.0
Iteration 550: Arm = 2, Reward = 0.0
Iteration 551: Arm = 2, Reward = 0.0
Iteration 552: Arm = 2, Reward = 1.0
Iteration 553: Arm = 2, Reward = 1.0
Iteration 554: Arm = 2, Reward = 0.0
Iteration 555: Arm = 2, Reward = 1.0
Iteration 556: Arm = 3, Reward = 0.0
Iteration 557: Arm = 2, Reward = 0.0
Iteration 558: Arm = 2, Reward = 0.0
Iteration 559: Arm = 2, Reward = 1.0
Iteration 560: Arm = 1, Reward = 0.0
Iteration 561: Arm = 2, Reward = 1.0
Iteration 562: Arm = 0, Reward = 0.0
Iteration 563: Arm = 2, Reward = 0.0
Iteration 564: Arm = 2, Reward = 1.0
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Iteration 565: Arm = 3, Reward = 0.0
Iteration 566: Arm = 2, Reward = 0.0
Iteration 567: Arm = 2, Reward = 0.0
Iteration 568: Arm = 0, Reward = 0.0
Iteration 569: Arm = 2, Reward = 1.0
Iteration 570: Arm = 2, Reward = 1.0
Iteration 571: Arm = 2, Reward = 0.0
Iteration 572: Arm = 2, Reward = 0.0
Iteration 573: Arm = 2, Reward = 0.0
Iteration 574: Arm = 1, Reward = 1.0
Iteration 575: Arm = 2, Reward = 1.0
Iteration 576: Arm = 0, Reward = 0.0
Iteration 577: Arm = 2, Reward = 0.0
Iteration 578: Arm = 3, Reward = 1.0
Iteration 579: Arm = 2, Reward = 1.0
Iteration 580: Arm = 2, Reward = 0.0
Iteration 581: Arm = 2, Reward = 0.0
Iteration 582: Arm = 2, Reward = 0.0
Iteration 583: Arm = 3, Reward = 1.0
Iteration 584: Arm = 2, Reward = 1.0
Iteration 585: Arm = 3, Reward = 0.0
Iteration 586: Arm = 2, Reward = 0.0
Iteration 587: Arm = 2, Reward = 0.0
Iteration 588: Arm = 2, Reward = 0.0
Iteration 589: Arm = 0, Reward = 0.0
Iteration 590: Arm = 2, Reward = 1.0
Iteration 591: Arm = 2, Reward = 1.0
Iteration 592: Arm = 2, Reward = 0.0
Iteration 593: Arm = 2, Reward = 1.0
Iteration 594: Arm = 2, Reward = 0.0
Iteration 595: Arm = 2, Reward = 1.0
Iteration 596: Arm = 2, Reward = 0.0
Iteration 597: Arm = 2, Reward = 1.0
Iteration 598: Arm = 2, Reward = 0.0
Iteration 599: Arm = 2, Reward = 0.0
Iteration 600: Arm = 2, Reward = 1.0
Iteration 601: Arm = 2, Reward = 1.0
Iteration 602: Arm = 2, Reward = 1.0
Iteration 603: Arm = 2, Reward = 0.0
Iteration 604: Arm = 2, Reward = 1.0
Iteration 605: Arm = 2, Reward = 0.0
Iteration 606: Arm = 2, Reward = 0.0
Iteration 607: Arm = 2, Reward = 1.0
Iteration 608: Arm = 2, Reward = 0.0
Iteration 609: Arm = 2, Reward = 0.0
Iteration 610: Arm = 2, Reward = 1.0
Iteration 611: Arm = 2, Reward = 0.0
Iteration 612: Arm = 2, Reward = 0.0
Iteration 613: Arm = 2, Reward = 0.0
Iteration 614: Arm = 2, Reward = 0.0
Iteration 615: Arm = 2, Reward = 0.0
Iteration 616: Arm = 2, Reward = 1.0
Iteration 617: Arm = 2, Reward = 1.0
Iteration 618: Arm = 2, Reward = 0.0
Iteration 619: Arm = 2, Reward = 1.0
Iteration 620: Arm = 2, Reward = 0.0
Iteration 621: Arm = 2, Reward = 1.0
Iteration 622: Arm = 2, Reward = 1.0
Iteration 623: Arm = 2, Reward = 0.0
Iteration 624: Arm = 2, Reward = 0.0
Iteration 625: Arm = 2, Reward = 0.0
Iteration 626: Arm = 2, Reward = 1.0
Iteration 627: Arm = 2, Reward = 1.0
Iteration 628: Arm = 1, Reward = 1.0
Iteration 629: Arm = 3, Reward = 1.0
Iteration 630: Arm = 2, Reward = 0.0
Iteration 631: Arm = 1, Reward = 1.0
Iteration 632: Arm = 3, Reward = 1.0
Iteration 633: Arm = 3, Reward = 1.0
Iteration 634: Arm = 3, Reward = 1.0
Iteration 635: Arm = 3, Reward = 0.0
Iteration 636: Arm = 2, Reward = 0.0
Iteration 637: Arm = 3, Reward = 0.0
Iteration 638: Arm = 3, Reward = 1.0
Iteration 639: Arm = 3, Reward = 0.0
Iteration 640: Arm = 3, Reward = 0.0
Iteration 641: Arm = 3, Reward = 1.0
Iteration 642: Arm = 3, Reward = 0.0
Iteration 643: Arm = 3, Reward = 1.0
Iteration 644: Arm = 3, Reward = 0.0
Iteration 645: Arm = 3, Reward = 0.0
Iteration 646: Arm = 3, Reward = 0.0
Iteration 647: Arm = 2, Reward = 0.0
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Iteration 648: Arm = 2, Reward = 1.0
Iteration 649: Arm = 2, Reward = 0.0
Iteration 650: Arm = 2, Reward = 1.0
Iteration 651: Arm = 2, Reward = 1.0
Iteration 652: Arm = 2, Reward = 0.0
Iteration 653: Arm = 2, Reward = 0.0
Iteration 654: Arm = 2, Reward = 1.0
Iteration 655: Arm = 2, Reward = 0.0
Iteration 656: Arm = 2, Reward = 1.0
Iteration 657: Arm = 2, Reward = 0.0
Iteration 658: Arm = 2, Reward = 0.0
Iteration 659: Arm = 2, Reward = 0.0
Iteration 660: Arm = 2, Reward = 1.0
Iteration 661: Arm = 2, Reward = 0.0
Iteration 662: Arm = 1, Reward = 0.0
Iteration 663: Arm = 2, Reward = 0.0
Iteration 664: Arm = 2, Reward = 0.0
Iteration 665: Arm = 2, Reward = 1.0
Iteration 666: Arm = 1, Reward = 1.0
Iteration 667: Arm = 2, Reward = 0.0
Iteration 668: Arm = 2, Reward = 0.0
Iteration 669: Arm = 3, Reward = 1.0
Iteration 670: Arm = 3, Reward = 0.0
Iteration 671: Arm = 2, Reward = 1.0
Iteration 672: Arm = 2, Reward = 0.0
Iteration 673: Arm = 2, Reward = 1.0
Iteration 674: Arm = 2, Reward = 0.0
Iteration 675: Arm = 2, Reward = 1.0
Iteration 676: Arm = 2, Reward = 0.0
Iteration 677: Arm = 2, Reward = 0.0
Iteration 678: Arm = 2, Reward = 1.0
Iteration 679: Arm = 2, Reward = 1.0
Iteration 680: Arm = 2, Reward = 1.0
Iteration 681: Arm = 2, Reward = 0.0
Iteration 682: Arm = 2, Reward = 1.0
Iteration 683: Arm = 2, Reward = 1.0
Iteration 684: Arm = 2, Reward = 0.0
Iteration 685: Arm = 1, Reward = 1.0
Iteration 686: Arm = 2, Reward = 1.0
Iteration 687: Arm = 2, Reward = 0.0
Iteration 688: Arm = 2, Reward = 1.0
Iteration 689: Arm = 2, Reward = 1.0
Iteration 690: Arm = 2, Reward = 1.0
Iteration 691: Arm = 2, Reward = 1.0
Iteration 692: Arm = 1, Reward = 0.0
Iteration 693: Arm = 2, Reward = 1.0
Iteration 694: Arm = 2, Reward = 1.0
Iteration 695: Arm = 2, Reward = 0.0
Iteration 696: Arm = 2, Reward = 0.0
Iteration 697: Arm = 2, Reward = 0.0
Iteration 698: Arm = 2, Reward = 1.0
Iteration 699: Arm = 2, Reward = 1.0
Iteration 700: Arm = 2, Reward = 1.0
Iteration 701: Arm = 1, Reward = 0.0
Iteration 702: Arm = 2, Reward = 1.0
Iteration 703: Arm = 2, Reward = 1.0
Iteration 704: Arm = 2, Reward = 1.0
Iteration 705: Arm = 2, Reward = 1.0
Iteration 706: Arm = 2, Reward = 1.0
Iteration 707: Arm = 2, Reward = 0.0
Iteration 708: Arm = 2, Reward = 0.0
Iteration 709: Arm = 2, Reward = 0.0
Iteration 710: Arm = 2, Reward = 0.0
Iteration 711: Arm = 2, Reward = 1.0
Iteration 712: Arm = 2, Reward = 1.0
Iteration 713: Arm = 2, Reward = 0.0
Iteration 714: Arm = 2, Reward = 1.0
Iteration 715: Arm = 2, Reward = 1.0
Iteration 716: Arm = 2, Reward = 1.0
Iteration 717: Arm = 2, Reward = 0.0
Iteration 718: Arm = 2, Reward = 1.0
Iteration 719: Arm = 3, Reward = 0.0
Iteration 720: Arm = 2, Reward = 1.0
Iteration 721: Arm = 3, Reward = 1.0
Iteration 722: Arm = 2, Reward = 1.0
Iteration 723: Arm = 2, Reward = 0.0
Iteration 724: Arm = 2, Reward = 1.0
Iteration 725: Arm = 2, Reward = 0.0
Iteration 726: Arm = 2, Reward = 1.0
Iteration 727: Arm = 1, Reward = 0.0
Iteration 728: Arm = 2, Reward = 1.0
Iteration 729: Arm = 0, Reward = 0.0
Iteration 730: Arm = 2, Reward = 1.0
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Iteration 731: Arm = 0, Reward = 1.0
Iteration 732: Arm = 2, Reward = 0.0
Iteration 733: Arm = 0, Reward = 0.0
Iteration 734: Arm = 0, Reward = 1.0
Iteration 735: Arm = 0, Reward = 0.0
Iteration 736: Arm = 2, Reward = 0.0
Iteration 737: Arm = 3, Reward = 1.0
Iteration 738: Arm = 2, Reward = 0.0
Iteration 739: Arm = 2, Reward = 0.0
Iteration 740: Arm = 0, Reward = 0.0
Iteration 741: Arm = 2, Reward = 0.0
Iteration 742: Arm = 2, Reward = 1.0
Iteration 743: Arm = 2, Reward = 1.0
Iteration 744: Arm = 2, Reward = 1.0
Iteration 745: Arm = 2, Reward = 0.0
Iteration 746: Arm = 2, Reward = 1.0
Iteration 747: Arm = 2, Reward = 0.0
Iteration 748: Arm = 2, Reward = 1.0
Iteration 749: Arm = 2, Reward = 0.0
Iteration 750: Arm = 2, Reward = 0.0
Iteration 751: Arm = 2, Reward = 0.0
Iteration 752: Arm = 2, Reward = 0.0
Iteration 753: Arm = 2, Reward = 0.0
Iteration 754: Arm = 2, Reward = 1.0
Iteration 755: Arm = 2, Reward = 1.0
Iteration 756: Arm = 2, Reward = 1.0
Iteration 757: Arm = 2, Reward = 0.0
Iteration 758: Arm = 2, Reward = 0.0
Iteration 759: Arm = 2, Reward = 1.0
Iteration 760: Arm = 0, Reward = 0.0
Iteration 761: Arm = 3, Reward = 1.0
Iteration 762: Arm = 2, Reward = 0.0
Iteration 763: Arm = 3, Reward = 1.0
Iteration 764: Arm = 3, Reward = 1.0
Iteration 765: Arm = 3, Reward = 0.0
Iteration 766: Arm = 3, Reward = 0.0
Iteration 767: Arm = 3, Reward = 1.0
Iteration 768: Arm = 3, Reward = 0.0
Iteration 769: Arm = 1, Reward = 1.0
Iteration 770: Arm = 3, Reward = 0.0
Iteration 771: Arm = 2, Reward = 1.0
Iteration 772: Arm = 2, Reward = 1.0
Iteration 773: Arm = 2, Reward = 0.0
Iteration 774: Arm = 2, Reward = 1.0
Iteration 775: Arm = 0, Reward = 0.0
Iteration 776: Arm = 2, Reward = 1.0
Iteration 777: Arm = 2, Reward = 1.0
Iteration 778: Arm = 2, Reward = 0.0
Iteration 779: Arm = 2, Reward = 0.0
Iteration 780: Arm = 2, Reward = 0.0
Iteration 781: Arm = 2, Reward = 1.0
Iteration 782: Arm = 2, Reward = 0.0
Iteration 783: Arm = 2, Reward = 1.0
Iteration 784: Arm = 2, Reward = 0.0
Iteration 785: Arm = 2, Reward = 1.0
Iteration 786: Arm = 2, Reward = 1.0
Iteration 787: Arm = 2, Reward = 0.0
Iteration 788: Arm = 2, Reward = 0.0
Iteration 789: Arm = 0, Reward = 0.0
Iteration 790: Arm = 2, Reward = 1.0
Iteration 791: Arm = 2, Reward = 0.0
Iteration 792: Arm = 2, Reward = 0.0
Iteration 793: Arm = 2, Reward = 0.0
Iteration 794: Arm = 2, Reward = 0.0
Iteration 795: Arm = 1, Reward = 1.0
Iteration 796: Arm = 3, Reward = 1.0
Iteration 797: Arm = 3, Reward = 0.0
Iteration 798: Arm = 3, Reward = 1.0
Iteration 799: Arm = 3, Reward = 1.0
Iteration 800: Arm = 3, Reward = 0.0
Iteration 801: Arm = 3, Reward = 0.0
Iteration 802: Arm = 3, Reward = 0.0
Iteration 803: Arm = 2, Reward = 0.0
Iteration 804: Arm = 2, Reward = 1.0
Iteration 805: Arm = 2, Reward = 0.0
Iteration 806: Arm = 2, Reward = 0.0
Iteration 807: Arm = 2, Reward = 0.0
Iteration 808: Arm = 2, Reward = 0.0
Iteration 809: Arm = 2, Reward = 1.0
Iteration 810: Arm = 2, Reward = 0.0
Iteration 811: Arm = 2, Reward = 0.0
Iteration 812: Arm = 2, Reward = 0.0
Iteration 813: Arm = 3, Reward = 1.0
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Iteration 814: Arm = 1, Reward = 1.0
Iteration 815: Arm = 3, Reward = 0.0
Iteration 816: Arm = 3, Reward = 1.0
Iteration 817: Arm = 3, Reward = 0.0
Iteration 818: Arm = 3, Reward = 0.0
Iteration 819: Arm = 2, Reward = 0.0
Iteration 820: Arm = 2, Reward = 0.0
Iteration 821: Arm = 2, Reward = 1.0
Iteration 822: Arm = 2, Reward = 1.0
Iteration 823: Arm = 2, Reward = 1.0
Iteration 824: Arm = 2, Reward = 0.0
Iteration 825: Arm = 2, Reward = 1.0
Iteration 826: Arm = 2, Reward = 0.0
Iteration 827: Arm = 2, Reward = 1.0
Iteration 828: Arm = 2, Reward = 0.0
Iteration 829: Arm = 3, Reward = 0.0
Iteration 830: Arm = 2, Reward = 1.0
Iteration 831: Arm = 3, Reward = 0.0
Iteration 832: Arm = 2, Reward = 1.0
Iteration 833: Arm = 2, Reward = 0.0
Iteration 834: Arm = 1, Reward = 1.0
Iteration 835: Arm = 0, Reward = 0.0
Iteration 836: Arm = 1, Reward = 1.0
Iteration 837: Arm = 2, Reward = 1.0
Iteration 838: Arm = 1, Reward = 0.0
Iteration 839: Arm = 1, Reward = 1.0
Iteration 840: Arm = 1, Reward = 0.0
Iteration 841: Arm = 1, Reward = 0.0
Iteration 842: Arm = 2, Reward = 0.0
Iteration 843: Arm = 2, Reward = 1.0
Iteration 844: Arm = 2, Reward = 0.0
Iteration 845: Arm = 0, Reward = 1.0
Iteration 846: Arm = 2, Reward = 0.0
Iteration 847: Arm = 2, Reward = 1.0
Iteration 848: Arm = 2, Reward = 1.0
Iteration 849: Arm = 2, Reward = 0.0
Iteration 850: Arm = 2, Reward = 0.0
Iteration 851: Arm = 2, Reward = 1.0
Iteration 852: Arm = 2, Reward = 0.0
Iteration 853: Arm = 2, Reward = 0.0
Iteration 854: Arm = 2, Reward = 0.0
Iteration 855: Arm = 1, Reward = 0.0
Iteration 856: Arm = 2, Reward = 0.0
Iteration 857: Arm = 2, Reward = 1.0
Iteration 858: Arm = 2, Reward = 0.0
Iteration 859: Arm = 2, Reward = 0.0
Iteration 860: Arm = 2, Reward = 0.0
Iteration 861: Arm = 2, Reward = 1.0
Iteration 862: Arm = 3, Reward = 0.0
Iteration 863: Arm = 2, Reward = 0.0
Iteration 864: Arm = 2, Reward = 0.0
Iteration 865: Arm = 2, Reward = 1.0
Iteration 866: Arm = 2, Reward = 0.0
Iteration 867: Arm = 2, Reward = 1.0
Iteration 868: Arm = 3, Reward = 0.0
Iteration 869: Arm = 2, Reward = 0.0
Iteration 870: Arm = 2, Reward = 0.0
Iteration 871: Arm = 2, Reward = 0.0
Iteration 872: Arm = 2, Reward = 1.0
Iteration 873: Arm = 2, Reward = 1.0
Iteration 874: Arm = 2, Reward = 0.0
Iteration 875: Arm = 2, Reward = 1.0
Iteration 876: Arm = 2, Reward = 0.0
Iteration 877: Arm = 2, Reward = 1.0
Iteration 878: Arm = 1, Reward = 1.0
Iteration 879: Arm = 1, Reward = 0.0
Iteration 880: Arm = 2, Reward = 1.0
Iteration 881: Arm = 2, Reward = 0.0
Iteration 882: Arm = 2, Reward = 0.0
Iteration 883: Arm = 2, Reward = 0.0
Iteration 884: Arm = 2, Reward = 0.0
Iteration 885: Arm = 2, Reward = 1.0
Iteration 886: Arm = 2, Reward = 0.0
Iteration 887: Arm = 2, Reward = 1.0
Iteration 888: Arm = 0, Reward = 0.0
Iteration 889: Arm = 1, Reward = 0.0
Iteration 890: Arm = 2, Reward = 0.0
Iteration 891: Arm = 2, Reward = 1.0
Iteration 892: Arm = 2, Reward = 0.0
Iteration 893: Arm = 3, Reward = 0.0
Iteration 894: Arm = 1, Reward = 1.0
Iteration 895: Arm = 2, Reward = 1.0
Iteration 896: Arm = 2, Reward = 1.0
```

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Iteration 897: Arm = 2, Reward = 0.0
Iteration 898: Arm = 2, Reward = 1.0
Iteration 899: Arm = 2, Reward = 0.0
Iteration 900: Arm = 2, Reward = 0.0
Iteration 901: Arm = 2, Reward = 0.0
Iteration 902: Arm = 2, Reward = 0.0
Iteration 903: Arm = 2, Reward = 0.0
Iteration 904: Arm = 1, Reward = 0.0
Iteration 905: Arm = 2, Reward = 0.0
Iteration 906: Arm = 1, Reward = 1.0
Iteration 907: Arm = 1, Reward = 1.0
Iteration 908: Arm = 1, Reward = 0.0
Iteration 909: Arm = 1, Reward = 1.0
Iteration 910: Arm = 1, Reward = 1.0
Iteration 911: Arm = 1, Reward = 1.0
Iteration 912: Arm = 1, Reward = 1.0
Iteration 913: Arm = 1, Reward = 0.0
Iteration 914: Arm = 1, Reward = 1.0
Iteration 915: Arm = 1, Reward = 0.0
Iteration 916: Arm = 0, Reward = 0.0
Iteration 917: Arm = 1, Reward = 1.0
Iteration 918: Arm = 1, Reward = 1.0
Iteration 919: Arm = 1, Reward = 1.0
Iteration 920: Arm = 1, Reward = 1.0
Iteration 921: Arm = 1, Reward = 1.0
Iteration 922: Arm = 1, Reward = 0.0
Iteration 923: Arm = 1, Reward = 1.0
Iteration 924: Arm = 1, Reward = 0.0
Iteration 925: Arm = 1, Reward = 1.0
Iteration 926: Arm = 1, Reward = 0.0
Iteration 927: Arm = 0, Reward = 1.0
Iteration 928: Arm = 3, Reward = 1.0
Iteration 929: Arm = 1, Reward = 0.0
Iteration 930: Arm = 1, Reward = 0.0
Iteration 931: Arm = 1, Reward = 1.0
Iteration 932: Arm = 1, Reward = 0.0
Iteration 933: Arm = 0, Reward = 1.0
Iteration 934: Arm = 0, Reward = 1.0
Iteration 935: Arm = 1, Reward = 1.0
Iteration 936: Arm = 1, Reward = 1.0
Iteration 937: Arm = 1, Reward = 1.0
Iteration 938: Arm = 1, Reward = 1.0
Iteration 939: Arm = 1, Reward = 1.0
Iteration 940: Arm = 1, Reward = 1.0
Iteration 941: Arm = 1, Reward = 1.0
Iteration 942: Arm = 1, Reward = 1.0
Iteration 943: Arm = 1, Reward = 1.0
Iteration 944: Arm = 1, Reward = 1.0
Iteration 945: Arm = 1, Reward = 1.0
Iteration 946: Arm = 1, Reward = 0.0
Iteration 947: Arm = 1, Reward = 0.0
Iteration 948: Arm = 1, Reward = 0.0
Iteration 949: Arm = 1, Reward = 0.0
Iteration 950: Arm = 1, Reward = 1.0
Iteration 951: Arm = 1, Reward = 0.0
Iteration 952: Arm = 1, Reward = 0.0
Iteration 953: Arm = 1, Reward = 0.0
Iteration 954: Arm = 1, Reward = 1.0
Iteration 955: Arm = 1, Reward = 0.0
Iteration 956: Arm = 1, Reward = 1.0
Iteration 957: Arm = 1, Reward = 1.0
Iteration 958: Arm = 1, Reward = 0.0
Iteration 959: Arm = 1, Reward = 1.0
Iteration 960: Arm = 0, Reward = 0.0
Iteration 961: Arm = 1, Reward = 0.0
Iteration 962: Arm = 1, Reward = 1.0
Iteration 963: Arm = 1, Reward = 1.0
Iteration 964: Arm = 1, Reward = 1.0
Iteration 965: Arm = 2, Reward = 0.0
Iteration 966: Arm = 1, Reward = 1.0
Iteration 967: Arm = 1, Reward = 0.0
Iteration 968: Arm = 1, Reward = 1.0
Iteration 969: Arm = 1, Reward = 1.0
Iteration 970: Arm = 1, Reward = 0.0
Iteration 971: Arm = 1, Reward = 1.0
Iteration 972: Arm = 1, Reward = 0.0
Iteration 973: Arm = 2, Reward = 1.0
Iteration 974: Arm = 1, Reward = 1.0
Iteration 975: Arm = 1, Reward = 0.0
Iteration 976: Arm = 1, Reward = 1.0
Iteration 977: Arm = 1, Reward = 0.0
Iteration 978: Arm = 1, Reward = 1.0
Iteration 979: Arm = 1, Reward = 1.0
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Iteration 980: Arm = 1, Reward = 0.0
Iteration 981: Arm = 2, Reward = 0.0
Iteration 982: Arm = 1, Reward = 0.0
Iteration 983: Arm = 1, Reward = 0.0
Iteration 984: Arm = 3, Reward = 1.0
Iteration 985: Arm = 3, Reward = 1.0
Iteration 986: Arm = 2, Reward = 1.0
Iteration 987: Arm = 1, Reward = 0.0
Iteration 988: Arm = 1, Reward = 1.0
Iteration 989: Arm = 1, Reward = 1.0
Iteration 990: Arm = 1, Reward = 1.0
Iteration 991: Arm = 1, Reward = 1.0
Iteration 992: Arm = 1, Reward = 1.0
Iteration 993: Arm = 1, Reward = 1.0
Iteration 994: Arm = 1, Reward = 0.0
Iteration 995: Arm = 1, Reward = 0.0
Iteration 996: Arm = 3, Reward = 1.0
Iteration 997: Arm = 1, Reward = 1.0
Iteration 998: Arm = 1, Reward = 1.0
Iteration 999: Arm = 3, Reward = 1.0
Iteration 1000: Arm = 1, Reward = 1.0

--- Epsilon-Greedy Policy (ε = 0.5) ---
Iteration 1: Arm = 0, Reward = 0.0
Iteration 2: Arm = 0, Reward = 0.0
Iteration 3: Arm = 0, Reward = 1.0
Iteration 4: Arm = 0, Reward = 0.0
Iteration 5: Arm = 0, Reward = 1.0
Iteration 6: Arm = 3, Reward = 0.0
Iteration 7: Arm = 0, Reward = 0.0
Iteration 8: Arm = 0, Reward = 1.0
Iteration 9: Arm = 0, Reward = 0.0
Iteration 10: Arm = 2, Reward = 1.0
Iteration 11: Arm = 2, Reward = 0.0
Iteration 12: Arm = 3, Reward = 1.0
Iteration 13: Arm = 2, Reward = 1.0
Iteration 14: Arm = 1, Reward = 0.0
Iteration 15: Arm = 2, Reward = 0.0
Iteration 16: Arm = 2, Reward = 0.0
Iteration 17: Arm = 3, Reward = 0.0
Iteration 18: Arm = 2, Reward = 1.0
Iteration 19: Arm = 0, Reward = 0.0
Iteration 20: Arm = 2, Reward = 0.0
Iteration 21: Arm = 2, Reward = 0.0
Iteration 22: Arm = 2, Reward = 1.0
Iteration 23: Arm = 0, Reward = 0.0
Iteration 24: Arm = 0, Reward = 1.0
Iteration 25: Arm = 2, Reward = 0.0
Iteration 26: Arm = 2, Reward = 1.0
Iteration 27: Arm = 0, Reward = 1.0
Iteration 28: Arm = 2, Reward = 0.0
Iteration 29: Arm = 0, Reward = 0.0
Iteration 30: Arm = 1, Reward = 0.0
Iteration 31: Arm = 2, Reward = 1.0
Iteration 32: Arm = 0, Reward = 0.0
Iteration 33: Arm = 2, Reward = 0.0
Iteration 34: Arm = 2, Reward = 1.0
Iteration 35: Arm = 2, Reward = 1.0
Iteration 36: Arm = 3, Reward = 0.0
Iteration 37: Arm = 1, Reward = 1.0
Iteration 38: Arm = 3, Reward = 0.0
Iteration 39: Arm = 2, Reward = 1.0
Iteration 40: Arm = 2, Reward = 1.0
Iteration 41: Arm = 2, Reward = 0.0
Iteration 42: Arm = 2, Reward = 0.0
Iteration 43: Arm = 2, Reward = 0.0
Iteration 44: Arm = 1, Reward = 0.0
Iteration 45: Arm = 2, Reward = 1.0
Iteration 46: Arm = 1, Reward = 0.0
Iteration 47: Arm = 3, Reward = 1.0
Iteration 48: Arm = 2, Reward = 1.0
Iteration 49: Arm = 2, Reward = 0.0
Iteration 50: Arm = 2, Reward = 0.0
Iteration 51: Arm = 2, Reward = 0.0
Iteration 52: Arm = 3, Reward = 0.0
Iteration 53: Arm = 2, Reward = 1.0
Iteration 54: Arm = 2, Reward = 0.0
Iteration 55: Arm = 2, Reward = 0.0
Iteration 56: Arm = 2, Reward = 1.0
Iteration 57: Arm = 3, Reward = 0.0
Iteration 58: Arm = 2, Reward = 0.0
Iteration 59: Arm = 2, Reward = 0.0
Iteration 60: Arm = 2, Reward = 1.0
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Iteration 61: Arm = 2, Reward = 1.0
Iteration 62: Arm = 2, Reward = 1.0
Iteration 63: Arm = 2, Reward = 1.0
Iteration 64: Arm = 3, Reward = 0.0
Iteration 65: Arm = 2, Reward = 1.0
Iteration 66: Arm = 2, Reward = 1.0
Iteration 67: Arm = 2, Reward = 1.0
Iteration 68: Arm = 2, Reward = 0.0
Iteration 69: Arm = 2, Reward = 1.0
Iteration 70: Arm = 3, Reward = 1.0
Iteration 71: Arm = 2, Reward = 1.0
Iteration 72: Arm = 0, Reward = 1.0
Iteration 73: Arm = 2, Reward = 0.0
Iteration 74: Arm = 2, Reward = 0.0
Iteration 75: Arm = 2, Reward = 1.0
Iteration 76: Arm = 2, Reward = 0.0
Iteration 77: Arm = 2, Reward = 0.0
Iteration 78: Arm = 2, Reward = 0.0
Iteration 79: Arm = 0, Reward = 1.0
Iteration 80: Arm = 2, Reward = 0.0
Iteration 81: Arm = 1, Reward = 1.0
Iteration 82: Arm = 2, Reward = 0.0
Iteration 83: Arm = 2, Reward = 1.0
Iteration 84: Arm = 1, Reward = 0.0
Iteration 85: Arm = 1, Reward = 1.0
Iteration 86: Arm = 3, Reward = 1.0
Iteration 87: Arm = 2, Reward = 0.0
Iteration 88: Arm = 2, Reward = 1.0
Iteration 89: Arm = 2, Reward = 0.0
Iteration 90: Arm = 2, Reward = 0.0
Iteration 91: Arm = 0, Reward = 0.0
Iteration 92: Arm = 1, Reward = 0.0
Iteration 93: Arm = 3, Reward = 0.0
Iteration 94: Arm = 2, Reward = 0.0
Iteration 95: Arm = 2, Reward = 1.0
Iteration 96: Arm = 2, Reward = 0.0
Iteration 97: Arm = 0, Reward = 1.0
Iteration 98: Arm = 2, Reward = 1.0
Iteration 99: Arm = 2, Reward = 0.0
Iteration 100: Arm = 0, Reward = 1.0
Iteration 101: Arm = 0, Reward = 0.0
Iteration 102: Arm = 2, Reward = 0.0
Iteration 103: Arm = 2, Reward = 0.0
Iteration 104: Arm = 2, Reward = 0.0
Iteration 105: Arm = 1, Reward = 1.0
Iteration 106: Arm = 1, Reward = 0.0
Iteration 107: Arm = 0, Reward = 1.0
Iteration 108: Arm = 0, Reward = 0.0
Iteration 109: Arm = 0, Reward = 0.0
Iteration 110: Arm = 2, Reward = 1.0
Iteration 111: Arm = 2, Reward = 1.0
Iteration 112: Arm = 2, Reward = 1.0
Iteration 113: Arm = 3, Reward = 1.0
Iteration 114: Arm = 2, Reward = 0.0
Iteration 115: Arm = 1, Reward = 1.0
Iteration 116: Arm = 3, Reward = 1.0
Iteration 117: Arm = 3, Reward = 1.0
Iteration 118: Arm = 1, Reward = 0.0
Iteration 119: Arm = 3, Reward = 0.0
Iteration 120: Arm = 2, Reward = 1.0
Iteration 121: Arm = 2, Reward = 1.0
Iteration 122: Arm = 2, Reward = 1.0
Iteration 123: Arm = 2, Reward = 1.0
Iteration 124: Arm = 2, Reward = 0.0
Iteration 125: Arm = 0, Reward = 0.0
Iteration 126: Arm = 2, Reward = 1.0
Iteration 127: Arm = 2, Reward = 1.0
Iteration 128: Arm = 0, Reward = 0.0
Iteration 129: Arm = 1, Reward = 1.0
Iteration 130: Arm = 2, Reward = 0.0
Iteration 131: Arm = 2, Reward = 0.0
Iteration 132: Arm = 2, Reward = 0.0
Iteration 133: Arm = 0, Reward = 0.0
Iteration 134: Arm = 2, Reward = 0.0
Iteration 135: Arm = 2, Reward = 1.0
Iteration 136: Arm = 2, Reward = 0.0
Iteration 137: Arm = 3, Reward = 1.0
Iteration 138: Arm = 2, Reward = 0.0
Iteration 139: Arm = 3, Reward = 1.0
Iteration 140: Arm = 2, Reward = 1.0
Iteration 141: Arm = 2, Reward = 1.0
Iteration 142: Arm = 3, Reward = 1.0
Iteration 143: Arm = 0, Reward = 0.0
```

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Iteration 144: Arm = 0, Reward = 0.0
Iteration 145: Arm = 3, Reward = 0.0
Iteration 146: Arm = 2, Reward = 0.0
Iteration 147: Arm = 2, Reward = 1.0
Iteration 148: Arm = 3, Reward = 1.0
Iteration 149: Arm = 3, Reward = 1.0
Iteration 150: Arm = 0, Reward = 0.0
Iteration 151: Arm = 3, Reward = 1.0
Iteration 152: Arm = 1, Reward = 1.0
Iteration 153: Arm = 3, Reward = 0.0
Iteration 154: Arm = 2, Reward = 0.0
Iteration 155: Arm = 1, Reward = 0.0
Iteration 156: Arm = 3, Reward = 1.0
Iteration 157: Arm = 3, Reward = 0.0
Iteration 158: Arm = 0, Reward = 0.0
Iteration 159: Arm = 3, Reward = 0.0
Iteration 160: Arm = 3, Reward = 0.0
Iteration 161: Arm = 3, Reward = 1.0
Iteration 162: Arm = 3, Reward = 0.0
Iteration 163: Arm = 2, Reward = 1.0
Iteration 164: Arm = 3, Reward = 0.0
Iteration 165: Arm = 3, Reward = 0.0
Iteration 166: Arm = 2, Reward = 1.0
Iteration 167: Arm = 2, Reward = 1.0
Iteration 168: Arm = 2, Reward = 1.0
Iteration 169: Arm = 0, Reward = 0.0
Iteration 170: Arm = 2, Reward = 1.0
Iteration 171: Arm = 3, Reward = 1.0
Iteration 172: Arm = 2, Reward = 1.0
Iteration 173: Arm = 2, Reward = 0.0
Iteration 174: Arm = 3, Reward = 1.0
Iteration 175: Arm = 2, Reward = 1.0
Iteration 176: Arm = 2, Reward = 0.0
Iteration 177: Arm = 1, Reward = 0.0
Iteration 178: Arm = 2, Reward = 1.0
Iteration 179: Arm = 3, Reward = 1.0
Iteration 180: Arm = 0, Reward = 0.0
Iteration 181: Arm = 3, Reward = 1.0
Iteration 182: Arm = 2, Reward = 1.0
Iteration 183: Arm = 3, Reward = 0.0
Iteration 184: Arm = 2, Reward = 1.0
Iteration 185: Arm = 2, Reward = 1.0
Iteration 186: Arm = 2, Reward = 1.0
Iteration 187: Arm = 2, Reward = 0.0
Iteration 188: Arm = 3, Reward = 0.0
Iteration 189: Arm = 2, Reward = 0.0
Iteration 190: Arm = 2, Reward = 0.0
Iteration 191: Arm = 2, Reward = 0.0
Iteration 192: Arm = 2, Reward = 0.0
Iteration 193: Arm = 2, Reward = 1.0
Iteration 194: Arm = 3, Reward = 0.0
Iteration 195: Arm = 3, Reward = 1.0
Iteration 196: Arm = 1, Reward = 0.0
Iteration 197: Arm = 2, Reward = 1.0
Iteration 198: Arm = 2, Reward = 0.0
Iteration 199: Arm = 2, Reward = 0.0
Iteration 200: Arm = 2, Reward = 1.0
Iteration 201: Arm = 2, Reward = 0.0
Iteration 202: Arm = 2, Reward = 1.0
Iteration 203: Arm = 1, Reward = 0.0
Iteration 204: Arm = 2, Reward = 0.0
Iteration 205: Arm = 2, Reward = 0.0
Iteration 206: Arm = 3, Reward = 1.0
Iteration 207: Arm = 2, Reward = 1.0
Iteration 208: Arm = 2, Reward = 0.0
Iteration 209: Arm = 1, Reward = 1.0
Iteration 210: Arm = 3, Reward = 1.0
Iteration 211: Arm = 2, Reward = 0.0
Iteration 212: Arm = 3, Reward = 1.0
Iteration 213: Arm = 3, Reward = 0.0
Iteration 214: Arm = 3, Reward = 1.0
Iteration 215: Arm = 3, Reward = 1.0
Iteration 216: Arm = 3, Reward = 0.0
Iteration 217: Arm = 3, Reward = 0.0
Iteration 218: Arm = 2, Reward = 1.0
Iteration 219: Arm = 3, Reward = 0.0
Iteration 220: Arm = 2, Reward = 0.0
Iteration 221: Arm = 3, Reward = 0.0
Iteration 222: Arm = 3, Reward = 0.0
Iteration 223: Arm = 2, Reward = 1.0
Iteration 224: Arm = 1, Reward = 1.0
Iteration 225: Arm = 2, Reward = 1.0
Iteration 226: Arm = 2, Reward = 1.0
```

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Iteration 227: Arm = 0, Reward = 0.0
Iteration 228: Arm = 1, Reward = 1.0
Iteration 229: Arm = 3, Reward = 0.0
Iteration 230: Arm = 2, Reward = 1.0
Iteration 231: Arm = 2, Reward = 1.0
Iteration 232: Arm = 2, Reward = 0.0
Iteration 233: Arm = 3, Reward = 1.0
Iteration 234: Arm = 2, Reward = 1.0
Iteration 235: Arm = 2, Reward = 0.0
Iteration 236: Arm = 2, Reward = 1.0
Iteration 237: Arm = 0, Reward = 0.0
Iteration 238: Arm = 2, Reward = 1.0
Iteration 239: Arm = 0, Reward = 1.0
Iteration 240: Arm = 0, Reward = 1.0
Iteration 241: Arm = 0, Reward = 0.0
Iteration 242: Arm = 1, Reward = 1.0
Iteration 243: Arm = 2, Reward = 0.0
Iteration 244: Arm = 2, Reward = 1.0
Iteration 245: Arm = 3, Reward = 0.0
Iteration 246: Arm = 2, Reward = 1.0
Iteration 247: Arm = 0, Reward = 0.0
Iteration 248: Arm = 1, Reward = 0.0
Iteration 249: Arm = 2, Reward = 1.0
Iteration 250: Arm = 0, Reward = 0.0
Iteration 251: Arm = 2, Reward = 1.0
Iteration 252: Arm = 3, Reward = 0.0
Iteration 253: Arm = 2, Reward = 1.0
Iteration 254: Arm = 2, Reward = 0.0
Iteration 255: Arm = 2, Reward = 0.0
Iteration 256: Arm = 1, Reward = 1.0
Iteration 257: Arm = 2, Reward = 1.0
Iteration 258: Arm = 2, Reward = 0.0
Iteration 259: Arm = 2, Reward = 1.0
Iteration 260: Arm = 1, Reward = 1.0
Iteration 261: Arm = 0, Reward = 0.0
Iteration 262: Arm = 3, Reward = 0.0
Iteration 263: Arm = 2, Reward = 1.0
Iteration 264: Arm = 2, Reward = 0.0
Iteration 265: Arm = 2, Reward = 1.0
Iteration 266: Arm = 2, Reward = 1.0
Iteration 267: Arm = 2, Reward = 1.0
Iteration 268: Arm = 1, Reward = 1.0
Iteration 269: Arm = 3, Reward = 0.0
Iteration 270: Arm = 0, Reward = 0.0
Iteration 271: Arm = 0, Reward = 1.0
Iteration 272: Arm = 2, Reward = 1.0
Iteration 273: Arm = 1, Reward = 0.0
Iteration 274: Arm = 3, Reward = 1.0
Iteration 275: Arm = 0, Reward = 0.0
Iteration 276: Arm = 2, Reward = 1.0
Iteration 277: Arm = 1, Reward = 0.0
Iteration 278: Arm = 2, Reward = 1.0
Iteration 279: Arm = 1, Reward = 1.0
Iteration 280: Arm = 2, Reward = 0.0
Iteration 281: Arm = 2, Reward = 1.0
Iteration 282: Arm = 3, Reward = 1.0
Iteration 283: Arm = 2, Reward = 0.0
Iteration 284: Arm = 1, Reward = 0.0
Iteration 285: Arm = 2, Reward = 1.0
Iteration 286: Arm = 2, Reward = 1.0
Iteration 287: Arm = 2, Reward = 1.0
Iteration 288: Arm = 2, Reward = 1.0
Iteration 289: Arm = 2, Reward = 0.0
Iteration 290: Arm = 3, Reward = 1.0
Iteration 291: Arm = 2, Reward = 1.0
Iteration 292: Arm = 3, Reward = 1.0
Iteration 293: Arm = 2, Reward = 1.0
Iteration 294: Arm = 2, Reward = 1.0
Iteration 295: Arm = 1, Reward = 0.0
Iteration 296: Arm = 3, Reward = 0.0
Iteration 297: Arm = 0, Reward = 0.0
Iteration 298: Arm = 1, Reward = 1.0
Iteration 299: Arm = 2, Reward = 0.0
Iteration 300: Arm = 0, Reward = 0.0
Iteration 301: Arm = 2, Reward = 0.0
Iteration 302: Arm = 2, Reward = 1.0
Iteration 303: Arm = 1, Reward = 0.0
Iteration 304: Arm = 1, Reward = 0.0
Iteration 305: Arm = 2, Reward = 0.0
Iteration 306: Arm = 2, Reward = 0.0
Iteration 307: Arm = 2, Reward = 0.0
Iteration 308: Arm = 3, Reward = 0.0
Iteration 309: Arm = 3, Reward = 1.0
```

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Iteration 310: Arm = 1, Reward = 0.0
Iteration 311: Arm = 1, Reward = 0.0
Iteration 312: Arm = 0, Reward = 0.0
Iteration 313: Arm = 2, Reward = 0.0
Iteration 314: Arm = 2, Reward = 0.0
Iteration 315: Arm = 2, Reward = 0.0
Iteration 316: Arm = 2, Reward = 1.0
Iteration 317: Arm = 0, Reward = 0.0
Iteration 318: Arm = 2, Reward = 1.0
Iteration 319: Arm = 2, Reward = 0.0
Iteration 320: Arm = 2, Reward = 1.0
Iteration 321: Arm = 2, Reward = 1.0
Iteration 322: Arm = 1, Reward = 0.0
Iteration 323: Arm = 2, Reward = 0.0
Iteration 324: Arm = 2, Reward = 0.0
Iteration 325: Arm = 2, Reward = 1.0
Iteration 326: Arm = 2, Reward = 1.0
Iteration 327: Arm = 2, Reward = 0.0
Iteration 328: Arm = 2, Reward = 0.0
Iteration 329: Arm = 2, Reward = 1.0
Iteration 330: Arm = 2, Reward = 0.0
Iteration 331: Arm = 3, Reward = 1.0
Iteration 332: Arm = 2, Reward = 1.0
Iteration 333: Arm = 2, Reward = 0.0
Iteration 334: Arm = 2, Reward = 1.0
Iteration 335: Arm = 0, Reward = 0.0
Iteration 336: Arm = 2, Reward = 1.0
Iteration 337: Arm = 3, Reward = 0.0
Iteration 338: Arm = 2, Reward = 0.0
Iteration 339: Arm = 2, Reward = 0.0
Iteration 340: Arm = 0, Reward = 0.0
Iteration 341: Arm = 2, Reward = 1.0
Iteration 342: Arm = 2, Reward = 0.0
Iteration 343: Arm = 0, Reward = 1.0
Iteration 344: Arm = 0, Reward = 1.0
Iteration 345: Arm = 2, Reward = 0.0
Iteration 346: Arm = 2, Reward = 0.0
Iteration 347: Arm = 2, Reward = 1.0
Iteration 348: Arm = 2, Reward = 1.0
Iteration 349: Arm = 2, Reward = 0.0
Iteration 350: Arm = 3, Reward = 0.0
Iteration 351: Arm = 1, Reward = 0.0
Iteration 352: Arm = 1, Reward = 1.0
Iteration 353: Arm = 3, Reward = 0.0
Iteration 354: Arm = 2, Reward = 1.0
Iteration 355: Arm = 2, Reward = 1.0
Iteration 356: Arm = 2, Reward = 1.0
Iteration 357: Arm = 2, Reward = 1.0
Iteration 358: Arm = 2, Reward = 0.0
Iteration 359: Arm = 2, Reward = 0.0
Iteration 360: Arm = 0, Reward = 0.0
Iteration 361: Arm = 2, Reward = 1.0
Iteration 362: Arm = 2, Reward = 1.0
Iteration 363: Arm = 2, Reward = 0.0
Iteration 364: Arm = 0, Reward = 0.0
Iteration 365: Arm = 1, Reward = 1.0
Iteration 366: Arm = 2, Reward = 1.0
Iteration 367: Arm = 0, Reward = 0.0
Iteration 368: Arm = 3, Reward = 0.0
Iteration 369: Arm = 2, Reward = 1.0
Iteration 370: Arm = 2, Reward = 0.0
Iteration 371: Arm = 2, Reward = 1.0
Iteration 372: Arm = 2, Reward = 0.0
Iteration 373: Arm = 1, Reward = 1.0
Iteration 374: Arm = 2, Reward = 1.0
Iteration 375: Arm = 2, Reward = 1.0
Iteration 376: Arm = 2, Reward = 1.0
Iteration 377: Arm = 0, Reward = 0.0
Iteration 378: Arm = 2, Reward = 0.0
Iteration 379: Arm = 2, Reward = 1.0
Iteration 380: Arm = 2, Reward = 0.0
Iteration 381: Arm = 2, Reward = 1.0
Iteration 382: Arm = 0, Reward = 0.0
Iteration 383: Arm = 1, Reward = 1.0
Iteration 384: Arm = 3, Reward = 1.0
Iteration 385: Arm = 2, Reward = 1.0
Iteration 386: Arm = 0, Reward = 0.0
Iteration 387: Arm = 2, Reward = 0.0
Iteration 388: Arm = 0, Reward = 1.0
Iteration 389: Arm = 2, Reward = 1.0
Iteration 390: Arm = 2, Reward = 0.0
Iteration 391: Arm = 2, Reward = 0.0
Iteration 392: Arm = 2, Reward = 1.0
```

```
Iteration 393: Arm = 2, Reward = 0.0
Iteration 394: Arm = 3, Reward = 1.0
Iteration 395: Arm = 2, Reward = 0.0
Iteration 396: Arm = 2, Reward = 1.0
Iteration 397: Arm = 3, Reward = 0.0
Iteration 398: Arm = 2, Reward = 0.0
Iteration 399: Arm = 0, Reward = 0.0
Iteration 400: Arm = 1, Reward = 1.0
Iteration 401: Arm = 0, Reward = 0.0
Iteration 402: Arm = 2, Reward = 0.0
Iteration 403: Arm = 2, Reward = 1.0
Iteration 404: Arm = 3, Reward = 1.0
Iteration 405: Arm = 2, Reward = 1.0
Iteration 406: Arm = 1, Reward = 1.0
Iteration 407: Arm = 2, Reward = 1.0
Iteration 408: Arm = 2, Reward = 1.0
Iteration 409: Arm = 1, Reward = 1.0
Iteration 410: Arm = 2, Reward = 1.0
Iteration 411: Arm = 2, Reward = 1.0
Iteration 412: Arm = 2, Reward = 0.0
Iteration 413: Arm = 3, Reward = 0.0
Iteration 414: Arm = 1, Reward = 1.0
Iteration 415: Arm = 1, Reward = 0.0
Iteration 416: Arm = 2, Reward = 1.0
Iteration 417: Arm = 1, Reward = 1.0
Iteration 418: Arm = 3, Reward = 0.0
Iteration 419: Arm = 2, Reward = 0.0
Iteration 420: Arm = 2, Reward = 0.0
Iteration 421: Arm = 0, Reward = 0.0
Iteration 422: Arm = 2, Reward = 0.0
Iteration 423: Arm = 2, Reward = 0.0
Iteration 424: Arm = 2, Reward = 0.0
Iteration 425: Arm = 0, Reward = 0.0
Iteration 426: Arm = 2, Reward = 1.0
Iteration 427: Arm = 2, Reward = 1.0
Iteration 428: Arm = 3, Reward = 0.0
Iteration 429: Arm = 2, Reward = 0.0
Iteration 430: Arm = 2, Reward = 1.0
Iteration 431: Arm = 2, Reward = 1.0
Iteration 432: Arm = 1, Reward = 0.0
Iteration 433: Arm = 2, Reward = 0.0
Iteration 434: Arm = 2, Reward = 0.0
Iteration 435: Arm = 1, Reward = 1.0
Iteration 436: Arm = 2, Reward = 1.0
Iteration 437: Arm = 2, Reward = 1.0
Iteration 438: Arm = 2, Reward = 1.0
Iteration 439: Arm = 2, Reward = 0.0
Iteration 440: Arm = 2, Reward = 0.0
Iteration 441: Arm = 3, Reward = 0.0
Iteration 442: Arm = 0, Reward = 0.0
Iteration 443: Arm = 2, Reward = 0.0
Iteration 444: Arm = 1, Reward = 1.0
Iteration 445: Arm = 0, Reward = 0.0
Iteration 446: Arm = 2, Reward = 1.0
Iteration 447: Arm = 2, Reward = 0.0
Iteration 448: Arm = 2, Reward = 1.0
Iteration 449: Arm = 2, Reward = 0.0
Iteration 450: Arm = 2, Reward = 0.0
Iteration 451: Arm = 2, Reward = 0.0
Iteration 452: Arm = 2, Reward = 1.0
Iteration 453: Arm = 2, Reward = 1.0
Iteration 454: Arm = 2, Reward = 0.0
Iteration 455: Arm = 2, Reward = 0.0
Iteration 456: Arm = 2, Reward = 1.0
Iteration 457: Arm = 2, Reward = 0.0
Iteration 458: Arm = 2, Reward = 0.0
Iteration 459: Arm = 1, Reward = 0.0
Iteration 460: Arm = 2, Reward = 1.0
Iteration 461: Arm = 3, Reward = 0.0
Iteration 462: Arm = 2, Reward = 0.0
Iteration 463: Arm = 2, Reward = 0.0
Iteration 464: Arm = 2, Reward = 1.0
Iteration 465: Arm = 0, Reward = 0.0
Iteration 466: Arm = 1, Reward = 1.0
Iteration 467: Arm = 2, Reward = 0.0
Iteration 468: Arm = 2, Reward = 0.0
Iteration 469: Arm = 3, Reward = 1.0
Iteration 470: Arm = 2, Reward = 0.0
Iteration 471: Arm = 2, Reward = 1.0
Iteration 472: Arm = 0, Reward = 1.0
Iteration 473: Arm = 1, Reward = 0.0
Iteration 474: Arm = 2, Reward = 1.0
Iteration 475: Arm = 3, Reward = 1.0
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Iteration 476: Arm = 2, Reward = 1.0
Iteration 477: Arm = 3, Reward = 0.0
Iteration 478: Arm = 1, Reward = 1.0
Iteration 479: Arm = 2, Reward = 1.0
Iteration 480: Arm = 2, Reward = 1.0
Iteration 481: Arm = 1, Reward = 1.0
Iteration 482: Arm = 0, Reward = 1.0
Iteration 483: Arm = 2, Reward = 0.0
Iteration 484: Arm = 3, Reward = 0.0
Iteration 485: Arm = 1, Reward = 0.0
Iteration 486: Arm = 2, Reward = 0.0
Iteration 487: Arm = 3, Reward = 0.0
Iteration 488: Arm = 0, Reward = 0.0
Iteration 489: Arm = 2, Reward = 1.0
Iteration 490: Arm = 2, Reward = 1.0
Iteration 491: Arm = 2, Reward = 1.0
Iteration 492: Arm = 2, Reward = 1.0
Iteration 493: Arm = 1, Reward = 0.0
Iteration 494: Arm = 3, Reward = 0.0
Iteration 495: Arm = 2, Reward = 0.0
Iteration 496: Arm = 2, Reward = 0.0
Iteration 497: Arm = 3, Reward = 1.0
Iteration 498: Arm = 2, Reward = 0.0
Iteration 499: Arm = 0, Reward = 1.0
Iteration 500: Arm = 1, Reward = 1.0
Iteration 501: Arm = 2, Reward = 0.0
Iteration 502: Arm = 1, Reward = 1.0
Iteration 503: Arm = 2, Reward = 0.0
Iteration 504: Arm = 1, Reward = 1.0
Iteration 505: Arm = 3, Reward = 0.0
Iteration 506: Arm = 1, Reward = 0.0
Iteration 507: Arm = 1, Reward = 1.0
Iteration 508: Arm = 1, Reward = 1.0
Iteration 509: Arm = 1, Reward = 0.0
Iteration 510: Arm = 0, Reward = 0.0
Iteration 511: Arm = 1, Reward = 1.0
Iteration 512: Arm = 1, Reward = 1.0
Iteration 513: Arm = 0, Reward = 0.0
Iteration 514: Arm = 0, Reward = 1.0
Iteration 515: Arm = 0, Reward = 0.0
Iteration 516: Arm = 2, Reward = 0.0
Iteration 517: Arm = 1, Reward = 1.0
Iteration 518: Arm = 2, Reward = 0.0
Iteration 519: Arm = 1, Reward = 0.0
Iteration 520: Arm = 1, Reward = 1.0
Iteration 521: Arm = 1, Reward = 0.0
Iteration 522: Arm = 1, Reward = 0.0
Iteration 523: Arm = 2, Reward = 0.0
Iteration 524: Arm = 1, Reward = 0.0
Iteration 525: Arm = 3, Reward = 0.0
Iteration 526: Arm = 1, Reward = 1.0
Iteration 527: Arm = 3, Reward = 0.0
Iteration 528: Arm = 3, Reward = 1.0
Iteration 529: Arm = 1, Reward = 0.0
Iteration 530: Arm = 0, Reward = 0.0
Iteration 531: Arm = 0, Reward = 1.0
Iteration 532: Arm = 1, Reward = 0.0
Iteration 533: Arm = 0, Reward = 0.0
Iteration 534: Arm = 1, Reward = 1.0
Iteration 535: Arm = 2, Reward = 0.0
Iteration 536: Arm = 0, Reward = 0.0
Iteration 537: Arm = 1, Reward = 0.0
Iteration 538: Arm = 3, Reward = 0.0
Iteration 539: Arm = 0, Reward = 0.0
Iteration 540: Arm = 1, Reward = 0.0
Iteration 541: Arm = 1, Reward = 1.0
Iteration 542: Arm = 2, Reward = 0.0
Iteration 543: Arm = 2, Reward = 0.0
Iteration 544: Arm = 1, Reward = 0.0
Iteration 545: Arm = 0, Reward = 0.0
Iteration 546: Arm = 2, Reward = 1.0
Iteration 547: Arm = 2, Reward = 1.0
Iteration 548: Arm = 1, Reward = 0.0
Iteration 549: Arm = 1, Reward = 0.0
Iteration 550: Arm = 3, Reward = 0.0
Iteration 551: Arm = 2, Reward = 1.0
Iteration 552: Arm = 2, Reward = 1.0
Iteration 553: Arm = 1, Reward = 0.0
Iteration 554: Arm = 1, Reward = 0.0
Iteration 555: Arm = 3, Reward = 0.0
Iteration 556: Arm = 3, Reward = 0.0
Iteration 557: Arm = 3, Reward = 0.0
Iteration 558: Arm = 0, Reward = 0.0
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Iteration 559: Arm = 2, Reward = 0.0
Iteration 560: Arm = 1, Reward = 0.0
Iteration 561: Arm = 3, Reward = 1.0
Iteration 562: Arm = 2, Reward = 0.0
Iteration 563: Arm = 1, Reward = 0.0
Iteration 564: Arm = 2, Reward = 1.0
Iteration 565: Arm = 0, Reward = 0.0
Iteration 566: Arm = 0, Reward = 1.0
Iteration 567: Arm = 2, Reward = 0.0
Iteration 568: Arm = 3, Reward = 0.0
Iteration 569: Arm = 2, Reward = 1.0
Iteration 570: Arm = 2, Reward = 1.0
Iteration 571: Arm = 3, Reward = 0.0
Iteration 572: Arm = 0, Reward = 0.0
Iteration 573: Arm = 1, Reward = 0.0
Iteration 574: Arm = 2, Reward = 0.0
Iteration 575: Arm = 2, Reward = 0.0
Iteration 576: Arm = 2, Reward = 1.0
Iteration 577: Arm = 0, Reward = 0.0
Iteration 578: Arm = 2, Reward = 1.0
Iteration 579: Arm = 2, Reward = 0.0
Iteration 580: Arm = 3, Reward = 1.0
Iteration 581: Arm = 1, Reward = 0.0
Iteration 582: Arm = 2, Reward = 1.0
Iteration 583: Arm = 1, Reward = 0.0
Iteration 584: Arm = 2, Reward = 0.0
Iteration 585: Arm = 2, Reward = 0.0
Iteration 586: Arm = 0, Reward = 1.0
Iteration 587: Arm = 0, Reward = 1.0
Iteration 588: Arm = 2, Reward = 0.0
Iteration 589: Arm = 2, Reward = 0.0
Iteration 590: Arm = 3, Reward = 0.0
Iteration 591: Arm = 0, Reward = 0.0
Iteration 592: Arm = 2, Reward = 1.0
Iteration 593: Arm = 2, Reward = 0.0
Iteration 594: Arm = 0, Reward = 0.0
Iteration 595: Arm = 2, Reward = 1.0
Iteration 596: Arm = 1, Reward = 0.0
Iteration 597: Arm = 3, Reward = 0.0
Iteration 598: Arm = 2, Reward = 1.0
Iteration 599: Arm = 2, Reward = 1.0
Iteration 600: Arm = 2, Reward = 0.0
Iteration 601: Arm = 2, Reward = 1.0
Iteration 602: Arm = 3, Reward = 1.0
Iteration 603: Arm = 2, Reward = 0.0
Iteration 604: Arm = 2, Reward = 1.0
Iteration 605: Arm = 2, Reward = 1.0
Iteration 606: Arm = 3, Reward = 0.0
Iteration 607: Arm = 3, Reward = 0.0
Iteration 608: Arm = 2, Reward = 0.0
Iteration 609: Arm = 2, Reward = 0.0
Iteration 610: Arm = 1, Reward = 0.0
Iteration 611: Arm = 1, Reward = 0.0
Iteration 612: Arm = 2, Reward = 1.0
Iteration 613: Arm = 2, Reward = 1.0
Iteration 614: Arm = 0, Reward = 0.0
Iteration 615: Arm = 2, Reward = 1.0
Iteration 616: Arm = 3, Reward = 0.0
Iteration 617: Arm = 2, Reward = 0.0
Iteration 618: Arm = 2, Reward = 0.0
Iteration 619: Arm = 0, Reward = 0.0
Iteration 620: Arm = 3, Reward = 0.0
Iteration 621: Arm = 2, Reward = 0.0
Iteration 622: Arm = 2, Reward = 0.0
Iteration 623: Arm = 2, Reward = 1.0
Iteration 624: Arm = 2, Reward = 0.0
Iteration 625: Arm = 2, Reward = 0.0
Iteration 626: Arm = 1, Reward = 1.0
Iteration 627: Arm = 2, Reward = 1.0
Iteration 628: Arm = 2, Reward = 0.0
Iteration 629: Arm = 1, Reward = 0.0
Iteration 630: Arm = 1, Reward = 0.0
Iteration 631: Arm = 2, Reward = 0.0
Iteration 632: Arm = 2, Reward = 0.0
Iteration 633: Arm = 1, Reward = 1.0
Iteration 634: Arm = 3, Reward = 0.0
Iteration 635: Arm = 2, Reward = 0.0
Iteration 636: Arm = 1, Reward = 1.0
Iteration 637: Arm = 0, Reward = 0.0
Iteration 638: Arm = 2, Reward = 1.0
Iteration 639: Arm = 2, Reward = 1.0
Iteration 640: Arm = 1, Reward = 0.0
Iteration 641: Arm = 1, Reward = 0.0
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Iteration 642: Arm = 2, Reward = 0.0
Iteration 643: Arm = 2, Reward = 1.0
Iteration 644: Arm = 2, Reward = 0.0
Iteration 645: Arm = 2, Reward = 0.0
Iteration 646: Arm = 2, Reward = 1.0
Iteration 647: Arm = 0, Reward = 1.0
Iteration 648: Arm = 0, Reward = 0.0
Iteration 649: Arm = 1, Reward = 1.0
Iteration 650: Arm = 2, Reward = 0.0
Iteration 651: Arm = 2, Reward = 0.0
Iteration 652: Arm = 2, Reward = 0.0
Iteration 653: Arm = 0, Reward = 1.0
Iteration 654: Arm = 3, Reward = 0.0
Iteration 655: Arm = 1, Reward = 1.0
Iteration 656: Arm = 2, Reward = 0.0
Iteration 657: Arm = 2, Reward = 1.0
Iteration 658: Arm = 1, Reward = 0.0
Iteration 659: Arm = 2, Reward = 1.0
Iteration 660: Arm = 2, Reward = 0.0
Iteration 661: Arm = 2, Reward = 0.0
Iteration 662: Arm = 2, Reward = 0.0
Iteration 663: Arm = 2, Reward = 0.0
Iteration 664: Arm = 2, Reward = 0.0
Iteration 665: Arm = 2, Reward = 1.0
Iteration 666: Arm = 2, Reward = 0.0
Iteration 667: Arm = 1, Reward = 1.0
Iteration 668: Arm = 2, Reward = 0.0
Iteration 669: Arm = 0, Reward = 0.0
Iteration 670: Arm = 2, Reward = 1.0
Iteration 671: Arm = 0, Reward = 1.0
Iteration 672: Arm = 2, Reward = 0.0
Iteration 673: Arm = 0, Reward = 1.0
Iteration 674: Arm = 2, Reward = 0.0
Iteration 675: Arm = 2, Reward = 1.0
Iteration 676: Arm = 2, Reward = 0.0
Iteration 677: Arm = 2, Reward = 1.0
Iteration 678: Arm = 2, Reward = 1.0
Iteration 679: Arm = 2, Reward = 0.0
Iteration 680: Arm = 2, Reward = 1.0
Iteration 681: Arm = 2, Reward = 0.0
Iteration 682: Arm = 2, Reward = 1.0
Iteration 683: Arm = 0, Reward = 0.0
Iteration 684: Arm = 3, Reward = 1.0
Iteration 685: Arm = 2, Reward = 1.0
Iteration 686: Arm = 2, Reward = 0.0
Iteration 687: Arm = 2, Reward = 0.0
Iteration 688: Arm = 2, Reward = 0.0
Iteration 689: Arm = 3, Reward = 0.0
Iteration 690: Arm = 0, Reward = 0.0
Iteration 691: Arm = 0, Reward = 0.0
Iteration 692: Arm = 2, Reward = 0.0
Iteration 693: Arm = 2, Reward = 1.0
Iteration 694: Arm = 2, Reward = 0.0
Iteration 695: Arm = 2, Reward = 1.0
Iteration 696: Arm = 3, Reward = 1.0
Iteration 697: Arm = 2, Reward = 1.0
Iteration 698: Arm = 2, Reward = 0.0
Iteration 699: Arm = 1, Reward = 1.0
Iteration 700: Arm = 3, Reward = 1.0
Iteration 701: Arm = 2, Reward = 0.0
Iteration 702: Arm = 0, Reward = 1.0
Iteration 703: Arm = 2, Reward = 1.0
Iteration 704: Arm = 2, Reward = 0.0
Iteration 705: Arm = 2, Reward = 1.0
Iteration 706: Arm = 2, Reward = 0.0
Iteration 707: Arm = 1, Reward = 0.0
Iteration 708: Arm = 3, Reward = 0.0
Iteration 709: Arm = 2, Reward = 1.0
Iteration 710: Arm = 2, Reward = 0.0
Iteration 711: Arm = 2, Reward = 0.0
Iteration 712: Arm = 3, Reward = 0.0
Iteration 713: Arm = 2, Reward = 0.0
Iteration 714: Arm = 2, Reward = 1.0
Iteration 715: Arm = 3, Reward = 1.0
Iteration 716: Arm = 2, Reward = 1.0
Iteration 717: Arm = 3, Reward = 0.0
Iteration 718: Arm = 2, Reward = 0.0
Iteration 719: Arm = 0, Reward = 0.0
Iteration 720: Arm = 2, Reward = 0.0
Iteration 721: Arm = 3, Reward = 1.0
Iteration 722: Arm = 2, Reward = 0.0
Iteration 723: Arm = 2, Reward = 0.0
Iteration 724: Arm = 1, Reward = 0.0
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Iteration 725: Arm = 2, Reward = 0.0
Iteration 726: Arm = 1, Reward = 0.0
Iteration 727: Arm = 1, Reward = 1.0
Iteration 728: Arm = 2, Reward = 1.0
Iteration 729: Arm = 0, Reward = 0.0
Iteration 730: Arm = 2, Reward = 0.0
Iteration 731: Arm = 3, Reward = 0.0
Iteration 732: Arm = 1, Reward = 1.0
Iteration 733: Arm = 0, Reward = 0.0
Iteration 734: Arm = 0, Reward = 0.0
Iteration 735: Arm = 2, Reward = 0.0
Iteration 736: Arm = 2, Reward = 1.0
Iteration 737: Arm = 2, Reward = 0.0
Iteration 738: Arm = 0, Reward = 0.0
Iteration 739: Arm = 2, Reward = 0.0
Iteration 740: Arm = 2, Reward = 0.0
Iteration 741: Arm = 2, Reward = 0.0
Iteration 742: Arm = 2, Reward = 0.0
Iteration 743: Arm = 2, Reward = 1.0
Iteration 744: Arm = 2, Reward = 1.0
Iteration 745: Arm = 3, Reward = 0.0
Iteration 746: Arm = 2, Reward = 1.0
Iteration 747: Arm = 3, Reward = 1.0
Iteration 748: Arm = 2, Reward = 1.0
Iteration 749: Arm = 2, Reward = 0.0
Iteration 750: Arm = 2, Reward = 0.0
Iteration 751: Arm = 2, Reward = 0.0
Iteration 752: Arm = 1, Reward = 1.0
Iteration 753: Arm = 2, Reward = 1.0
Iteration 754: Arm = 2, Reward = 0.0
Iteration 755: Arm = 1, Reward = 1.0
Iteration 756: Arm = 2, Reward = 1.0
Iteration 757: Arm = 0, Reward = 0.0
Iteration 758: Arm = 2, Reward = 1.0
Iteration 759: Arm = 2, Reward = 1.0
Iteration 760: Arm = 2, Reward = 1.0
Iteration 761: Arm = 2, Reward = 1.0
Iteration 762: Arm = 3, Reward = 0.0
Iteration 763: Arm = 1, Reward = 0.0
Iteration 764: Arm = 2, Reward = 0.0
Iteration 765: Arm = 2, Reward = 1.0
Iteration 766: Arm = 3, Reward = 0.0
Iteration 767: Arm = 2, Reward = 0.0
Iteration 768: Arm = 2, Reward = 1.0
Iteration 769: Arm = 2, Reward = 0.0
Iteration 770: Arm = 3, Reward = 0.0
Iteration 771: Arm = 2, Reward = 1.0
Iteration 772: Arm = 2, Reward = 0.0
Iteration 773: Arm = 2, Reward = 1.0
Iteration 774: Arm = 2, Reward = 0.0
Iteration 775: Arm = 2, Reward = 0.0
Iteration 776: Arm = 0, Reward = 0.0
Iteration 777: Arm = 2, Reward = 1.0
Iteration 778: Arm = 2, Reward = 1.0
Iteration 779: Arm = 2, Reward = 1.0
Iteration 780: Arm = 3, Reward = 0.0
Iteration 781: Arm = 3, Reward = 1.0
Iteration 782: Arm = 2, Reward = 0.0
Iteration 783: Arm = 2, Reward = 1.0
Iteration 784: Arm = 2, Reward = 1.0
Iteration 785: Arm = 0, Reward = 1.0
Iteration 786: Arm = 0, Reward = 0.0
Iteration 787: Arm = 2, Reward = 0.0
Iteration 788: Arm = 2, Reward = 0.0
Iteration 789: Arm = 1, Reward = 0.0
Iteration 790: Arm = 3, Reward = 1.0
Iteration 791: Arm = 2, Reward = 0.0
Iteration 792: Arm = 2, Reward = 0.0
Iteration 793: Arm = 1, Reward = 1.0
Iteration 794: Arm = 2, Reward = 0.0
Iteration 795: Arm = 2, Reward = 1.0
Iteration 796: Arm = 2, Reward = 1.0
Iteration 797: Arm = 0, Reward = 0.0
Iteration 798: Arm = 0, Reward = 0.0
Iteration 799: Arm = 3, Reward = 0.0
Iteration 800: Arm = 2, Reward = 0.0
Iteration 801: Arm = 2, Reward = 1.0
Iteration 802: Arm = 3, Reward = 0.0
Iteration 803: Arm = 3, Reward = 1.0
Iteration 804: Arm = 2, Reward = 0.0
Iteration 805: Arm = 2, Reward = 0.0
Iteration 806: Arm = 2, Reward = 1.0
Iteration 807: Arm = 2, Reward = 1.0
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Iteration 808: Arm = 0, Reward = 1.0
Iteration 809: Arm = 2, Reward = 0.0
Iteration 810: Arm = 1, Reward = 0.0
Iteration 811: Arm = 2, Reward = 0.0
Iteration 812: Arm = 2, Reward = 1.0
Iteration 813: Arm = 2, Reward = 1.0
Iteration 814: Arm = 3, Reward = 1.0
Iteration 815: Arm = 2, Reward = 1.0
Iteration 816: Arm = 2, Reward = 1.0
Iteration 817: Arm = 2, Reward = 0.0
Iteration 818: Arm = 1, Reward = 1.0
Iteration 819: Arm = 2, Reward = 0.0
Iteration 820: Arm = 3, Reward = 1.0
Iteration 821: Arm = 2, Reward = 1.0
Iteration 822: Arm = 0, Reward = 0.0
Iteration 823: Arm = 2, Reward = 0.0
Iteration 824: Arm = 3, Reward = 0.0
Iteration 825: Arm = 2, Reward = 1.0
Iteration 826: Arm = 2, Reward = 1.0
Iteration 827: Arm = 2, Reward = 0.0
Iteration 828: Arm = 2, Reward = 1.0
Iteration 829: Arm = 2, Reward = 1.0
Iteration 830: Arm = 2, Reward = 1.0
Iteration 831: Arm = 3, Reward = 0.0
Iteration 832: Arm = 2, Reward = 0.0
Iteration 833: Arm = 2, Reward = 1.0
Iteration 834: Arm = 2, Reward = 0.0
Iteration 835: Arm = 2, Reward = 1.0
Iteration 836: Arm = 2, Reward = 0.0
Iteration 837: Arm = 1, Reward = 0.0
Iteration 838: Arm = 2, Reward = 0.0
Iteration 839: Arm = 1, Reward = 1.0
Iteration 840: Arm = 2, Reward = 1.0
Iteration 841: Arm = 2, Reward = 1.0
Iteration 842: Arm = 3, Reward = 1.0
Iteration 843: Arm = 2, Reward = 0.0
Iteration 844: Arm = 0, Reward = 0.0
Iteration 845: Arm = 3, Reward = 0.0
Iteration 846: Arm = 2, Reward = 1.0
Iteration 847: Arm = 2, Reward = 0.0
Iteration 848: Arm = 2, Reward = 1.0
Iteration 849: Arm = 2, Reward = 1.0
Iteration 850: Arm = 2, Reward = 1.0
Iteration 851: Arm = 1, Reward = 0.0
Iteration 852: Arm = 3, Reward = 1.0
Iteration 853: Arm = 2, Reward = 1.0
Iteration 854: Arm = 3, Reward = 1.0
Iteration 855: Arm = 2, Reward = 0.0
Iteration 856: Arm = 2, Reward = 1.0
Iteration 857: Arm = 2, Reward = 0.0
Iteration 858: Arm = 0, Reward = 1.0
Iteration 859: Arm = 2, Reward = 1.0
Iteration 860: Arm = 2, Reward = 1.0
Iteration 861: Arm = 2, Reward = 0.0
Iteration 862: Arm = 0, Reward = 0.0
Iteration 863: Arm = 1, Reward = 1.0
Iteration 864: Arm = 1, Reward = 1.0
Iteration 865: Arm = 2, Reward = 0.0
Iteration 866: Arm = 0, Reward = 0.0
Iteration 867: Arm = 2, Reward = 1.0
Iteration 868: Arm = 0, Reward = 0.0
Iteration 869: Arm = 2, Reward = 0.0
Iteration 870: Arm = 3, Reward = 0.0
Iteration 871: Arm = 3, Reward = 1.0
Iteration 872: Arm = 2, Reward = 1.0
Iteration 873: Arm = 1, Reward = 1.0
Iteration 874: Arm = 2, Reward = 0.0
Iteration 875: Arm = 0, Reward = 0.0
Iteration 876: Arm = 2, Reward = 0.0
Iteration 877: Arm = 3, Reward = 0.0
Iteration 878: Arm = 2, Reward = 1.0
Iteration 879: Arm = 2, Reward = 1.0
Iteration 880: Arm = 2, Reward = 0.0
Iteration 881: Arm = 2, Reward = 0.0
Iteration 882: Arm = 2, Reward = 1.0
Iteration 883: Arm = 3, Reward = 1.0
Iteration 884: Arm = 2, Reward = 1.0
Iteration 885: Arm = 0, Reward = 0.0
Iteration 886: Arm = 2, Reward = 1.0
Iteration 887: Arm = 2, Reward = 1.0
Iteration 888: Arm = 2, Reward = 1.0
Iteration 889: Arm = 2, Reward = 1.0
Iteration 890: Arm = 0, Reward = 0.0
```

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Iteration 891: Arm = 2, Reward = 0.0
Iteration 892: Arm = 2, Reward = 1.0
Iteration 893: Arm = 0, Reward = 0.0
Iteration 894: Arm = 2, Reward = 1.0
Iteration 895: Arm = 2, Reward = 1.0
Iteration 896: Arm = 1, Reward = 1.0
Iteration 897: Arm = 3, Reward = 0.0
Iteration 898: Arm = 2, Reward = 0.0
Iteration 899: Arm = 3, Reward = 0.0
Iteration 900: Arm = 0, Reward = 1.0
Iteration 901: Arm = 2, Reward = 1.0
Iteration 902: Arm = 2, Reward = 0.0
Iteration 903: Arm = 2, Reward = 1.0
Iteration 904: Arm = 2, Reward = 1.0
Iteration 905: Arm = 3, Reward = 0.0
Iteration 906: Arm = 2, Reward = 1.0
Iteration 907: Arm = 2, Reward = 0.0
Iteration 908: Arm = 2, Reward = 0.0
Iteration 909: Arm = 3, Reward = 0.0
Iteration 910: Arm = 0, Reward = 0.0
Iteration 911: Arm = 2, Reward = 1.0
Iteration 912: Arm = 2, Reward = 0.0
Iteration 913: Arm = 2, Reward = 0.0
Iteration 914: Arm = 2, Reward = 0.0
Iteration 915: Arm = 2, Reward = 0.0
Iteration 916: Arm = 0, Reward = 0.0
Iteration 917: Arm = 2, Reward = 1.0
Iteration 918: Arm = 0, Reward = 0.0
Iteration 919: Arm = 1, Reward = 1.0
Iteration 920: Arm = 1, Reward = 1.0
Iteration 921: Arm = 2, Reward = 1.0
Iteration 922: Arm = 2, Reward = 1.0
Iteration 923: Arm = 2, Reward = 1.0
Iteration 924: Arm = 2, Reward = 0.0
Iteration 925: Arm = 2, Reward = 1.0
Iteration 926: Arm = 2, Reward = 1.0
Iteration 927: Arm = 0, Reward = 0.0
Iteration 928: Arm = 2, Reward = 1.0
Iteration 929: Arm = 2, Reward = 0.0
Iteration 930: Arm = 2, Reward = 1.0
Iteration 931: Arm = 2, Reward = 1.0
Iteration 932: Arm = 2, Reward = 1.0
Iteration 933: Arm = 2, Reward = 0.0
Iteration 934: Arm = 2, Reward = 0.0
Iteration 935: Arm = 2, Reward = 0.0
Iteration 936: Arm = 1, Reward = 1.0
Iteration 937: Arm = 0, Reward = 1.0
Iteration 938: Arm = 1, Reward = 1.0
Iteration 939: Arm = 3, Reward = 1.0
Iteration 940: Arm = 0, Reward = 0.0
Iteration 941: Arm = 2, Reward = 0.0
Iteration 942: Arm = 2, Reward = 0.0
Iteration 943: Arm = 2, Reward = 0.0
Iteration 944: Arm = 2, Reward = 1.0
Iteration 945: Arm = 2, Reward = 0.0
Iteration 946: Arm = 0, Reward = 0.0
Iteration 947: Arm = 1, Reward = 1.0
Iteration 948: Arm = 2, Reward = 0.0
Iteration 949: Arm = 0, Reward = 0.0
Iteration 950: Arm = 2, Reward = 1.0
Iteration 951: Arm = 2, Reward = 0.0
Iteration 952: Arm = 2, Reward = 0.0
Iteration 953: Arm = 2, Reward = 1.0
Iteration 954: Arm = 2, Reward = 1.0
Iteration 955: Arm = 1, Reward = 1.0
Iteration 956: Arm = 1, Reward = 1.0
Iteration 957: Arm = 3, Reward = 1.0
Iteration 958: Arm = 2, Reward = 0.0
Iteration 959: Arm = 1, Reward = 1.0
Iteration 960: Arm = 1, Reward = 1.0
Iteration 961: Arm = 3, Reward = 1.0
Iteration 962: Arm = 0, Reward = 0.0
Iteration 963: Arm = 1, Reward = 0.0
Iteration 964: Arm = 1, Reward = 1.0
Iteration 965: Arm = 1, Reward = 1.0
Iteration 966: Arm = 1, Reward = 0.0
Iteration 967: Arm = 1, Reward = 0.0
Iteration 968: Arm = 1, Reward = 0.0
Iteration 969: Arm = 0, Reward = 0.0
Iteration 970: Arm = 1, Reward = 1.0
Iteration 971: Arm = 3, Reward = 0.0
Iteration 972: Arm = 2, Reward = 0.0
Iteration 973: Arm = 1, Reward = 1.0
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Iteration 974: Arm = 1, Reward = 1.0
Iteration 975: Arm = 3, Reward = 1.0
Iteration 976: Arm = 1, Reward = 1.0
Iteration 977: Arm = 1, Reward = 1.0
Iteration 978: Arm = 1, Reward = 0.0
Iteration 979: Arm = 1, Reward = 1.0
Iteration 980: Arm = 1, Reward = 1.0
Iteration 981: Arm = 1, Reward = 1.0
Iteration 982: Arm = 1, Reward = 1.0
Iteration 983: Arm = 3, Reward = 1.0
Iteration 984: Arm = 1, Reward = 1.0
Iteration 985: Arm = 2, Reward = 1.0
Iteration 986: Arm = 1, Reward = 1.0
Iteration 987: Arm = 1, Reward = 1.0
Iteration 988: Arm = 3, Reward = 0.0
Iteration 989: Arm = 0, Reward = 0.0
Iteration 990: Arm = 0, Reward = 0.0
Iteration 991: Arm = 1, Reward = 1.0
Iteration 992: Arm = 1, Reward = 1.0
Iteration 993: Arm = 0, Reward = 0.0
Iteration 994: Arm = 3, Reward = 1.0
Iteration 995: Arm = 3, Reward = 1.0
Iteration 996: Arm = 2, Reward = 1.0
Iteration 997: Arm = 0, Reward = 0.0
Iteration 998: Arm = 3, Reward = 1.0
Iteration 999: Arm = 1, Reward = 0.0
Iteration 1000: Arm = 3, Reward = 1.0
```

In [4]:

```python
# --- UCB POLICY ---
print("\n--- UCB Policy ---")
ucb_rewards = []
ucb_counts = np.zeros(num_arms)
ucb_values = np.zeros(num_arms)

for i in range(iterations):
    if i < num_arms:
        arm = i
    else:
        confidence_bounds = ucb_values + np.sqrt(2 * np.log(i + 1) / (ucb_counts + 1e-5))
        arm = np.argmax(confidence_bounds)
    reward = df[df['trt'] == arm].sample(1).iloc[0]['reward']
    ucb_counts[arm] += 1
    ucb_values[arm] += (reward - ucb_values[arm]) / ucb_counts[arm]
    ucb_rewards.append(reward)
    print(f"Iteration {i+1}: Arm = {arm}, Reward = {reward}")
policies_rewards["UCB"] = ucb_rewards
arm_counts["UCB"] = ucb_counts
```

```
--- UCB Policy ---
Iteration 1: Arm = 0, Reward = 1.0
Iteration 2: Arm = 1, Reward = 0.0
Iteration 3: Arm = 2, Reward = 0.0
Iteration 4: Arm = 3, Reward = 1.0
Iteration 5: Arm = 0, Reward = 1.0
Iteration 6: Arm = 3, Reward = 0.0
Iteration 7: Arm = 0, Reward = 0.0
Iteration 8: Arm = 1, Reward = 1.0
Iteration 9: Arm = 2, Reward = 1.0
Iteration 10: Arm = 1, Reward = 0.0
Iteration 11: Arm = 2, Reward = 0.0
Iteration 12: Arm = 3, Reward = 1.0
Iteration 13: Arm = 0, Reward = 1.0
Iteration 14: Arm = 3, Reward = 1.0
Iteration 15: Arm = 0, Reward = 1.0
Iteration 16: Arm = 3, Reward = 1.0
Iteration 17: Arm = 0, Reward = 1.0
Iteration 18: Arm = 3, Reward = 1.0
Iteration 19: Arm = 0, Reward = 0.0
Iteration 20: Arm = 3, Reward = 1.0
Iteration 21: Arm = 3, Reward = 1.0
Iteration 22: Arm = 1, Reward = 1.0
Iteration 23: Arm = 2, Reward = 1.0
Iteration 24: Arm = 3, Reward = 0.0
Iteration 25: Arm = 1, Reward = 1.0
Iteration 26: Arm = 2, Reward = 1.0
Iteration 27: Arm = 1, Reward = 0.0
Iteration 28: Arm = 2, Reward = 0.0
Iteration 29: Arm = 0, Reward = 1.0
Iteration 30: Arm = 0, Reward = 1.0
Iteration 31: Arm = 0, Reward = 1.0
Iteration 32: Arm = 3, Reward = 1.0
Iteration 33: Arm = 0, Reward = 0.0
Iteration 34: Arm = 3, Reward = 0.0
Iteration 35: Arm = 1, Reward = 0.0
```

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Iteration 36: Arm = 2, Reward = 0.0
Iteration 37: Arm = 0, Reward = 1.0
Iteration 38: Arm = 3, Reward = 0.0
Iteration 39: Arm = 0, Reward = 0.0
Iteration 40: Arm = 1, Reward = 0.0
Iteration 41: Arm = 2, Reward = 0.0
Iteration 42: Arm = 3, Reward = 0.0
Iteration 43: Arm = 0, Reward = 1.0
Iteration 44: Arm = 0, Reward = 1.0
Iteration 45: Arm = 0, Reward = 0.0
Iteration 46: Arm = 3, Reward = 1.0
Iteration 47: Arm = 3, Reward = 1.0
Iteration 48: Arm = 3, Reward = 0.0
Iteration 49: Arm = 0, Reward = 0.0
Iteration 50: Arm = 1, Reward = 1.0
Iteration 51: Arm = 1, Reward = 0.0
Iteration 52: Arm = 2, Reward = 0.0
Iteration 53: Arm = 0, Reward = 0.0
Iteration 54: Arm = 3, Reward = 0.0
Iteration 55: Arm = 1, Reward = 1.0
Iteration 56: Arm = 1, Reward = 1.0
Iteration 57: Arm = 1, Reward = 1.0
Iteration 58: Arm = 1, Reward = 1.0
Iteration 59: Arm = 1, Reward = 0.0
Iteration 60: Arm = 2, Reward = 0.0
Iteration 61: Arm = 0, Reward = 1.0
Iteration 62: Arm = 0, Reward = 1.0
Iteration 63: Arm = 0, Reward = 0.0
Iteration 64: Arm = 3, Reward = 0.0
Iteration 65: Arm = 1, Reward = 1.0
Iteration 66: Arm = 1, Reward = 1.0
Iteration 67: Arm = 1, Reward = 1.0
Iteration 68: Arm = 1, Reward = 1.0
Iteration 69: Arm = 1, Reward = 0.0
Iteration 70: Arm = 0, Reward = 1.0
Iteration 71: Arm = 0, Reward = 0.0
Iteration 72: Arm = 1, Reward = 1.0
Iteration 73: Arm = 1, Reward = 1.0
Iteration 74: Arm = 1, Reward = 1.0
Iteration 75: Arm = 1, Reward = 1.0
Iteration 76: Arm = 1, Reward = 1.0
Iteration 77: Arm = 1, Reward = 1.0
Iteration 78: Arm = 1, Reward = 1.0
Iteration 79: Arm = 1, Reward = 0.0
Iteration 80: Arm = 3, Reward = 0.0
Iteration 81: Arm = 1, Reward = 1.0
Iteration 82: Arm = 1, Reward = 0.0
Iteration 83: Arm = 2, Reward = 1.0
Iteration 84: Arm = 2, Reward = 0.0
Iteration 85: Arm = 0, Reward = 1.0
Iteration 86: Arm = 0, Reward = 0.0
Iteration 87: Arm = 1, Reward = 0.0
Iteration 88: Arm = 3, Reward = 0.0
Iteration 89: Arm = 0, Reward = 1.0
Iteration 90: Arm = 0, Reward = 0.0
Iteration 91: Arm = 2, Reward = 1.0
Iteration 92: Arm = 2, Reward = 1.0
Iteration 93: Arm = 2, Reward = 0.0
Iteration 94: Arm = 1, Reward = 1.0
Iteration 95: Arm = 1, Reward = 0.0
Iteration 96: Arm = 2, Reward = 0.0
Iteration 97: Arm = 3, Reward = 0.0
Iteration 98: Arm = 0, Reward = 0.0
Iteration 99: Arm = 1, Reward = 1.0
Iteration 100: Arm = 1, Reward = 0.0
Iteration 101: Arm = 0, Reward = 0.0
Iteration 102: Arm = 1, Reward = 1.0
Iteration 103: Arm = 1, Reward = 1.0
Iteration 104: Arm = 1, Reward = 0.0
Iteration 105: Arm = 3, Reward = 1.0
Iteration 106: Arm = 3, Reward = 0.0
Iteration 107: Arm = 2, Reward = 0.0
Iteration 108: Arm = 1, Reward = 0.0
Iteration 109: Arm = 0, Reward = 0.0
Iteration 110: Arm = 3, Reward = 1.0
Iteration 111: Arm = 3, Reward = 1.0
Iteration 112: Arm = 3, Reward = 1.0
Iteration 113: Arm = 3, Reward = 0.0
Iteration 114: Arm = 3, Reward = 0.0
Iteration 115: Arm = 1, Reward = 1.0
Iteration 116: Arm = 1, Reward = 0.0
Iteration 117: Arm = 2, Reward = 1.0
Iteration 118: Arm = 2, Reward = 0.0
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Iteration 119: Arm = 0, Reward = 1.0
Iteration 120: Arm = 0, Reward = 0.0
Iteration 121: Arm = 1, Reward = 1.0
Iteration 122: Arm = 1, Reward = 0.0
Iteration 123: Arm = 3, Reward = 1.0
Iteration 124: Arm = 3, Reward = 0.0
Iteration 125: Arm = 2, Reward = 0.0
Iteration 126: Arm = 0, Reward = 1.0
Iteration 127: Arm = 0, Reward = 1.0
Iteration 128: Arm = 0, Reward = 0.0
Iteration 129: Arm = 1, Reward = 1.0
Iteration 130: Arm = 1, Reward = 0.0
Iteration 131: Arm = 0, Reward = 0.0
Iteration 132: Arm = 3, Reward = 0.0
Iteration 133: Arm = 1, Reward = 0.0
Iteration 134: Arm = 2, Reward = 1.0
Iteration 135: Arm = 2, Reward = 0.0
Iteration 136: Arm = 0, Reward = 1.0
Iteration 137: Arm = 0, Reward = 0.0
Iteration 138: Arm = 1, Reward = 1.0
Iteration 139: Arm = 1, Reward = 1.0
Iteration 140: Arm = 1, Reward = 1.0
Iteration 141: Arm = 1, Reward = 1.0
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Iteration 143: Arm = 1, Reward = 0.0
Iteration 144: Arm = 1, Reward = 0.0
Iteration 145: Arm = 3, Reward = 1.0
Iteration 146: Arm = 3, Reward = 0.0
Iteration 147: Arm = 0, Reward = 0.0
Iteration 148: Arm = 1, Reward = 0.0
Iteration 149: Arm = 2, Reward = 1.0
Iteration 150: Arm = 2, Reward = 0.0
Iteration 151: Arm = 3, Reward = 1.0
Iteration 152: Arm = 3, Reward = 1.0
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Iteration 156: Arm = 3, Reward = 1.0
Iteration 157: Arm = 3, Reward = 0.0
Iteration 158: Arm = 3, Reward = 0.0
Iteration 159: Arm = 1, Reward = 1.0
Iteration 160: Arm = 1, Reward = 1.0
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Iteration 167: Arm = 1, Reward = 0.0
Iteration 168: Arm = 2, Reward = 0.0
Iteration 169: Arm = 0, Reward = 0.0
Iteration 170: Arm = 1, Reward = 1.0
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Iteration 175: Arm = 1, Reward = 0.0
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Iteration 177: Arm = 3, Reward = 0.0
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Iteration 201: Arm = 3, Reward = 0.0
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Iteration 202: Arm = 1, Reward = 1.0
Iteration 203: Arm = 1, Reward = 1.0
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Iteration 221: Arm = 2, Reward = 0.0
Iteration 222: Arm = 1, Reward = 0.0
Iteration 223: Arm = 3, Reward = 0.0
Iteration 224: Arm = 0, Reward = 1.0
Iteration 225: Arm = 0, Reward = 0.0
Iteration 226: Arm = 2, Reward = 0.0
Iteration 227: Arm = 1, Reward = 1.0
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Iteration 232: Arm = 1, Reward = 0.0
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Iteration 234: Arm = 3, Reward = 0.0
Iteration 235: Arm = 1, Reward = 0.0
Iteration 236: Arm = 2, Reward = 1.0
Iteration 237: Arm = 2, Reward = 1.0
Iteration 238: Arm = 2, Reward = 0.0
Iteration 239: Arm = 2, Reward = 1.0
Iteration 240: Arm = 2, Reward = 0.0
Iteration 241: Arm = 3, Reward = 0.0
Iteration 242: Arm = 0, Reward = 0.0
Iteration 243: Arm = 1, Reward = 1.0
Iteration 244: Arm = 1, Reward = 1.0
Iteration 245: Arm = 1, Reward = 0.0
Iteration 246: Arm = 2, Reward = 1.0
Iteration 247: Arm = 2, Reward = 0.0
Iteration 248: Arm = 1, Reward = 0.0
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Iteration 250: Arm = 3, Reward = 1.0
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Iteration 259: Arm = 0, Reward = 1.0
Iteration 260: Arm = 0, Reward = 0.0
Iteration 261: Arm = 1, Reward = 1.0
Iteration 262: Arm = 1, Reward = 0.0
Iteration 263: Arm = 3, Reward = 0.0
Iteration 264: Arm = 1, Reward = 0.0
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Iteration 266: Arm = 1, Reward = 1.0
Iteration 267: Arm = 1, Reward = 0.0
Iteration 268: Arm = 2, Reward = 1.0
Iteration 269: Arm = 2, Reward = 0.0
Iteration 270: Arm = 3, Reward = 1.0
Iteration 271: Arm = 3, Reward = 1.0
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Iteration 282: Arm = 2, Reward = 1.0
Iteration 283: Arm = 2, Reward = 0.0
Iteration 284: Arm = 3, Reward = 0.0
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Iteration 285: Arm = 1, Reward = 0.0
Iteration 286: Arm = 0, Reward = 0.0
Iteration 287: Arm = 2, Reward = 0.0
Iteration 288: Arm = 1, Reward = 0.0
Iteration 289: Arm = 3, Reward = 0.0
Iteration 290: Arm = 0, Reward = 1.0
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Iteration 301: Arm = 3, Reward = 1.0
Iteration 302: Arm = 3, Reward = 1.0
Iteration 303: Arm = 3, Reward = 1.0
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Iteration 314: Arm = 3, Reward = 0.0
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Iteration 328: Arm = 3, Reward = 1.0
Iteration 329: Arm = 3, Reward = 0.0
Iteration 330: Arm = 3, Reward = 0.0
Iteration 331: Arm = 3, Reward = 1.0
Iteration 332: Arm = 3, Reward = 1.0
Iteration 333: Arm = 3, Reward = 0.0
Iteration 334: Arm = 3, Reward = 0.0
Iteration 335: Arm = 2, Reward = 0.0
Iteration 336: Arm = 0, Reward = 1.0
Iteration 337: Arm = 0, Reward = 0.0
Iteration 338: Arm = 1, Reward = 1.0
Iteration 339: Arm = 1, Reward = 1.0
Iteration 340: Arm = 1, Reward = 1.0
Iteration 341: Arm = 1, Reward = 0.0
Iteration 342: Arm = 1, Reward = 1.0
Iteration 343: Arm = 1, Reward = 0.0
Iteration 344: Arm = 3, Reward = 0.0
Iteration 345: Arm = 1, Reward = 1.0
Iteration 346: Arm = 1, Reward = 1.0
Iteration 347: Arm = 1, Reward = 1.0
Iteration 348: Arm = 1, Reward = 1.0
Iteration 349: Arm = 1, Reward = 1.0
Iteration 350: Arm = 1, Reward = 1.0
Iteration 351: Arm = 1, Reward = 1.0
Iteration 352: Arm = 1, Reward = 0.0
Iteration 353: Arm = 1, Reward = 1.0
Iteration 354: Arm = 1, Reward = 0.0
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Iteration 360: Arm = 1, Reward = 1.0
Iteration 361: Arm = 1, Reward = 1.0
Iteration 362: Arm = 1, Reward = 1.0
Iteration 363: Arm = 1, Reward = 1.0
Iteration 364: Arm = 1, Reward = 0.0
Iteration 365: Arm = 1, Reward = 0.0
Iteration 366: Arm = 3, Reward = 1.0
Iteration 367: Arm = 3, Reward = 0.0
```

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Iteration 368: Arm = 1, Reward = 1.0
Iteration 369: Arm = 1, Reward = 0.0
Iteration 370: Arm = 2, Reward = 1.0
Iteration 371: Arm = 2, Reward = 0.0
Iteration 372: Arm = 3, Reward = 1.0
Iteration 373: Arm = 3, Reward = 1.0
Iteration 374: Arm = 3, Reward = 0.0
Iteration 375: Arm = 3, Reward = 1.0
Iteration 376: Arm = 3, Reward = 0.0
Iteration 377: Arm = 1, Reward = 0.0
Iteration 378: Arm = 0, Reward = 0.0
Iteration 379: Arm = 2, Reward = 1.0
Iteration 380: Arm = 2, Reward = 0.0
Iteration 381: Arm = 3, Reward = 1.0
Iteration 382: Arm = 3, Reward = 1.0
Iteration 383: Arm = 3, Reward = 1.0
Iteration 384: Arm = 3, Reward = 0.0
Iteration 385: Arm = 3, Reward = 1.0
Iteration 386: Arm = 3, Reward = 0.0
Iteration 387: Arm = 1, Reward = 0.0
Iteration 388: Arm = 3, Reward = 0.0
Iteration 389: Arm = 2, Reward = 1.0
Iteration 390: Arm = 2, Reward = 1.0
Iteration 391: Arm = 2, Reward = 1.0
Iteration 392: Arm = 2, Reward = 0.0
Iteration 393: Arm = 2, Reward = 1.0
Iteration 394: Arm = 2, Reward = 1.0
Iteration 395: Arm = 2, Reward = 1.0
Iteration 396: Arm = 2, Reward = 0.0
Iteration 397: Arm = 2, Reward = 0.0
Iteration 398: Arm = 2, Reward = 1.0
Iteration 399: Arm = 2, Reward = 1.0
Iteration 400: Arm = 2, Reward = 0.0
Iteration 401: Arm = 2, Reward = 0.0
Iteration 402: Arm = 1, Reward = 1.0
Iteration 403: Arm = 1, Reward = 1.0
Iteration 404: Arm = 1, Reward = 0.0
Iteration 405: Arm = 0, Reward = 1.0
Iteration 406: Arm = 0, Reward = 1.0
Iteration 407: Arm = 0, Reward = 1.0
Iteration 408: Arm = 0, Reward = 0.0
Iteration 409: Arm = 0, Reward = 0.0
Iteration 410: Arm = 3, Reward = 1.0
Iteration 411: Arm = 3, Reward = 1.0
Iteration 412: Arm = 3, Reward = 1.0
Iteration 413: Arm = 3, Reward = 0.0
Iteration 414: Arm = 3, Reward = 1.0
Iteration 415: Arm = 3, Reward = 0.0
Iteration 416: Arm = 1, Reward = 1.0
Iteration 417: Arm = 1, Reward = 0.0
Iteration 418: Arm = 2, Reward = 0.0
Iteration 419: Arm = 3, Reward = 0.0
Iteration 420: Arm = 1, Reward = 1.0
Iteration 421: Arm = 1, Reward = 1.0
Iteration 422: Arm = 1, Reward = 0.0
Iteration 423: Arm = 0, Reward = 0.0
Iteration 424: Arm = 1, Reward = 1.0
Iteration 425: Arm = 1, Reward = 0.0
Iteration 426: Arm = 1, Reward = 1.0
Iteration 427: Arm = 1, Reward = 0.0
Iteration 428: Arm = 3, Reward = 1.0
Iteration 429: Arm = 3, Reward = 0.0
Iteration 430: Arm = 1, Reward = 0.0
Iteration 431: Arm = 3, Reward = 0.0
Iteration 432: Arm = 2, Reward = 0.0
Iteration 433: Arm = 0, Reward = 1.0
Iteration 434: Arm = 0, Reward = 0.0
Iteration 435: Arm = 1, Reward = 0.0
Iteration 436: Arm = 3, Reward = 1.0
Iteration 437: Arm = 3, Reward = 0.0
Iteration 438: Arm = 0, Reward = 0.0
Iteration 439: Arm = 3, Reward = 1.0
Iteration 440: Arm = 3, Reward = 0.0
Iteration 441: Arm = 1, Reward = 1.0
Iteration 442: Arm = 1, Reward = 0.0
Iteration 443: Arm = 2, Reward = 0.0
Iteration 444: Arm = 3, Reward = 1.0
Iteration 445: Arm = 3, Reward = 1.0
Iteration 446: Arm = 3, Reward = 1.0
Iteration 447: Arm = 3, Reward = 0.0
Iteration 448: Arm = 3, Reward = 0.0
Iteration 449: Arm = 1, Reward = 0.0
Iteration 450: Arm = 0, Reward = 0.0
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Iteration 451: Arm = 3, Reward = 1.0
Iteration 452: Arm = 3, Reward = 0.0
Iteration 453: Arm = 1, Reward = 1.0
Iteration 454: Arm = 1, Reward = 1.0
Iteration 455: Arm = 1, Reward = 1.0
Iteration 456: Arm = 1, Reward = 0.0
Iteration 457: Arm = 1, Reward = 1.0
Iteration 458: Arm = 1, Reward = 0.0
Iteration 459: Arm = 3, Reward = 1.0
Iteration 460: Arm = 3, Reward = 0.0
Iteration 461: Arm = 1, Reward = 1.0
Iteration 462: Arm = 1, Reward = 1.0
Iteration 463: Arm = 1, Reward = 1.0
Iteration 464: Arm = 1, Reward = 0.0
Iteration 465: Arm = 1, Reward = 1.0
Iteration 466: Arm = 1, Reward = 0.0
Iteration 467: Arm = 1, Reward = 1.0
Iteration 468: Arm = 1, Reward = 0.0
Iteration 469: Arm = 2, Reward = 1.0
Iteration 470: Arm = 2, Reward = 0.0
Iteration 471: Arm = 3, Reward = 0.0
Iteration 472: Arm = 1, Reward = 1.0
Iteration 473: Arm = 1, Reward = 1.0
Iteration 474: Arm = 1, Reward = 1.0
Iteration 475: Arm = 1, Reward = 0.0
Iteration 476: Arm = 1, Reward = 0.0
Iteration 477: Arm = 0, Reward = 0.0
Iteration 478: Arm = 2, Reward = 0.0
Iteration 479: Arm = 1, Reward = 1.0
Iteration 480: Arm = 1, Reward = 1.0
Iteration 481: Arm = 1, Reward = 1.0
Iteration 482: Arm = 1, Reward = 0.0
Iteration 483: Arm = 1, Reward = 1.0
Iteration 484: Arm = 1, Reward = 1.0
Iteration 485: Arm = 1, Reward = 1.0
Iteration 486: Arm = 1, Reward = 1.0
Iteration 487: Arm = 1, Reward = 0.0
Iteration 488: Arm = 1, Reward = 1.0
Iteration 489: Arm = 1, Reward = 1.0
Iteration 490: Arm = 1, Reward = 0.0
Iteration 491: Arm = 1, Reward = 1.0
Iteration 492: Arm = 1, Reward = 0.0
Iteration 493: Arm = 1, Reward = 1.0
Iteration 494: Arm = 1, Reward = 1.0
Iteration 495: Arm = 1, Reward = 0.0
Iteration 496: Arm = 3, Reward = 1.0
Iteration 497: Arm = 3, Reward = 1.0
Iteration 498: Arm = 3, Reward = 0.0
Iteration 499: Arm = 1, Reward = 1.0
Iteration 500: Arm = 1, Reward = 1.0
Iteration 501: Arm = 1, Reward = 0.0
Iteration 502: Arm = 3, Reward = 0.0
Iteration 503: Arm = 1, Reward = 0.0
Iteration 504: Arm = 1, Reward = 1.0
Iteration 505: Arm = 1, Reward = 1.0
Iteration 506: Arm = 1, Reward = 1.0
Iteration 507: Arm = 1, Reward = 1.0
Iteration 508: Arm = 1, Reward = 1.0
Iteration 509: Arm = 1, Reward = 1.0
Iteration 510: Arm = 1, Reward = 1.0
Iteration 511: Arm = 1, Reward = 0.0
Iteration 512: Arm = 1, Reward = 0.0
Iteration 513: Arm = 1, Reward = 0.0
Iteration 514: Arm = 3, Reward = 1.0
Iteration 515: Arm = 3, Reward = 1.0
Iteration 516: Arm = 3, Reward = 1.0
Iteration 517: Arm = 3, Reward = 0.0
Iteration 518: Arm = 3, Reward = 1.0
Iteration 519: Arm = 3, Reward = 0.0
Iteration 520: Arm = 0, Reward = 1.0
Iteration 521: Arm = 0, Reward = 0.0
Iteration 522: Arm = 1, Reward = 0.0
Iteration 523: Arm = 3, Reward = 0.0
Iteration 524: Arm = 2, Reward = 0.0
Iteration 525: Arm = 0, Reward = 0.0
Iteration 526: Arm = 1, Reward = 0.0
Iteration 527: Arm = 3, Reward = 1.0
Iteration 528: Arm = 3, Reward = 0.0
Iteration 529: Arm = 1, Reward = 1.0
Iteration 530: Arm = 1, Reward = 1.0
Iteration 531: Arm = 1, Reward = 0.0
Iteration 532: Arm = 3, Reward = 0.0
Iteration 533: Arm = 1, Reward = 0.0
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Iteration 534: Arm = 2, Reward = 0.0
Iteration 535: Arm = 0, Reward = 0.0
Iteration 536: Arm = 1, Reward = 1.0
Iteration 537: Arm = 1, Reward = 1.0
Iteration 538: Arm = 1, Reward = 0.0
Iteration 539: Arm = 1, Reward = 1.0
Iteration 540: Arm = 1, Reward = 1.0
Iteration 541: Arm = 1, Reward = 0.0
Iteration 542: Arm = 3, Reward = 0.0
Iteration 543: Arm = 1, Reward = 0.0
Iteration 544: Arm = 1, Reward = 0.0
Iteration 545: Arm = 3, Reward = 1.0
Iteration 546: Arm = 3, Reward = 0.0
Iteration 547: Arm = 3, Reward = 0.0
Iteration 548: Arm = 0, Reward = 1.0
Iteration 549: Arm = 0, Reward = 0.0
Iteration 550: Arm = 1, Reward = 1.0
Iteration 551: Arm = 1, Reward = 0.0
Iteration 552: Arm = 2, Reward = 1.0
Iteration 553: Arm = 2, Reward = 0.0
Iteration 554: Arm = 1, Reward = 1.0
Iteration 555: Arm = 1, Reward = 1.0
Iteration 556: Arm = 1, Reward = 1.0
Iteration 557: Arm = 1, Reward = 0.0
Iteration 558: Arm = 1, Reward = 1.0
Iteration 559: Arm = 1, Reward = 1.0
Iteration 560: Arm = 1, Reward = 0.0
Iteration 561: Arm = 1, Reward = 0.0
Iteration 562: Arm = 0, Reward = 0.0
Iteration 563: Arm = 2, Reward = 0.0
Iteration 564: Arm = 3, Reward = 0.0
Iteration 565: Arm = 1, Reward = 0.0
Iteration 566: Arm = 1, Reward = 1.0
Iteration 567: Arm = 1, Reward = 1.0
Iteration 568: Arm = 1, Reward = 0.0
Iteration 569: Arm = 3, Reward = 1.0
Iteration 570: Arm = 3, Reward = 1.0
Iteration 571: Arm = 3, Reward = 0.0
Iteration 572: Arm = 3, Reward = 0.0
Iteration 573: Arm = 1, Reward = 0.0
Iteration 574: Arm = 0, Reward = 1.0
Iteration 575: Arm = 0, Reward = 0.0
Iteration 576: Arm = 1, Reward = 1.0
Iteration 577: Arm = 1, Reward = 0.0
Iteration 578: Arm = 2, Reward = 1.0
Iteration 579: Arm = 2, Reward = 0.0
Iteration 580: Arm = 3, Reward = 0.0
Iteration 581: Arm = 1, Reward = 1.0
Iteration 582: Arm = 1, Reward = 1.0
Iteration 583: Arm = 1, Reward = 1.0
Iteration 584: Arm = 1, Reward = 0.0
Iteration 585: Arm = 1, Reward = 1.0
Iteration 586: Arm = 1, Reward = 0.0
Iteration 587: Arm = 0, Reward = 0.0
Iteration 588: Arm = 1, Reward = 0.0
Iteration 589: Arm = 2, Reward = 1.0
Iteration 590: Arm = 2, Reward = 1.0
Iteration 591: Arm = 2, Reward = 1.0
Iteration 592: Arm = 2, Reward = 0.0
Iteration 593: Arm = 2, Reward = 0.0
Iteration 594: Arm = 2, Reward = 1.0
Iteration 595: Arm = 2, Reward = 0.0
Iteration 596: Arm = 3, Reward = 0.0
Iteration 597: Arm = 1, Reward = 0.0
Iteration 598: Arm = 2, Reward = 1.0
Iteration 599: Arm = 2, Reward = 1.0
Iteration 600: Arm = 2, Reward = 0.0
Iteration 601: Arm = 2, Reward = 0.0
Iteration 602: Arm = 1, Reward = 0.0
Iteration 603: Arm = 3, Reward = 1.0
Iteration 604: Arm = 3, Reward = 1.0
Iteration 605: Arm = 3, Reward = 1.0
Iteration 606: Arm = 3, Reward = 0.0
Iteration 607: Arm = 3, Reward = 0.0
Iteration 608: Arm = 0, Reward = 1.0
Iteration 609: Arm = 0, Reward = 0.0
Iteration 610: Arm = 3, Reward = 0.0
Iteration 611: Arm = 2, Reward = 0.0
Iteration 612: Arm = 0, Reward = 0.0
Iteration 613: Arm = 1, Reward = 0.0
Iteration 614: Arm = 1, Reward = 0.0
Iteration 615: Arm = 3, Reward = 1.0
Iteration 616: Arm = 3, Reward = 1.0
```

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Iteration 617: Arm = 3, Reward = 0.0
Iteration 618: Arm = 3, Reward = 1.0
Iteration 619: Arm = 3, Reward = 1.0
Iteration 620: Arm = 3, Reward = 0.0
Iteration 621: Arm = 3, Reward = 0.0
Iteration 622: Arm = 1, Reward = 1.0
Iteration 623: Arm = 1, Reward = 1.0
Iteration 624: Arm = 1, Reward = 1.0
Iteration 625: Arm = 1, Reward = 0.0
Iteration 626: Arm = 1, Reward = 0.0
Iteration 627: Arm = 2, Reward = 0.0
Iteration 628: Arm = 0, Reward = 1.0
Iteration 629: Arm = 0, Reward = 1.0
Iteration 630: Arm = 0, Reward = 0.0
Iteration 631: Arm = 0, Reward = 0.0
Iteration 632: Arm = 3, Reward = 1.0
Iteration 633: Arm = 3, Reward = 1.0
Iteration 634: Arm = 3, Reward = 0.0
Iteration 635: Arm = 3, Reward = 1.0
Iteration 636: Arm = 3, Reward = 0.0
Iteration 637: Arm = 1, Reward = 1.0
Iteration 638: Arm = 1, Reward = 1.0
Iteration 639: Arm = 1, Reward = 1.0
Iteration 640: Arm = 1, Reward = 0.0
Iteration 641: Arm = 1, Reward = 1.0
Iteration 642: Arm = 1, Reward = 0.0
Iteration 643: Arm = 1, Reward = 1.0
Iteration 644: Arm = 1, Reward = 0.0
Iteration 645: Arm = 3, Reward = 0.0
Iteration 646: Arm = 1, Reward = 0.0
Iteration 647: Arm = 0, Reward = 0.0
Iteration 648: Arm = 1, Reward = 0.0
Iteration 649: Arm = 3, Reward = 0.0
Iteration 650: Arm = 2, Reward = 0.0
Iteration 651: Arm = 1, Reward = 0.0
Iteration 652: Arm = 3, Reward = 0.0
Iteration 653: Arm = 1, Reward = 1.0
Iteration 654: Arm = 1, Reward = 0.0
Iteration 655: Arm = 0, Reward = 0.0
Iteration 656: Arm = 1, Reward = 1.0
Iteration 657: Arm = 1, Reward = 0.0
Iteration 658: Arm = 3, Reward = 0.0
Iteration 659: Arm = 1, Reward = 0.0
Iteration 660: Arm = 2, Reward = 1.0
Iteration 661: Arm = 2, Reward = 0.0
Iteration 662: Arm = 2, Reward = 0.0
Iteration 663: Arm = 3, Reward = 1.0
Iteration 664: Arm = 3, Reward = 0.0
Iteration 665: Arm = 1, Reward = 0.0
Iteration 666: Arm = 0, Reward = 0.0
Iteration 667: Arm = 3, Reward = 0.0
Iteration 668: Arm = 1, Reward = 0.0
Iteration 669: Arm = 3, Reward = 0.0
Iteration 670: Arm = 1, Reward = 0.0
Iteration 671: Arm = 2, Reward = 1.0
Iteration 672: Arm = 2, Reward = 0.0
Iteration 673: Arm = 1, Reward = 0.0
Iteration 674: Arm = 0, Reward = 1.0
Iteration 675: Arm = 0, Reward = 1.0
Iteration 676: Arm = 0, Reward = 1.0
Iteration 677: Arm = 0, Reward = 0.0
Iteration 678: Arm = 0, Reward = 1.0
Iteration 679: Arm = 0, Reward = 1.0
Iteration 680: Arm = 0, Reward = 0.0
Iteration 681: Arm = 0, Reward = 1.0
Iteration 682: Arm = 0, Reward = 0.0
Iteration 683: Arm = 0, Reward = 0.0
Iteration 684: Arm = 0, Reward = 0.0
Iteration 685: Arm = 3, Reward = 1.0
Iteration 686: Arm = 3, Reward = 1.0
Iteration 687: Arm = 3, Reward = 1.0
Iteration 688: Arm = 3, Reward = 1.0
Iteration 689: Arm = 3, Reward = 1.0
Iteration 690: Arm = 3, Reward = 1.0
Iteration 691: Arm = 3, Reward = 0.0
Iteration 692: Arm = 3, Reward = 0.0
Iteration 693: Arm = 3, Reward = 0.0
Iteration 694: Arm = 3, Reward = 0.0
Iteration 695: Arm = 2, Reward = 1.0
Iteration 696: Arm = 2, Reward = 0.0
Iteration 697: Arm = 2, Reward = 0.0
Iteration 698: Arm = 1, Reward = 1.0
Iteration 699: Arm = 1, Reward = 0.0
```

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Iteration 700: Arm = 3, Reward = 0.0
Iteration 701: Arm = 1, Reward = 0.0
Iteration 702: Arm = 0, Reward = 0.0
Iteration 703: Arm = 3, Reward = 1.0
Iteration 704: Arm = 3, Reward = 0.0
Iteration 705: Arm = 1, Reward = 1.0
Iteration 706: Arm = 1, Reward = 0.0
Iteration 707: Arm = 3, Reward = 1.0
Iteration 708: Arm = 3, Reward = 1.0
Iteration 709: Arm = 3, Reward = 0.0
Iteration 710: Arm = 3, Reward = 0.0
Iteration 711: Arm = 1, Reward = 0.0
Iteration 712: Arm = 2, Reward = 0.0
Iteration 713: Arm = 1, Reward = 0.0
Iteration 714: Arm = 3, Reward = 0.0
Iteration 715: Arm = 0, Reward = 0.0
Iteration 716: Arm = 1, Reward = 1.0
Iteration 717: Arm = 1, Reward = 0.0
Iteration 718: Arm = 3, Reward = 1.0
Iteration 719: Arm = 3, Reward = 0.0
Iteration 720: Arm = 1, Reward = 1.0
Iteration 721: Arm = 1, Reward = 1.0
Iteration 722: Arm = 1, Reward = 1.0
Iteration 723: Arm = 1, Reward = 0.0
Iteration 724: Arm = 1, Reward = 0.0
Iteration 725: Arm = 3, Reward = 0.0
Iteration 726: Arm = 1, Reward = 0.0
Iteration 727: Arm = 2, Reward = 0.0
Iteration 728: Arm = 0, Reward = 1.0
Iteration 729: Arm = 0, Reward = 1.0
Iteration 730: Arm = 0, Reward = 0.0
Iteration 731: Arm = 0, Reward = 0.0
Iteration 732: Arm = 1, Reward = 1.0
Iteration 733: Arm = 1, Reward = 0.0
Iteration 734: Arm = 3, Reward = 0.0
Iteration 735: Arm = 1, Reward = 0.0
Iteration 736: Arm = 0, Reward = 0.0
Iteration 737: Arm = 1, Reward = 0.0
Iteration 738: Arm = 3, Reward = 0.0
Iteration 739: Arm = 2, Reward = 0.0
Iteration 740: Arm = 1, Reward = 1.0
Iteration 741: Arm = 1, Reward = 0.0
Iteration 742: Arm = 3, Reward = 1.0
Iteration 743: Arm = 3, Reward = 0.0
Iteration 744: Arm = 1, Reward = 0.0
Iteration 745: Arm = 0, Reward = 0.0
Iteration 746: Arm = 3, Reward = 0.0
Iteration 747: Arm = 1, Reward = 0.0
Iteration 748: Arm = 2, Reward = 0.0
Iteration 749: Arm = 3, Reward = 0.0
Iteration 750: Arm = 1, Reward = 1.0
Iteration 751: Arm = 1, Reward = 0.0
Iteration 752: Arm = 1, Reward = 1.0
Iteration 753: Arm = 1, Reward = 0.0
Iteration 754: Arm = 0, Reward = 1.0
Iteration 755: Arm = 0, Reward = 0.0
Iteration 756: Arm = 1, Reward = 1.0
Iteration 757: Arm = 1, Reward = 0.0
Iteration 758: Arm = 3, Reward = 1.0
Iteration 759: Arm = 3, Reward = 0.0
Iteration 760: Arm = 0, Reward = 0.0
Iteration 761: Arm = 1, Reward = 1.0
Iteration 762: Arm = 1, Reward = 1.0
Iteration 763: Arm = 1, Reward = 1.0
Iteration 764: Arm = 1, Reward = 0.0
Iteration 765: Arm = 1, Reward = 1.0
Iteration 766: Arm = 1, Reward = 1.0
Iteration 767: Arm = 1, Reward = 1.0
Iteration 768: Arm = 1, Reward = 1.0
Iteration 769: Arm = 1, Reward = 1.0
Iteration 770: Arm = 1, Reward = 0.0
Iteration 771: Arm = 1, Reward = 1.0
Iteration 772: Arm = 1, Reward = 1.0
Iteration 773: Arm = 1, Reward = 1.0
Iteration 774: Arm = 1, Reward = 0.0
Iteration 775: Arm = 1, Reward = 1.0
Iteration 776: Arm = 1, Reward = 1.0
Iteration 777: Arm = 1, Reward = 1.0
Iteration 778: Arm = 1, Reward = 0.0
Iteration 779: Arm = 1, Reward = 0.0
Iteration 780: Arm = 1, Reward = 1.0
Iteration 781: Arm = 1, Reward = 0.0
Iteration 782: Arm = 1, Reward = 0.0
```

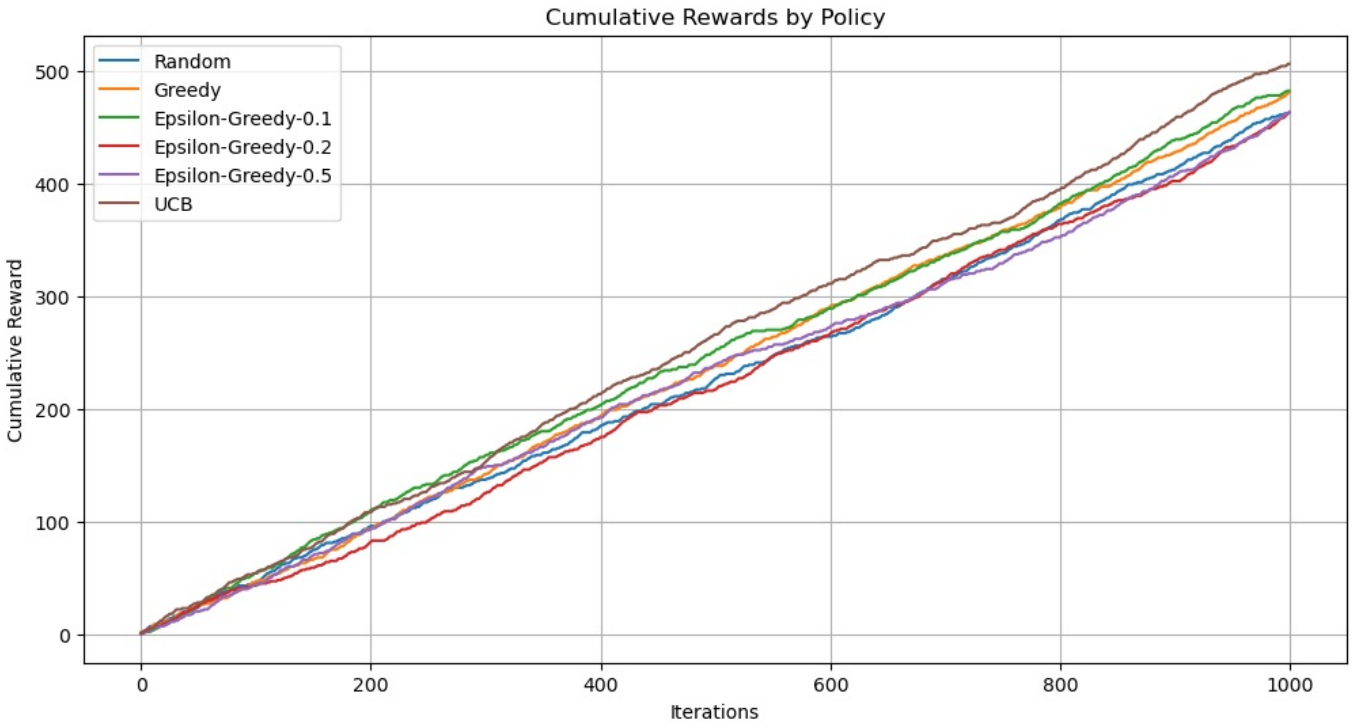
```
Iteration 783: Arm = 1, Reward = 1.0
Iteration 784: Arm = 1, Reward = 0.0
Iteration 785: Arm = 1, Reward = 0.0
Iteration 786: Arm = 3, Reward = 0.0
Iteration 787: Arm = 1, Reward = 1.0
Iteration 788: Arm = 1, Reward = 1.0
Iteration 789: Arm = 1, Reward = 0.0
Iteration 790: Arm = 1, Reward = 1.0
Iteration 791: Arm = 1, Reward = 1.0
Iteration 792: Arm = 1, Reward = 1.0
Iteration 793: Arm = 1, Reward = 0.0
Iteration 794: Arm = 1, Reward = 0.0
Iteration 795: Arm = 2, Reward = 1.0
Iteration 796: Arm = 2, Reward = 0.0
Iteration 797: Arm = 1, Reward = 1.0
Iteration 798: Arm = 1, Reward = 1.0
Iteration 799: Arm = 1, Reward = 0.0
Iteration 800: Arm = 1, Reward = 0.0
Iteration 801: Arm = 2, Reward = 1.0
Iteration 802: Arm = 2, Reward = 1.0
Iteration 803: Arm = 2, Reward = 0.0
Iteration 804: Arm = 2, Reward = 0.0
Iteration 805: Arm = 1, Reward = 1.0
Iteration 806: Arm = 1, Reward = 1.0
Iteration 807: Arm = 1, Reward = 1.0
Iteration 808: Arm = 1, Reward = 0.0
Iteration 809: Arm = 1, Reward = 1.0
Iteration 810: Arm = 1, Reward = 1.0
Iteration 811: Arm = 1, Reward = 1.0
Iteration 812: Arm = 1, Reward = 0.0
Iteration 813: Arm = 1, Reward = 1.0
Iteration 814: Arm = 1, Reward = 1.0
Iteration 815: Arm = 1, Reward = 1.0
Iteration 816: Arm = 1, Reward = 0.0
Iteration 817: Arm = 1, Reward = 0.0
Iteration 818: Arm = 1, Reward = 1.0
Iteration 819: Arm = 1, Reward = 0.0
Iteration 820: Arm = 1, Reward = 1.0
Iteration 821: Arm = 1, Reward = 1.0
Iteration 822: Arm = 1, Reward = 1.0
Iteration 823: Arm = 1, Reward = 1.0
Iteration 824: Arm = 1, Reward = 0.0
Iteration 825: Arm = 1, Reward = 0.0
Iteration 826: Arm = 1, Reward = 1.0
Iteration 827: Arm = 1, Reward = 0.0
Iteration 828: Arm = 1, Reward = 1.0
Iteration 829: Arm = 1, Reward = 0.0
Iteration 830: Arm = 3, Reward = 0.0
Iteration 831: Arm = 1, Reward = 0.0
Iteration 832: Arm = 0, Reward = 0.0
Iteration 833: Arm = 2, Reward = 1.0
Iteration 834: Arm = 2, Reward = 1.0
Iteration 835: Arm = 2, Reward = 0.0
Iteration 836: Arm = 2, Reward = 0.0
Iteration 837: Arm = 1, Reward = 1.0
Iteration 838: Arm = 1, Reward = 1.0
Iteration 839: Arm = 1, Reward = 0.0
Iteration 840: Arm = 1, Reward = 1.0
Iteration 841: Arm = 1, Reward = 0.0
Iteration 842: Arm = 3, Reward = 0.0
Iteration 843: Arm = 1, Reward = 1.0
Iteration 844: Arm = 1, Reward = 1.0
Iteration 845: Arm = 1, Reward = 1.0
Iteration 846: Arm = 1, Reward = 1.0
Iteration 847: Arm = 1, Reward = 1.0
Iteration 848: Arm = 1, Reward = 0.0
Iteration 849: Arm = 1, Reward = 0.0
Iteration 850: Arm = 1, Reward = 1.0
Iteration 851: Arm = 1, Reward = 0.0
Iteration 852: Arm = 1, Reward = 1.0
Iteration 853: Arm = 1, Reward = 1.0
Iteration 854: Arm = 1, Reward = 1.0
Iteration 855: Arm = 1, Reward = 0.0
Iteration 856: Arm = 1, Reward = 0.0
Iteration 857: Arm = 1, Reward = 1.0
Iteration 858: Arm = 1, Reward = 1.0
Iteration 859: Arm = 1, Reward = 1.0
Iteration 860: Arm = 1, Reward = 1.0
Iteration 861: Arm = 1, Reward = 0.0
Iteration 862: Arm = 1, Reward = 1.0
Iteration 863: Arm = 1, Reward = 1.0
Iteration 864: Arm = 1, Reward = 1.0
Iteration 865: Arm = 1, Reward = 1.0
```

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Iteration 866: Arm = 1, Reward = 1.0
Iteration 867: Arm = 1, Reward = 1.0
Iteration 868: Arm = 1, Reward = 1.0
Iteration 869: Arm = 1, Reward = 1.0
Iteration 870: Arm = 1, Reward = 1.0
Iteration 871: Arm = 1, Reward = 0.0
Iteration 872: Arm = 1, Reward = 0.0
Iteration 873: Arm = 1, Reward = 0.0
Iteration 874: Arm = 1, Reward = 1.0
Iteration 875: Arm = 1, Reward = 1.0
Iteration 876: Arm = 1, Reward = 1.0
Iteration 877: Arm = 1, Reward = 0.0
Iteration 878: Arm = 1, Reward = 0.0
Iteration 879: Arm = 1, Reward = 1.0
Iteration 880: Arm = 1, Reward = 1.0
Iteration 881: Arm = 1, Reward = 1.0
Iteration 882: Arm = 1, Reward = 0.0
Iteration 883: Arm = 1, Reward = 0.0
Iteration 884: Arm = 1, Reward = 0.0
Iteration 885: Arm = 1, Reward = 1.0
Iteration 886: Arm = 1, Reward = 1.0
Iteration 887: Arm = 1, Reward = 1.0
Iteration 888: Arm = 1, Reward = 1.0
Iteration 889: Arm = 1, Reward = 1.0
Iteration 890: Arm = 1, Reward = 0.0
Iteration 891: Arm = 1, Reward = 0.0
Iteration 892: Arm = 1, Reward = 1.0
Iteration 893: Arm = 1, Reward = 1.0
Iteration 894: Arm = 1, Reward = 1.0
Iteration 895: Arm = 1, Reward = 0.0
Iteration 896: Arm = 1, Reward = 1.0
Iteration 897: Arm = 1, Reward = 1.0
Iteration 898: Arm = 1, Reward = 1.0
Iteration 899: Arm = 1, Reward = 0.0
Iteration 900: Arm = 1, Reward = 1.0
Iteration 901: Arm = 1, Reward = 1.0
Iteration 902: Arm = 1, Reward = 1.0
Iteration 903: Arm = 1, Reward = 0.0
Iteration 904: Arm = 1, Reward = 0.0
Iteration 905: Arm = 1, Reward = 0.0
Iteration 906: Arm = 1, Reward = 0.0
Iteration 907: Arm = 1, Reward = 1.0
Iteration 908: Arm = 1, Reward = 1.0
Iteration 909: Arm = 1, Reward = 1.0
Iteration 910: Arm = 1, Reward = 0.0
Iteration 911: Arm = 1, Reward = 1.0
Iteration 912: Arm = 1, Reward = 1.0
Iteration 913: Arm = 1, Reward = 0.0
Iteration 914: Arm = 1, Reward = 0.0
Iteration 915: Arm = 2, Reward = 1.0
Iteration 916: Arm = 2, Reward = 0.0
Iteration 917: Arm = 1, Reward = 1.0
Iteration 918: Arm = 1, Reward = 1.0
Iteration 919: Arm = 1, Reward = 1.0
Iteration 920: Arm = 1, Reward = 1.0
Iteration 921: Arm = 1, Reward = 1.0
Iteration 922: Arm = 1, Reward = 0.0
Iteration 923: Arm = 1, Reward = 1.0
Iteration 924: Arm = 1, Reward = 0.0
Iteration 925: Arm = 1, Reward = 1.0
Iteration 926: Arm = 1, Reward = 1.0
Iteration 927: Arm = 1, Reward = 0.0
Iteration 928: Arm = 1, Reward = 1.0
Iteration 929: Arm = 1, Reward = 1.0
Iteration 930: Arm = 1, Reward = 1.0
Iteration 931: Arm = 1, Reward = 1.0
Iteration 932: Arm = 1, Reward = 1.0
Iteration 933: Arm = 1, Reward = 1.0
Iteration 934: Arm = 1, Reward = 0.0
Iteration 935: Arm = 1, Reward = 1.0
Iteration 936: Arm = 1, Reward = 0.0
Iteration 937: Arm = 1, Reward = 1.0
Iteration 938: Arm = 1, Reward = 0.0
Iteration 939: Arm = 1, Reward = 1.0
Iteration 940: Arm = 1, Reward = 0.0
Iteration 941: Arm = 1, Reward = 0.0
Iteration 942: Arm = 1, Reward = 1.0
Iteration 943: Arm = 1, Reward = 1.0
Iteration 944: Arm = 1, Reward = 0.0
Iteration 945: Arm = 1, Reward = 0.0
Iteration 946: Arm = 1, Reward = 1.0
Iteration 947: Arm = 1, Reward = 1.0
Iteration 948: Arm = 1, Reward = 0.0
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Iteration 949: Arm = 1, Reward = 1.0
Iteration 950: Arm = 1, Reward = 0.0
Iteration 951: Arm = 2, Reward = 1.0
Iteration 952: Arm = 2, Reward = 0.0
Iteration 953: Arm = 1, Reward = 0.0
Iteration 954: Arm = 0, Reward = 1.0
Iteration 955: Arm = 0, Reward = 0.0
Iteration 956: Arm = 2, Reward = 1.0
Iteration 957: Arm = 2, Reward = 0.0
Iteration 958: Arm = 1, Reward = 1.0
Iteration 959: Arm = 1, Reward = 0.0
Iteration 960: Arm = 3, Reward = 0.0
Iteration 961: Arm = 1, Reward = 1.0
Iteration 962: Arm = 1, Reward = 1.0
Iteration 963: Arm = 1, Reward = 0.0
Iteration 964: Arm = 1, Reward = 0.0
Iteration 965: Arm = 0, Reward = 1.0
Iteration 966: Arm = 0, Reward = 0.0
Iteration 967: Arm = 2, Reward = 0.0
Iteration 968: Arm = 0, Reward = 1.0
Iteration 969: Arm = 0, Reward = 1.0
Iteration 970: Arm = 0, Reward = 1.0
Iteration 971: Arm = 0, Reward = 0.0
Iteration 972: Arm = 0, Reward = 0.0
Iteration 973: Arm = 0, Reward = 0.0
Iteration 974: Arm = 1, Reward = 0.0
Iteration 975: Arm = 3, Reward = 0.0
Iteration 976: Arm = 1, Reward = 1.0
Iteration 977: Arm = 1, Reward = 0.0
Iteration 978: Arm = 1, Reward = 0.0
Iteration 979: Arm = 0, Reward = 0.0
Iteration 980: Arm = 1, Reward = 0.0
Iteration 981: Arm = 3, Reward = 0.0
Iteration 982: Arm = 2, Reward = 1.0
Iteration 983: Arm = 2, Reward = 1.0
Iteration 984: Arm = 2, Reward = 0.0
Iteration 985: Arm = 2, Reward = 0.0
Iteration 986: Arm = 1, Reward = 1.0
Iteration 987: Arm = 1, Reward = 0.0
Iteration 988: Arm = 1, Reward = 0.0
Iteration 989: Arm = 2, Reward = 1.0
Iteration 990: Arm = 2, Reward = 0.0
Iteration 991: Arm = 3, Reward = 1.0
Iteration 992: Arm = 3, Reward = 0.0
Iteration 993: Arm = 2, Reward = 1.0
Iteration 994: Arm = 2, Reward = 0.0
Iteration 995: Arm = 1, Reward = 0.0
Iteration 996: Arm = 3, Reward = 0.0
Iteration 997: Arm = 0, Reward = 0.0
Iteration 998: Arm = 2, Reward = 1.0
Iteration 999: Arm = 2, Reward = 1.0
Iteration 1000: Arm = 2, Reward = 0.0
```

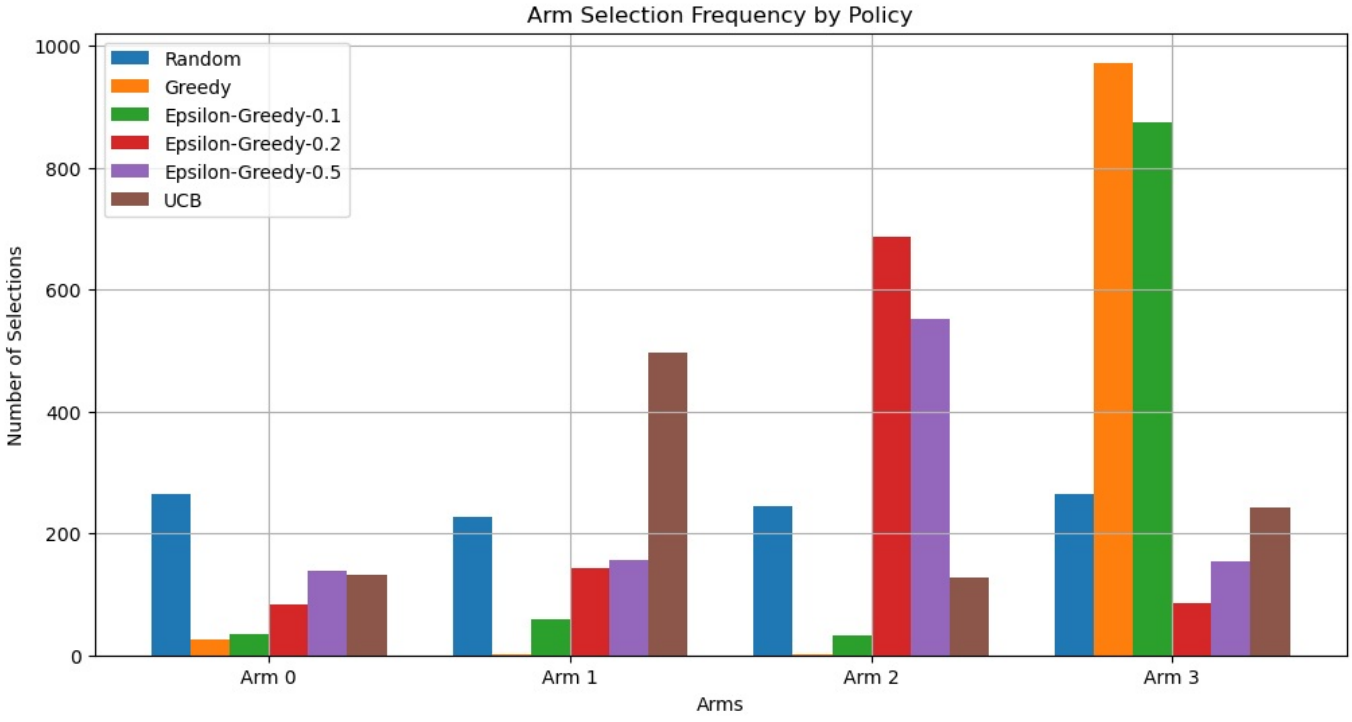
In [5]:

```python
# --- PLOTTING CUMULATIVE REWARDS ---
plt.figure(figsize=(12, 6))
for policy, rewards in policies_rewards.items():
    cumulative = np.cumsum(rewards)
    plt.plot(cumulative, label=policy)
plt.xlabel("Iterations")
plt.ylabel("Cumulative Reward")
plt.title("Cumulative Rewards by Policy")
plt.legend()
plt.grid(True)
plt.show()
```



In [6]:
```python
# --- PLOTTING ARM SELECTION FREQUENCY ---
labels = [f"Arm {i}" for i in range(num_arms)]
x = np.arange(num_arms)
width = 0.13

plt.figure(figsize=(12, 6))
for idx, (policy, counts) in enumerate(arm_counts.items()):
    plt.bar(x + idx * width, counts, width, label=policy)
plt.xlabel("Arms")
plt.ylabel("Number of Selections")
plt.title("Arm Selection Frequency by Policy")
plt.xticks(x + width * 2.5, labels)
plt.legend()
plt.grid(True)
plt.show()
```



Conclusion

After implementing and comparing multiple Multi-Armed Bandit (MAB) strategies—Greedy, Epsilon-Greedy (ε = 0.1, 0.2, 0.5), and Upper Confidence Bound (UCB)—the following insights were drawn:

Greedy Policy tends to get stuck in suboptimal arms because it lacks exploration.

- Epsilon-Greedy Policies strike a better balance between exploration and exploitation. Among the variations, ε = 0.1 gave consistently

higher cumulative rewards due to controlled exploration.

- UCB Policy performed very well overall by using confidence bounds to manage exploration. It dynamically reduces exploration as confidence increases.

Recommended Policy: UCB demonstrated the best cumulative performance, making it the most effective strategy for this simulated clinical trial scenario.